

CHAPTER 4

LEAST-SQUARES ADJUSTMENTS

Least-squares adjustment is a device for carrying out objective quality control of measurements by processing sets of redundant observations according to mathematically well-defined rules. The objectivity of least-squares quality control is especially useful when depositing or exchanging observations. Least-squares solutions require redundant observations, i.e., more observations are required than are necessary to determine a set of unknowns exactly. Details will be given as to what constitutes optimal redundancy. This chapter contains compact but complete derivations of least-squares algorithms.

First, the statistical nature of measurements is analyzed, followed by a discussion of stochastic and mathematical models and the law of variance-covariance propagation of random variables. The mixed adjustment model is derived in detail, and the observation equation and the condition equation models are deduced from the mixed model through appropriate specification. The cases of observed and weighted parameters are presented as well. A special section is devoted to minimal and inner constraint solutions and to those quantities that remain invariant with respect to a change in minimal constraints. Whenever the goal is to perform quality control on the observations, minimal or inner constraint solutions are especially relevant. Statistical testing is important for judging the quality of observations or the outcome of an adjustment. A separate section deals with statistics in least-squares adjustments. The chapter ends with a presentation of additional quality measures, such as internal and external reliability and blunder detection and a brief exposition of Kalman filtering.

4.1 ELEMENTS

Objective quality control of observations is necessary when dealing with any kind of measurements such as angles, distances, pseudoranges, carrier phases, and the geopotential. It is best to separate conceptually quality control of observations and precision or accuracy of parameters. It is unfortunate that least-squares adjustment is most often associated only with high-precision surveying. It may be as important to discover and remove a 10 m blunder in a low-precision survey as a 1 cm blunder in a high-precision survey.

Least-squares adjustment allows the combination of different types of observations (such as angles, distances, and height differences) into one solution and permits simultaneous statistical analysis. For example, there is no need to treat traverses, intersections, and resections separately. Since these geometric figures consist of angle and distance measurements, the least-square rules apply to all of them, regardless of the specific arrangements of the observations or the geometric shape they represent.

Least-squares adjustment simulation is a useful tool to plan a survey and to ensure that accuracy specifications will be met once the actual observations have been made. Simulations allow the observation selection to be optimized when alternatives exist. For example, should one primarily measure angles or rely on distances? Considering the available instrumentation, what is the optimal use of the equipment under the constraints of the project? Experienced surveyors often answer these questions intuitively. Even in these cases, an objective verification using least-squares simulation and the concept of internal and external reliability of networks is a welcome assurance to those who carry responsibility for the project.

4.1.1 Statistical Nature of Surveying Measurement

Assume that a distance of 100 m is measured repeatedly with a tape that has centimeter divisions. A likely outcome of these measurements could be 99.99, 100.02, 100.00, 100.01, etc. Because of the centimeter subdivision of the tape, the surveyor is likely to record the observations to two decimal places. The result therefore is a series of numbers ending with two decimal places. One could wrongly conclude that this measurement process belongs to the realm of discrete statistics yielding discrete outcomes with two decimal places. In reality, however, the series is given two decimal places because of the centimeter division of the tape and the fact that the surveyor did not choose to estimate the millimeters. Imagining a reading device that allows us to read the tape to as many decimal places as desired, we readily see that the process of measuring a distance belongs to the realm of continuous statistics. The same is true for other types of measurements typically used in positioning. A classic textbook case for a discrete statistical process is the throwing of a die in which case the outcome is limited to integers.

When measuring the distance we recognize that any value x_i could be obtained, although experience tells us that values close to 100.00 are most likely. Values such as 99.90 or 100.25 are very unlikely when measured with care. Assume that n

measurements have been made and that they have been grouped into bins of length Δx , with bin i containing n_i observations. Graphing the bins in a coordinate system of relative frequency n_i/n versus x_i gives the histogram. For surveying measurements, the smoothed step function of the rectangular bins typically has a bell-like shape. The maximum occurs around the sample mean. The larger the deviation from the mean, the smaller the relative frequency, i.e., the probability that such a measurement will actually be obtained. A goodness-of-fit test would normally confirm the hypothesis that the observations have a normal distribution. Thus, the typical measurement process in surveying follows the statistical law of normal distribution.

4.1.2 Elementary Statistical Concepts

Several concepts from statistics are required in least-squares adjustment. The following is a partial listing of frequently used concepts and terminology:

- **Observation:** An observation, or a statistical event, is the outcome of a statistical experiment, e.g., throwing a dice or measuring an angle or a distance.
- **Random Variable:** A random variable is the outcome of an event. The random variable is denoted by a tilde. Thus, $\tilde{x}$ is a random variable and $\tilde{\mathbf{x}}$ is a vector of random variables. However, we will frequently not use the tilde to simplify the notation when it is unambiguous which symbol represents the random variable.
- **Population:** The population is the totality of all events. It includes all possible values that the random variable can have. The population is described by a finite set of parameters, called the population parameters. The normal distribution, e.g., describes such a population and is completely specified by the mean and the variance.
- **Sample:** A sample is a subset of the population. For example, if the same distance is measured ten times, then these ten measurements are a sample of all the possible measurements.
- **A statistic represents an estimate of the population parameters or functions of these parameters.** It is computed from a sample. For example, the ten measurements of the same distance can be used to estimate the mean and the variance of the normal distribution.
- **Probability:** Probability is related to the frequency of occurrence of a specific event. Each value of the random variable has an associated probability.
- **Probability Density Function:** The probability density function relates the probability to the possible values of the random variable. If $f(x)$ denotes the probability density function, then

$$P(a \leq \tilde{x} \leq b) = \int_a^b f(x) dx \quad (4.1)$$

is the probability that the random variable $\tilde{x}$ assumes a value in the interval $[a, b]$.

4.1.3 Observational Errors

Field observations are not perfect, and neither are the recordings and management of observations. The measurement process suffers from several error sources. Repeated measurements do not yield identical numerical values because of random measurement errors. These errors are usually small, and the probability of a positive or a negative error of a given magnitude is the same (equal frequency of occurrence). Random errors are inherent in the nature of measurements and can never be completely overcome. Random errors are dealt with in least-squares adjustment.

Systematic errors are errors that vary systematically in sign and/or magnitude. Examples are a tape that is 10 cm too short or the failure to correct for vertical or lateral refraction in angular measurement. Systematic errors are particularly dangerous because they tend to accumulate. Adequate instrument calibration, care when observing, such as double centering, and observing under various external conditions help avoid systematic errors. If the errors are known, the observations can be corrected before making the adjustment; otherwise, one might attempt to model and estimate these errors. Discovering and dealing with systematic errors requires a great deal of experience with the data. Success is not at all guaranteed.

Blunders are usually large errors resulting from carelessness. Examples of blunders are counting errors in a whole tape length, transposing digits when recording field observations, continuing measurements after upsetting the tripod, and so on. Blunders can largely be avoided through careful observation, although there can never be absolute certainty that all blunders have been avoided or eliminated. Therefore, an important part of least-squares adjustment is to discover and remove remaining blunders in the observations.

4.1.4 Accuracy and Precision

Accuracy refers to the closeness of the observations (or the quantities derived from the observations) to the true value. Precision refers to the closeness of repeated observations (or quantities derived from repeated sets of observations) to the sample mean. Figure 4.1 shows four density functions that represent four distinctly different measurement processes of the same quantity. Curves 1 and 2 are symmetric with respect to the true value x_T . These measurements have a high accuracy, because the sample mean coincides or is very close to the true value. However, the shapes of both curves are quite different. Curve 1 is tall and narrow, whereas curve 2 is short and broad. The observations of process 1 are clustered closely around the mean (true value), whereas the spread of observations around the mean is larger for process 2. Larger deviations from the true value occur more frequently for process 2 than for process 1. Thus, process 1 is more precise than process 2; however, both processes are equally accurate. Curves 3 and 4 are symmetric with respect to the sample mean x_S , which differs from the true value x_T . Both sequences have equally low accuracy, but the precision of process 3 is higher than that of process 4. The difference $x_T - x_S$ is caused by a systematic error. An increase in the number of observations does not reduce this difference.

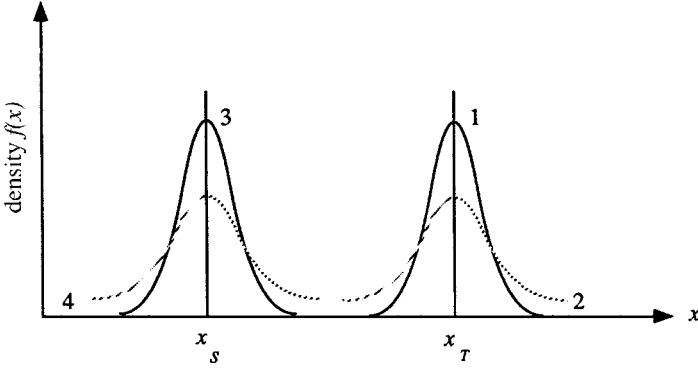


Figure 4.1 Accuracy and precision.

4.2 STOCHASTIC AND MATHEMATICAL MODELS

Least-squares adjustment deals with two equally important components: the stochastic model and the mathematical model. Both components are indispensable and contribute to the adjustment algorithm (see Figure 4.2). We denote the vector of observation with ℓ_b , and the number of observations by n . The observations are random variables; thus the complete notation for the $n \times 1$ vector of observations is $\tilde{\ell}_b$. To simplify the notation, we do not use the tilde in connection with ℓ_b . The true value of the observations, i.e., the means of the populations, are estimated from the sample measurements. Since each observation belongs to a different population, the sample size is usually 1. The variances of these distributions comprise the stochastic model. It introduces information about the precision of the observations (or accuracy if only random errors are present). The variance-covariance matrix Σ_{ℓ_b} expresses the stochastic model. In many cases, the observations are not correlated and the variance-covariance matrix is diagonal. Occasionally, when so-called derived observations are used which are the outcome from a previous adjustment, or when linear combinations of original observations are adjusted, the variance-covariance matrix contains off-diagonal elements. Because in surveying the observations are normal distributed, the vector of observations has a multivariate normal distribution. We use the notation

$$\ell_b \sim N(\ell_T, \Sigma_{\ell_b}) \quad (4.2)$$

where ℓ_T is the vector mean of the population. The cofactor matrix of the observations $\mathbf{Q}_{\ell_b}$ and the weight matrix $\mathbf{P}$ are defined by

$$\mathbf{Q}_{\ell_b} = \frac{1}{\sigma_0^2} \Sigma_{\ell_b} \quad (4.3)$$

$$\mathbf{P} = \mathbf{Q}_{\ell_b}^{-1} = \sigma_0^2 \Sigma_{\ell_b}^{-1} \quad (4.4)$$

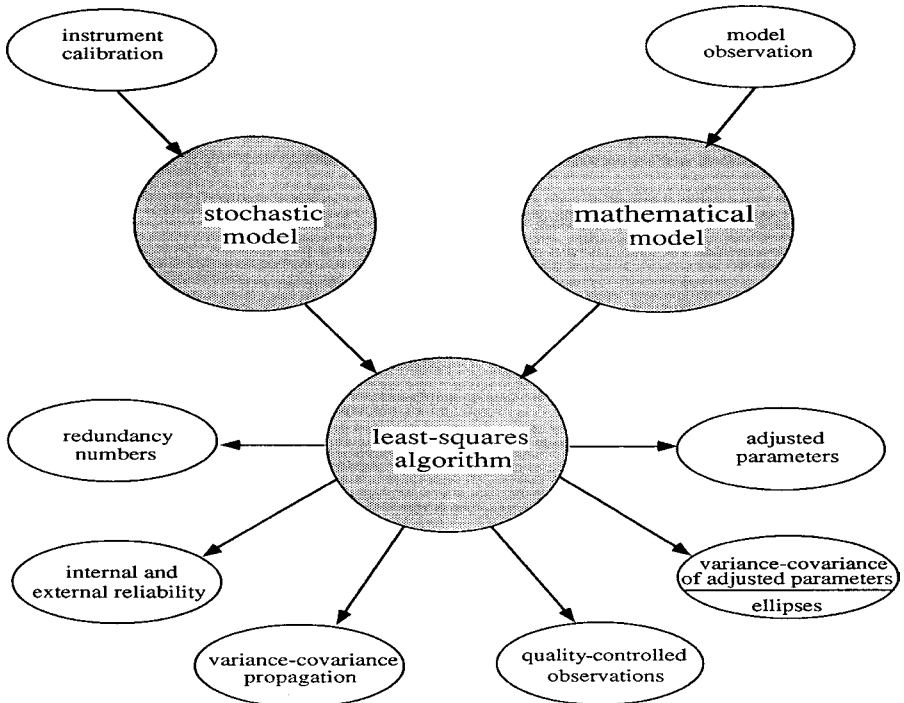


Figure 4.2 Elements of least-squares adjustment.

Typically we do not use a subscript to identify $\mathbf{P}$ as the weight matrix of the observations. The symbol σ_0^2 denotes the a priori variance of unit weight. It relates the weight matrix and the inverted covariance matrix. An important capability of least-squares adjustment is the estimation of σ_0^2 from observations. We denote that estimate by $\hat{\sigma}_0^2$ and it is the a posteriori variance of unit weight. If the a priori and a posteriori variances of unit weight are statistically equal, the adjustment is said to be correct. More on this fundamental statistical test and its implications will follow in later sections. In general, the a priori variance of unit weight σ_0^2 is set to 1; i.e., the weight matrix is equated with the inverse of the variance-covariance matrix of the observations. The term *variance of unit weight* is derived from the fact that if the variance of an observation equals σ_0^2 , then the weight for this observation equals unity. The special cases that $\mathbf{P}$ equals the identity matrix, $\mathbf{P} = \mathbf{I}$, frequently allow a simple and geometrically intuitive interpretation of the minimization.

The mathematical model expresses a simplification of existing physical reality. It attempts to express mathematically the relations between observations and parameters (unknowns) such as coordinates, heights, and refraction coefficients. Least-squares adjustment is a very general tool that can be used whenever a relationship between observations and parameters has been established. Even though the mathematical model is well known for many routine applications, there are always new

cases that require a new mathematical model. Finding the right mathematical model can be a challenge.

Much research has gone into establishing a mathematical formulation that is general enough to deal with all types of globally distributed measurement in a unified model. The collection of observations might include distances, angles, heights, gravity anomalies, gravity gradients, geopotential differences, astronomical observations, and GPS observations. The mathematical models become simpler if one does not deal with all types of observations at the same time but instead uses additional external information. See Chapter 2 for a detailed discussion on the 3D geodetic model.

A popular approach is to reduce (modify) the original observations to be compatible with the mathematical model. These are the model observations. For example, if measured vertical angles are used, the mathematical model must include refraction parameters. On the other hand, the original measurements can be corrected for refraction using an atmospheric refraction model. The thus reduced observations refer to a simpler model that does not require refraction parameters. The more reductions are applied to the original observation, the less general the respective mathematical model is. The final form of the model also depends on the purpose of the adjustment. For example, if the objective is to study refraction, one needs refraction parameters in the model. In surveying applications where the objective typically is to determine location, one prefers not to deal with refraction parameters explicitly. The relation between observations and parameterization is central to the success of estimation and at times requires much attention.

In the most general case, the observations and the parameters are related by an implicit nonlinear function:

$$\mathbf{f}(\mathbf{x}_a, \ell_a) = \mathbf{0} \quad (4.5)$$

This is the mixed adjustment model. The subscript a is to be read as “adjusted.” The symbol ℓ_a denotes the $n \times 1$ vector of adjusted observations, and the vector $\mathbf{x}_a$ contains u adjusted parameters. There are r nonlinear mathematical functions in $\mathbf{f}$. Often the observations are explicitly related to the parameters, such as in

$$\ell_a = \mathbf{f}(\mathbf{x}_a) \quad (4.6)$$

This is the observation equation model. A further variation is the absence of any parameters as in

$$\mathbf{f}(\ell_a) = \mathbf{0} \quad (4.7)$$

This is the condition equation model.

The application usually dictates which model might be preferred. Selecting another model might require a mathematically more involved formulation. In the case of a leveling network, e.g., the observation equation model and the condition equation model can be applied with equal ease.

The observation equation model has the major advantage in that each observation adds one equation. This allows the observation equation model to be implemented

relatively easily and generally in software. One does not have to identify independent loop closures, etc.

Figure 4.2 indicates some of the outcomes from the adjustment. Statistical tests are available to verify the acceptance of the adjustment or aid in discovering and removing blunders. The adjustment provides probability regions for the estimated parameters and allows variance-covariance propagation to determine functions of the estimated parameters and the respective standard deviations. Of particular interest is the ability of the least-squares adjustment to perform internal and external reliability analysis, in order to quantify marginally detectable blunders and to determine their potential influence on the estimated parameters.

Statistical concepts enter the least-squares adjustment in two distinct ways. The actual least-squares solution merely requires the existence of the variance-covariance matrix; there is no need to specify a particular distribution for the observations. If statistical tests are required, then the distribution of the observations must be known. In most cases, one indeed desires to carry out some statistical testing.

4.3 VARIANCE-COVARIANCE PROPAGATION

The purpose of variance-covariance propagation is to compute the variances and covariances of linear functions of random variables. Nonlinear functions must first be linearized. Variance-covariance propagation is applicable to single random variables or to vectors of random variables.

Probability Density and Accumulative Probability For $f(x)$ to be a probability function of the random variable $\tilde{x}$, it has to fulfill certain conditions. First, $f(x)$ must be a nonnegative function, because there is *always* an outcome of an experiment; i.e., the observation can be positive, negative, or even zero. Second, the probability that a sample (observation) is one of all possible outcomes should be 1. Thus the density function $f(x)$ must fulfill the following conditions:

$$f(x) \geq 0 \quad (4.8)$$

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad (4.9)$$

The integration is taken over the whole range (population) of the random variable. Conditions (4.8) and (4.9) imply that the density function is zero at minus infinity and plus infinity. The probability

$$P(\tilde{x} \leq x) = F(x) = \int_{-\infty}^x f(t) dt \quad (4.10)$$

is called the cumulative distribution function. It is a nondecreasing function because of condition (4.8).

Mean The mean, also called the expected value of a continuously distributed random variable, is defined as

$$\mu_x = E(\tilde{x}) = \int_{-\infty}^{\infty} x f(x) dx \quad (4.11)$$

The mean is a function of the density function of the random variable. The integration is extended over the whole population. Equation (4.11) is the analogy to the weighted mean in the case of discrete distributions.

Variance The variance is defined by

$$\sigma_x^2 = E(\tilde{x} - \mu_x)^2 = \int_{-\infty}^{\infty} (x - \mu_x)^2 f(x) dx \quad (4.12)$$

The variance measures the spread of the probability density in the sense that it gives the expected value of the squared deviations from the mean. A small variance therefore indicates that most of the probability density is located around the mean.

Multivariate Distribution Any function $f(x_1, x_2, \dots, x_n)$ of n continuous variables $\tilde{x}_i$ can be a joint density function provided that

$$f(x_1, x_2, \dots, x_n) \geq 0 \quad (4.13)$$

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_1 \cdots dx_n = 1 \quad (4.14)$$

It follows as a natural extension from (4.10) that

$$P(\tilde{x}_1 < a_1, \dots, \tilde{x}_n < a_n) = \int_{-\infty}^{a_1} \cdots \int_{-\infty}^{a_n} f(x_1, x_2, \dots, x_n) dx_1 \cdots dx_n \quad (4.15)$$

The marginal density of a subset of random variables $(x_1, x_2, \dots, x_p)$ is

$$g(x_1, x_2, \dots, x_p) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_{p+1} dx_{p+2} \cdots dx_n \quad (4.16)$$

Stochastic Independence The concept of stochastic independence is required when dealing with multivariate distributions. Two sets of random variables, $(\tilde{x}_1, \dots, \tilde{x}_p)$ and $(\tilde{x}_{p+1}, \dots, \tilde{x}_n)$, are stochastically independent if the joint density function can be written as a product of the two respective marginal density functions, e.g.,

$$f(x_1, x_2, \dots, x_n) = g_1(x_1, x_2, \dots, x_p) g_2(x_{p+1}, x_{p+2}, \dots, x_n) \quad (4.17)$$

Vector of Means The expected value for the individual parameter x_i is

$$\mu_{x_i} = E(\tilde{x}_i) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} x_i f(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n \quad (4.18)$$

In vector notation the expected values of all parameters are

$$E(\tilde{\mathbf{x}}) = [E(\tilde{x}_1) \quad \dots \quad E(\tilde{x}_n)]^T \quad (4.19)$$

Variance The variance of an individual parameter is given by

$$\sigma_{x_i}^2 = E(\tilde{x}_i - \mu_{x_i})^2 = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x_i - \mu_{x_i})^2 f(x_1, x_2, \dots, x_n) dx_1 \dots dx_n \quad (4.20)$$

Covariance For multivariate distributions, another quantity called the covariance becomes important. The covariance describes the statistical relationship between two random variables. The covariance is

$$\begin{aligned} \sigma_{x_i, x_j} &= E[(x_i - \mu_{x_i})(x_j - \mu_{x_j})] \\ &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x_i - \mu_{x_i})(x_j - \mu_{x_j}) f(x_1, x_2, \dots, x_n) dx_1 \dots dx_n \end{aligned} \quad (4.21)$$

Whereas the variance is always larger than or equal to zero, the covariance can be negative, positive, or even zero.

Correlation Coefficients The correlation coefficient of two random variables is defined as

$$\rho_{x_i, x_j} = \frac{E[(\tilde{x}_i - \mu_{x_i})(\tilde{x}_j - \mu_{x_j})]}{\sigma_{x_i} \sigma_{x_j}} = \frac{\sigma_{x_i, x_j}}{\sigma_{x_i} \sigma_{x_j}} \quad (4.22)$$

Therefore, the correlation coefficient equals the covariance divided by the respective standard deviations. An important property of the correlation coefficient is that

$$-1 \leq \rho_{x_i, x_j} \leq 1 \quad (4.23)$$

If two random variables are stochastically independent, then the covariance (and thus the correlation coefficient) is zero. By making use of (4.17) for the density function of stochastically independent random variables, we can write (4.21) as

$$\begin{aligned} \sigma_{x_i, x_j} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x_i - \mu_{x_i})(x_j - \mu_{x_j}) g_i(x_i) g_j(x_j) dx_i dx_j \\ &= \int_{-\infty}^{\infty} (x_i - \mu_{x_i}) g_i(x_i) dx_i \int_{-\infty}^{\infty} (x_j - \mu_{x_j}) g_j(x_j) dx_j \end{aligned} \quad (4.24)$$

These integrals are zero because of the definition of the mean. The converse, i.e., zero correlation, implies stochastic independence is valid only for the multivariate normal distribution.

Variance-Covariance Matrix Equations (4.20) and (4.21) can be used to compute the variances and covariances for all components in the random vector $\tilde{\mathbf{x}}$. Arranging the result in the form of a matrix yields the variance-covariance matrix. Thus, for the random vector

$$\tilde{\mathbf{x}} - \boldsymbol{\mu}_x = [\tilde{x}_1 - \mu_{x_1} \quad \cdots \quad \tilde{x}_n - \mu_{x_n}]^T \quad (4.25)$$

the $(n \times n)$ variance-covariance matrix becomes

$$\boldsymbol{\Sigma}_x = E \left[(\tilde{\mathbf{x}} - \boldsymbol{\mu}_x) (\tilde{\mathbf{x}} - \boldsymbol{\mu}_x)^T \right] = \begin{bmatrix} \sigma_{x_1}^2 & \sigma_{x_1, x_2} & \cdots & \sigma_{x_1, x_n} \\ & & \ddots & \sigma_{x_2, x_n} \\ & & & \vdots \\ \text{sym} & & & \sigma_{x_n}^2 \end{bmatrix} \quad (4.26)$$

The variance-covariance matrix is symmetric because of (4.21). The expectation operator E is applied to each matrix element. The variance-covariance matrix is simply called the covariance matrix for the sake of brevity. The correlations are computed according to Equation (4.22) and can be arranged in the same order. Thus, the correlation matrix is

$$\mathbf{C} = \begin{bmatrix} 1 & \rho_{x_1, x_2} & \cdots & \rho_{x_1, x_n} \\ & & \ddots & \rho_{x_2, x_n} \\ & & & \vdots \\ \text{sym} & & & 1 \end{bmatrix} \quad (4.27)$$

The correlation matrix is symmetric, the diagonal elements equal 1, and the off-diagonal elements are between -1 and $+1$.

Propagation Usually we are more interested in a linear function of the random variables than in the random variables themselves. Typical examples are the adjusted coordinates used to compute distances and angles. From the definition of the mean (4.11), it follows that for a constant c

$$E(c) = c \int_{-\infty}^{\infty} f(x) dx = c \quad (4.28)$$

and

$$E(c\tilde{x}) = cE(\tilde{x}) \quad (4.29)$$

The expected value (mean) of a constant equals the constant. Because the mean is a constant, it follows that

$$E[E(\tilde{x})] = \mu_x \quad (4.30)$$

Relations (4.28) and (4.29) also hold for multivariate density functions, as can be seen from (4.18). Let $\tilde{y} = \tilde{x}_1 + \tilde{x}_2$ be a linear function of random variables, then

$$\begin{aligned} E(\tilde{x}_1 + \tilde{x}_2) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x_1 + x_2) f(x_1, x_2) dx_1 dx_2 \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 f(x_1, x_2) dx_1 dx_2 + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_2 f(x_1, x_2) dx_1 dx_2 \\ &= E(\tilde{x}_1) + E(\tilde{x}_2) \end{aligned} \quad (4.31)$$

Thus, the expected value of the sum of two random variables equals the sum of the individual expected values. By combining (4.28) and (4.31), we can compute the expected value of a general linear function of random variables. Thus, if the elements of the $n \times u$ matrix $\mathbf{A}$ and the $n \times 1$ vector $\mathbf{a}_0$ are constants and

$$\tilde{\mathbf{y}} = \mathbf{a}_0 + \mathbf{A}\tilde{\mathbf{x}} \quad (4.32)$$

then the expected value is

$$E(\tilde{\mathbf{y}}) = \mathbf{a}_0 + \mathbf{A}E(\tilde{\mathbf{x}}) \quad (4.33)$$

This is the law for propagating the mean. The law of variance-covariance propagation is as follows:

$$\begin{aligned} \Sigma_y &\equiv E[(\tilde{\mathbf{y}} - \mu_y)(\tilde{\mathbf{y}} - \mu_y)^T] \\ &= E\{[\tilde{\mathbf{y}} - E(\tilde{\mathbf{y}})][\tilde{\mathbf{y}} - E(\tilde{\mathbf{y}})]^T\} \\ &= E\{[\tilde{\mathbf{y}} - \mathbf{a}_0 - \mathbf{A}E(\tilde{\mathbf{x}})][\tilde{\mathbf{y}} - \mathbf{a}_0 - \mathbf{A}E(\tilde{\mathbf{x}})]^T\} \\ &= E\{[\mathbf{A}\tilde{\mathbf{x}} - \mathbf{A}E(\tilde{\mathbf{x}})][\mathbf{A}\tilde{\mathbf{x}} - \mathbf{A}E(\tilde{\mathbf{x}})]^T\} \\ &= \mathbf{A}E\{[\tilde{\mathbf{x}} - E(\tilde{\mathbf{x}})][\tilde{\mathbf{x}} - E(\tilde{\mathbf{x}})]^T\}\mathbf{A}^T \\ &= \mathbf{A}\Sigma_x\mathbf{A}^T \end{aligned} \quad (4.34)$$

The first line in Expression (4.34) is the general expression for the variance-covariance matrix of the random variable $\tilde{\mathbf{y}}$ according to definition (4.26); μ_y is the expected

value of $\tilde{\mathbf{y}}$. The third line follows by substituting (4.33) for the expected value of $\tilde{\mathbf{y}}$. Equation (4.32) has been substituted in the third line for $\tilde{\mathbf{y}}$, and, finally, the $\mathbf{A}$ matrix has been factored out. Thus the variance-covariance matrix of the random variable $\tilde{\mathbf{y}}$ is obtained by pre- and postmultiplying the variance-covariance matrix of the original random variable $\tilde{\mathbf{x}}$ by the coefficient matrix $\mathbf{A}$ and its transpose. The constant term $\mathbf{a}_0$ cancels. This is the law of variance-covariance propagation for linear functions of random variables. The covariance matrix $\Sigma_{\mathbf{y}}$ is a full matrix in general.

For later reference, the expression for the covariance matrix (4.26) can be rewritten as

$$\begin{aligned}\Sigma_{\mathbf{x}} &= E \left[(\tilde{\mathbf{x}} - \mu_{\mathbf{x}}) (\tilde{\mathbf{x}} - \mu_{\mathbf{x}})^T \right] \\ &= E \left[\tilde{\mathbf{x}} \tilde{\mathbf{x}}^T - \mu_{\mathbf{x}} \mu_{\mathbf{x}}^T \right]\end{aligned}\tag{4.35}$$

4.4 MIXED ADJUSTMENT MODEL

To simplify the notation, the tilde will not be used in this section to identify random variables. Observations or functions of observations are always random variables. A caret is used to identify quantities estimated by least-squares, i.e., those quantities that are a solution of a specific minimization. Caret quantities are always random variables because they are functions of observations. To simplify the notation even further, the caret symbol is used consistently only in connection with the parameters $\mathbf{x}$.

In the mixed adjustment model, the observations and the parameters are implicitly related. If ℓ_a denotes the vector of n adjusted observations and $\mathbf{x}_a$ denotes u adjusted parameters (unknowns), the mathematical model is given by

$$\mathbf{f}(\ell_a, \mathbf{x}_a) = \mathbf{0}\tag{4.36}$$

The total number of equations in (4.36) is denoted by r . The stochastic model is

$$\mathbf{P} = \sigma_0^2 \Sigma_{\ell_b}^{-1}\tag{4.37}$$

where $\mathbf{P}$ denotes the $n \times n$ weight matrix, and Σ_{ℓ_b} denotes the covariance matrix of the observations. The objective is to estimate the parameters. It should be noted that the observations are stochastic (random) variables and that the parameters are deterministic quantities. The parameters exist, but their values are unknown. The estimated parameters, however, will be functions of the observations and therefore random variables.

4.4.1 Linearization

If we let $\mathbf{x}_0$ denote a vector of known approximate values of the parameters, then the parameter corrections $\mathbf{x}$ are

$$\mathbf{x} = \mathbf{x}_a - \mathbf{x}_0 \quad (4.38)$$

If ℓ_b denotes the vector of observations, then the residuals are defined by

$$\mathbf{v} = \ell_a - \ell_b \quad (4.39)$$

With (4.38) and (4.39) the mathematical model can be written as

$$\mathbf{f}(\ell_b + \mathbf{v}, \mathbf{x}_0 + \mathbf{x}) = \mathbf{o} \quad (4.40)$$

The nonlinear mathematical model is linearized around the known point of expansion $(\ell_b, \mathbf{x}_0)$, giving

$${}_r\mathbf{B}_{nn}\mathbf{v}_1 + {}_r\mathbf{A}_{uu}\mathbf{x}_1 + {}_r\mathbf{w}_1 = {}_r\mathbf{o}_1 \quad (4.41)$$

where

$$\mathbf{B} = \left. \frac{\partial \mathbf{f}}{\partial \ell} \right|_{\mathbf{x}_0, \ell_b} \quad (4.42)$$

$$\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\mathbf{x}_0, \ell_b} \quad (4.43)$$

$$\mathbf{w} = \mathbf{f}(\ell_b, \mathbf{x}_0) \quad (4.44)$$

See Appendix A for linearization of multivariable functions. The coefficient matrices must be evaluated at the point of expansion, which consists of observations and approximate parameters. The discrepancies $\mathbf{w}$ must be evaluated for the same point of expansion. The better the approximate values $\mathbf{x}_0$, the smaller the parameter corrections $\mathbf{x}$.

4.4.2 Minimization and Solution

The least-squares estimate $\hat{\mathbf{x}}$ is based on the minimization of the function $\mathbf{v}^T \mathbf{P} \mathbf{v}$. A solution is obtained by introducing a vector of Lagrange multipliers, $\mathbf{k}$, and minimizing the function

$$\phi(\mathbf{v}, \mathbf{k}, \mathbf{x}) = \mathbf{v}^T \mathbf{P} \mathbf{v} - 2\mathbf{k}^T (\mathbf{B}\mathbf{v} + \mathbf{A}\mathbf{x} + \mathbf{w}) \quad (4.45)$$

Equation (4.45) is a function of three variables, namely, $\mathbf{v}$, $\mathbf{k}$, and $\mathbf{x}$. A necessary condition for the minimum is that the partial derivatives must be zero. It can be readily shown that this condition is also sufficient. Differentiating (4.45) following the rules of Appendix A and setting the partial derivatives to zero gives

$$\frac{1}{2} \frac{\partial \phi}{\partial \mathbf{v}} = \mathbf{P}\hat{\mathbf{v}} - \mathbf{B}^T \hat{\mathbf{k}} = \mathbf{o} \quad (4.46)$$

$$\frac{1}{2} \frac{\partial \phi}{\partial \mathbf{k}} = \mathbf{B}\hat{\mathbf{v}} + \mathbf{A}\hat{\mathbf{x}} + \mathbf{w} = \mathbf{o} \quad (4.47)$$

$$\frac{1}{2} \frac{\partial \phi}{\partial \mathbf{x}} = -\mathbf{A}^T \hat{\mathbf{k}} = \mathbf{o} \quad (4.48)$$

The solution of (4.46) to (4.48) starts with the recognition that $\mathbf{P}$ is a square matrix and can be inverted. Thus, the expression for the residuals follows from (4.46):

$$\hat{\mathbf{v}} = \mathbf{P}^{-1} \mathbf{B}^T \hat{\mathbf{k}} \quad (4.49)$$

Substituting (4.49) into (4.47), we obtain the solution for the Lagrange multiplier:

$$\hat{\mathbf{k}} = -\mathbf{M}^{-1}(\mathbf{A}\hat{\mathbf{x}} + \mathbf{w}) \quad (4.50)$$

with

$${}_r\mathbf{M}_r = {}_r\mathbf{B}_{nn} \mathbf{P}_{nn}^{-1} {}_n\mathbf{B}_r^T \quad (4.51)$$

Finally, the estimate $\hat{\mathbf{x}}$ follows from (4.48) and (4.50)

$$\hat{\mathbf{x}} = -(\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{M}^{-1} \mathbf{w} \quad (4.52)$$

The estimates $\hat{\mathbf{x}}$ and $\hat{\mathbf{v}}$ are independent of the a priori variance of unit weight. The first step is to compute the parameters $\hat{\mathbf{x}}$ from (4.52), then the Lagrange multipliers $\hat{\mathbf{k}}$ from (4.50), followed by the residuals $\hat{\mathbf{v}}$ (4.49). The adjusted parameters and adjusted observations follow from (4.38) and (4.39).

The caret symbol in $\hat{\mathbf{v}}$, $\hat{\mathbf{k}}$, and $\hat{\mathbf{x}}$ indicates that all three estimated values follow from minimizing of $\mathbf{v}^T \mathbf{P} \mathbf{v}$. However, as stated earlier, the caret is only used consistently for the estimated parameters $\hat{\mathbf{x}}$ in order to simplify the notation.

4.4.3 Cofactor Matrices

Equation (4.44) shows that $\mathbf{w}$ is a random variable because it is a function of the observation ℓ_b . With (4.37), the law of variance-covariance propagation (4.34), and the use of $\mathbf{B}$ in (4.42), the cofactor matrix $\mathbf{Q}_w$ becomes

$$\mathbf{Q}_w = \mathbf{B} \mathbf{P}^{-1} \mathbf{B}^T = \mathbf{M} \quad (4.53)$$

From (4.53) and (4.52) it follows that

$$\mathbf{Q}_x = (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \quad (4.54)$$

Combining (4.49) through (4.52) the expression for the residuals becomes

$$\mathbf{v} = [\mathbf{P}^{-1} \mathbf{B}^T \mathbf{M}^{-1} \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{M}^{-1} - \mathbf{P}^{-1} \mathbf{B}^T \mathbf{M}^{-1}] \mathbf{w} \quad (4.55)$$

It follows from the law of variance propagation (4.34) and (4.53) that

$$\mathbf{Q}_v = \mathbf{P}^{-1} \mathbf{B}^T \mathbf{M}^{-1} [\mathbf{M} - \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T] \mathbf{M}^{-1} \mathbf{B} \mathbf{P}^{-1} \quad (4.56)$$

The adjusted observations are

$$\begin{aligned} \ell_a &= \ell_b + \mathbf{v} \\ &= \ell_b + [\mathbf{P}^{-1} \mathbf{B}^T \mathbf{M}^{-1} \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{M}^{-1} - \mathbf{P}^{-1} \mathbf{B}^T \mathbf{M}^{-1}] \mathbf{w} \end{aligned} \quad (4.57)$$

Because

$$\frac{\partial \ell_a}{\partial \ell_b} = \mathbf{I} + \mathbf{P}^{-1} \mathbf{B}^T \mathbf{M}^{-1} \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{M}^{-1} \mathbf{B} - \mathbf{P}^{-1} \mathbf{B}^T \mathbf{M}^{-1} \mathbf{B} \quad (4.58)$$

it follows that

$$\mathbf{Q}_{\ell_a} = \mathbf{Q}_{\ell_b} - \mathbf{Q}_v \quad (4.59)$$

where the inverse of $\mathbf{P}$ has been replaced by $\mathbf{Q}_{\ell_b}$ according to (4.4)

4.4.4 A Posteriori Variance of Unit Weight

The minimum of $\mathbf{v}^T \mathbf{P} \mathbf{v}$ follows from (4.49), (4.50), and (4.52) as

$$\mathbf{v}^T \mathbf{P} \mathbf{v} = \mathbf{w}^T [\mathbf{M}^{-1} - \mathbf{M}^{-1} \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{M}^{-1}] \mathbf{w} \quad (4.60)$$

The expected value of this random variable is

$$\begin{aligned} E(\mathbf{v}^T \mathbf{P} \mathbf{v}) &= E(\text{Tr } \mathbf{v}^T \mathbf{P} \mathbf{v}) \\ &= E \left\{ \text{Tr} [\mathbf{w}^T (\mathbf{M}^{-1} - \mathbf{M}^{-1} \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{M}^{-1}) \mathbf{w}] \right\} \\ &= E \left\{ \text{Tr} [(\mathbf{M}^{-1} - \mathbf{M}^{-1} \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{M}^{-1}) \mathbf{w} \mathbf{w}^T] \right\} \\ &= \text{Tr} \left\{ [\mathbf{M}^{-1} - \mathbf{M}^{-1} \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{M}^{-1}] E(\mathbf{w} \mathbf{w}^T) \right\} \end{aligned} \quad (4.61)$$

The trace (Tr) of a matrix equals the sum of its diagonal elements. In the first part of (4.61), the property that the trace of a 1×1 matrix equals the matrix element itself is used. Next, the matrix products are switched, leaving the trace invariant. In the last part of the equation, the expectation operator and the trace are switched. The expected value $E(\mathbf{w} \mathbf{w}^T)$ can be readily computed. Per definition, the expected value of the residuals

$$E(\mathbf{v}) = \mathbf{0} \quad (4.62)$$

is zero because the residuals represent random errors for which positive and negative errors of the same magnitude occur with the same probability. It follows from (4.41) that

$$E(\mathbf{w}) = -\mathbf{A}\mathbf{x} \quad (4.63)$$

Note that $\mathbf{x}$ in (4.63) or (4.41) is not a random variable. In this expression, $\mathbf{x}$ simply denotes the vector of unknown parameters that have fixed values, even though the values are not known. The estimate $\hat{\mathbf{x}}$ is a random variable because it is a function of the observations. By using (4.35) for the definition of the covariance matrix (4.53) and using (4.63), it follows that

$$\begin{aligned} E(\mathbf{w}\mathbf{w}^T) &= \Sigma_w + E(\mathbf{w})E(\mathbf{w})^T \\ &= \sigma_0^2 \mathbf{M} + \mathbf{A}\mathbf{x}\mathbf{x}^T \mathbf{A}^T \end{aligned} \quad (4.64)$$

Substituting (4.64) into (4.61) yields the expected value for $\mathbf{v}^T \mathbf{P}\mathbf{v}$:

$$\begin{aligned} E(\mathbf{v}^T \mathbf{P}\mathbf{v}) &= \sigma_0^2 \text{Tr} \left\{ {}_r \mathbf{I}_r - \mathbf{M}^{-1} \mathbf{A} (\mathbf{A}^T \mathbf{M}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \right\} \\ &= \sigma_0^2 (r - u) \end{aligned} \quad (4.65)$$

The difference $r - u$ is called the degree of freedom and equals the number of redundant equations in the model (4.36). Strictly, the degree of freedom is $r - R(\mathbf{A})$ because the second matrix in (4.65) is idempotent. The symbol $R(\mathbf{A})$ denotes the rank of the matrix $\mathbf{A}$. The a posteriori variance of unit weight is computed from

$$\hat{\sigma}_0^2 = \frac{\hat{\mathbf{v}}^T \mathbf{P} \hat{\mathbf{v}}}{r - u} \quad (4.66)$$

Using (4.65), we see that

$$E(\hat{\sigma}_0^2) = \sigma_0^2 \quad (4.67)$$

The expected value of the a posteriori variance of unit weight equals the a priori variance of unit weight.

Finally, the estimated covariance matrices are

$$\Sigma_x = \hat{\sigma}_0^2 \mathbf{Q}_x \quad (4.68)$$

$$\Sigma_v = \hat{\sigma}_0^2 \mathbf{Q}_v \quad (4.69)$$

$$\Sigma_{\ell_a} = \hat{\sigma}_0^2 \mathbf{Q}_{\ell_a} \quad (4.70)$$

With Equation (4.59) it follows that

$$\Sigma_{\ell_a} = \Sigma_{\ell_b} - \Sigma_v \quad (4.71)$$

Because the diagonal elements of all three covariance matrices in (4.71) are positive, it follows that the variances of the adjusted observations are smaller than those of the original observations. The difference is a function of the geometry of the adjustment, as implied by the covariance matrix Σ_v .

4.4.5 Iterations

Because the mathematical model is generally nonlinear, the least-squares solution must be iterated. Recall that (4.36) is true only for $(\ell_a, \mathbf{x}_a)$. Since neither of these quantities is known before the adjustment, the initial point of expansion is chosen as $(\ell_b, \mathbf{x}_0)$. For the i th iteration, the linearized model can be written

$$\mathbf{B}_{x_{0i}, \ell_{0i}} \bar{\mathbf{v}}_i + \mathbf{A}_{x_{0i}, \ell_{0i}} \mathbf{x}_i + \mathbf{w}_{x_{0i}, \ell_{0i}} = \mathbf{0} \quad (4.72)$$

where the point of expansion $(\ell_{0i}, \mathbf{x}_{0i})$ represents the previous solution. The symbols ℓ_{ai} and $\mathbf{x}_{ai}$ denote the adjusted observations and adjusted parameters for the current (i th) solution. They are computed from

$$\bar{\mathbf{v}}_i = \ell_{ai} - \ell_{0i} \quad (4.73)$$

$$\mathbf{x}_i = \mathbf{x}_{ai} - \mathbf{x}_{0i} \quad (4.74)$$

once the least-squares solution of (4.72) has been obtained. The iteration starts with $\ell_{01} = \ell_b$ and $\mathbf{x}_{01} = \mathbf{x}_0$. If the adjustment converges properly, then both $\bar{\mathbf{v}}_i$ and $\mathbf{x}_i$ converge to zero, or, stated differently, ℓ_{ai} and $\mathbf{x}_{ai}$ converge toward ℓ_a and $\mathbf{x}_a$, respectively. The quantity $\bar{\mathbf{v}}_i$ does not equal the residuals. The residuals express the random difference between the adjusted observations and the original observations according to Equation (4.39). Defining

$$\mathbf{v}_i = \ell_{ai} - \ell_b \quad (4.75)$$

it follows from (4.73) that

$$\bar{\mathbf{v}}_i = \mathbf{v}_i + (\ell_b - \ell_{0i}) \quad (4.76)$$

Substituting this expression into (4.72) gives

$$\mathbf{B}_{x_{0i}, \ell_{01}} \mathbf{v}_i + \mathbf{A}_{x_{0i}, \ell_{01}} \mathbf{x}_i + \mathbf{w}_{x_{0i}, \ell_{01}} + \mathbf{B}_{x_{0i}, \ell_{01}} (\ell_b - \ell_{0i}) = \mathbf{0} \quad (4.77)$$

The formulation (4.77) assures that the vector $\mathbf{v}_i$ converges toward the vector of residuals $\mathbf{v}$. The last term in (4.77) will be zero for the first iteration when $\ell_{0i} = \ell_b$. The iteration has converged if

$$|\mathbf{v}^T \mathbf{P} \mathbf{v}_i - \mathbf{v}^T \mathbf{P} \mathbf{v}_{i-1}| < \varepsilon \quad (4.78)$$

where ε is a small positive number.

4.5 OBSERVATION AND CONDITION EQUATION MODELS

Often there is an explicit relationship between the observations and the parameters, such as

$$\ell_a = \mathbf{f}(\mathbf{x}_a) \quad (4.79)$$

This is the observation equation model. Comparing both mathematical models (4.36) and (4.79), and taking the definition of the matrix $\mathbf{B}$ (4.42) into account, we see that the observation equation model follows from the mixed model using the specification

$$\mathbf{B} \equiv -\mathbf{I} \quad (4.80)$$

$$\ell \equiv \mathbf{w} = \mathbf{f}(\mathbf{x}_0) - \ell_b = \ell_0 - \ell_b \quad (4.81)$$

It is customary to denote the discrepancy by ℓ instead of $\mathbf{w}$ when dealing with the observation equation model. The symbol ℓ_0 equals the value of the observations as computed from the approximate parameters $\mathbf{x}_0$. The point of expansion for the linearization is $\mathbf{x}_0$; the observation vector is not involved in the iteration because of the explicit form of (4.79). The linearized equations

$${}_n\mathbf{v}_1 = {}_n\mathbf{A}_{u \ u}\mathbf{x}_1 + {}_n\ell_1 \quad (4.82)$$

are the *observation equations*. There is one equation for each observation in (4.82).

If the observations are related by a nonlinear function without use of parameters, we speak of the condition equation model. It is written as

$$\mathbf{f}(\ell_a) = \mathbf{o} \quad (4.83)$$

By comparing this with the mixed model (4.36), and applying the definition of the $\mathbf{A}$ matrix (4.43) we see that the condition equation model follows upon the specification

$$\mathbf{A} = \mathbf{O} \quad (4.84)$$

The linear equations

$${}_r\mathbf{B}_n {}_n\mathbf{v}_1 + {}_r\mathbf{w}_1 = \mathbf{o} \quad (4.85)$$

are called the condition equations. The iteration for the model (4.85) is analogous to a mixed model with the added simplification that there is no $\mathbf{A}$ matrix and no parameter vector $\mathbf{x}$.

The significance of these three models (observation, condition, and mixed) is that a specific adjustment problem can usually be formulated more easily in one of the models. Clearly, that model should be chosen. There are situations in which it is equally easy to use any of the models. A typical example is the adjustment of a level

TABLE 4.1 Three Adjustment Models

	Mixed Model	Observation Model	Condition Model
Nonlinear model	$\mathbf{f}(\ell_a, \mathbf{x}_a) = \mathbf{o}$	$\ell_a = \mathbf{f}(\mathbf{x}_a)$	$\mathbf{f}(\ell_a) = \mathbf{o}$
Specifications		$\mathbf{B} = -\mathbf{I}, \quad \ell = \mathbf{w},$ $r = n$	$\mathbf{A} = \mathbf{O}$
Linear model	$\mathbf{B}\mathbf{v} + \mathbf{A}\mathbf{x} + \mathbf{w} = \mathbf{o}$	$\mathbf{v} = \mathbf{A}\mathbf{x} + \ell$	$\mathbf{B}\mathbf{v} + \mathbf{w} = \mathbf{o}$
Normal equation elements	$\mathbf{M} = \mathbf{B}\mathbf{P}^{-1}\mathbf{B}^T$ $\mathbf{N} = \mathbf{A}^T\mathbf{M}^{-1}\mathbf{A}$ $\mathbf{u} = \mathbf{A}^T\mathbf{M}^{-1}\mathbf{w}$	$\mathbf{M} = \mathbf{P}^{-1}$ $\mathbf{N} = \mathbf{A}^T\mathbf{P}\mathbf{A}$ $\mathbf{u} = \mathbf{A}^T\mathbf{P}\ell$	$\mathbf{M} = \mathbf{B}\mathbf{P}^{-1}\mathbf{B}^T$
Normal equations	$\mathbf{N}\hat{\mathbf{x}} = -\mathbf{u}$	$\mathbf{N}\hat{\mathbf{x}} = -\mathbf{u}$	
Minimum $\mathbf{v}^T\mathbf{P}\mathbf{v}$	$\mathbf{v}^T\mathbf{P}\mathbf{v} = -\mathbf{u}^T\mathbf{N}^{-1}\mathbf{u} + \mathbf{w}^T\mathbf{M}^{-1}\mathbf{w}$	$\mathbf{v}^T\mathbf{P}\mathbf{v} = -\mathbf{u}^T\mathbf{N}^{-1}\mathbf{u} + \ell^T\mathbf{P}\ell$	$\mathbf{v}^T\mathbf{P}\mathbf{v} = \mathbf{w}^T\mathbf{M}^{-1}\mathbf{w}$
Estimated parameters	$\hat{\mathbf{x}} = -\mathbf{N}^{-1}\mathbf{u}$	$\hat{\mathbf{x}} = -\mathbf{N}^{-1}\mathbf{u}$	—
Estimated residuals	$\hat{\mathbf{v}} = \mathbf{P}^{-1}\mathbf{B}^T\hat{\mathbf{k}}$	$\hat{\mathbf{v}} = \mathbf{A}\hat{\mathbf{x}} + \ell$	$\hat{\mathbf{v}} = \mathbf{P}^{-1}\mathbf{B}^T\hat{\mathbf{k}}$
Estimated variance of unit weight	$\hat{\sigma}_0^2 = \frac{\hat{\mathbf{v}}^T\mathbf{P}\hat{\mathbf{v}}}{r - u}$	$\hat{\sigma}_0^2 = \frac{\hat{\mathbf{v}}^T\mathbf{P}\hat{\mathbf{v}}}{n - u}$	$\hat{\sigma}_0^2 = \frac{\hat{\mathbf{v}}^T\mathbf{P}\hat{\mathbf{v}}}{r}$
Estimated parameter cofactor matrix	$\mathbf{Q}_x = \mathbf{N}^{-1}$	$\mathbf{Q}_x = \mathbf{N}^{-1}$	—
Estimated residual cofactor matrix	$\mathbf{Q}_v = \mathbf{P}^{-1}\mathbf{B}^T\mathbf{M}^{-1}(\mathbf{M} - \mathbf{A}\mathbf{N}^{-1}\mathbf{A}^T)\mathbf{M}^{-1}\mathbf{B}\mathbf{P}^{-1}$	$\mathbf{Q}_v = \mathbf{P}^{-1} - \mathbf{A}\mathbf{N}^{-1}\mathbf{A}^T$	$\mathbf{Q}_v = \mathbf{P}^{-1}\mathbf{B}^T\mathbf{M}^{-1}\mathbf{B}\mathbf{P}^{-1}$
Adjusted observation cofactor matrix	$\mathbf{Q}_{\ell_a} = \mathbf{Q}_{\ell_b} - \mathbf{Q}_v$	$\mathbf{Q}_{\ell_a} = \mathbf{Q}_{\ell_b} - \mathbf{Q}_v$	$\mathbf{Q}_{\ell_a} = \mathbf{Q}_{\ell_b} - \mathbf{Q}_v$

network. Most of the time, however, the observation equation model is preferred, because the simple rule “one observation, one equation” is suitable for setting up general software. Table 4.1 lists the important expressions for all three models.

4.6 SEQUENTIAL SOLUTION

Assume that observations are made in two groups, with the second group consisting of one or several observations. Both groups have a common set of parameters. The two mixed adjustment models can be written as

$$\mathbf{f}_1(\ell_{1a}, \mathbf{x}_a) = \mathbf{0} \quad (4.86)$$

$$\mathbf{f}_2(\ell_{2a}, \mathbf{x}_a) = \mathbf{0} \quad (4.87)$$

Both sets of observations should be uncorrelated, and the a priori variance of unit weight should be the same for both groups; i.e.,

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{P}_2 \end{bmatrix} = \sigma_0^2 \begin{bmatrix} \boldsymbol{\Sigma}_1^{-1} & \mathbf{O} \\ \mathbf{O} & \boldsymbol{\Sigma}_2^{-1} \end{bmatrix} \quad (4.88)$$

The number of observations in ℓ_{1a} and ℓ_{2a} are n_1 and n_2 , respectively; and r_1 and r_2 are the number of equations in the models $\mathbf{f}_1$ and $\mathbf{f}_2$, respectively. The linearization of (4.86) and (4.87) yields

$$\mathbf{B}_1 \mathbf{v}_1 + \mathbf{A}_1 \mathbf{x} + \mathbf{w}_1 = \mathbf{0} \quad (4.89)$$

$$\mathbf{B}_2 \mathbf{v}_2 + \mathbf{A}_2 \mathbf{x} + \mathbf{w}_2 = \mathbf{0} \quad (4.90)$$

where

$$\left. \begin{aligned} \mathbf{B}_1 &= \left. \frac{\partial \mathbf{f}_1}{\partial \ell_1} \right|_{\ell_{1b}, \mathbf{x}_0} & \mathbf{A}_1 &= \left. \frac{\partial \mathbf{f}_1}{\partial \mathbf{x}} \right|_{\ell_{1b}, \mathbf{x}_0} & \mathbf{w}_1 &= \mathbf{f}_1(\ell_{1b}, \mathbf{x}_0) \\ \mathbf{B}_2 &= \left. \frac{\partial \mathbf{f}_2}{\partial \ell_2} \right|_{\ell_{2b}, \mathbf{x}_0} & \mathbf{A}_2 &= \left. \frac{\partial \mathbf{f}_2}{\partial \mathbf{x}} \right|_{\ell_{2b}, \mathbf{x}_0} & \mathbf{w}_2 &= \mathbf{f}_2(\ell_{2b}, \mathbf{x}_0) \end{aligned} \right\} \quad (4.91)$$

The function to be minimized is

$$\begin{aligned} \phi(\mathbf{v}_1, \mathbf{v}_2, \mathbf{k}_1, \mathbf{k}_2, \mathbf{x}) &= \mathbf{v}_1^T \mathbf{P}_1 \mathbf{v}_1 + \mathbf{v}_2^T \mathbf{P}_2 \mathbf{v}_2 - 2\mathbf{k}_1^T (\mathbf{B}_1 \mathbf{v}_1 + \mathbf{A}_1 \mathbf{x} + \mathbf{w}_1) \\ &\quad - 2\mathbf{k}_2^T (\mathbf{B}_2 \mathbf{v}_2 + \mathbf{A}_2 \mathbf{x} + \mathbf{w}_2) \end{aligned} \quad (4.92)$$

The solution is obtained by setting the partial derivatives of (4.92) to zero,

$$\frac{1}{2} \frac{\partial \phi}{\partial \mathbf{v}_1} = \mathbf{P}_1 \mathbf{v}_1 - \mathbf{B}_1^T \mathbf{k}_1 = \mathbf{o} \quad (4.93)$$

$$\frac{1}{2} \frac{\partial \phi}{\partial \mathbf{v}_2} = \mathbf{P}_2 \mathbf{v}_2 - \mathbf{B}_2^T \mathbf{k}_2 = \mathbf{o} \quad (4.94)$$

$$\frac{1}{2} \frac{\partial \phi}{\partial \mathbf{x}} = -\mathbf{A}_1^T \mathbf{k}_1 - \mathbf{A}_2^T \mathbf{k}_2 = \mathbf{o} \quad (4.95)$$

$$\frac{1}{2} \frac{\partial \phi}{\partial \mathbf{k}_1} = \mathbf{B}_1 \mathbf{v}_1 + \mathbf{A}_1 \hat{\mathbf{x}} + \mathbf{w}_1 = \mathbf{o} \quad (4.96)$$

$$\frac{1}{2} \frac{\partial \phi}{\partial \mathbf{k}_2} = \mathbf{B}_2 \mathbf{v}_2 + \mathbf{A}_2 \hat{\mathbf{x}} + \mathbf{w}_2 = \mathbf{o} \quad (4.97)$$

and solving for $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{k}_1$, $\mathbf{k}_2$ and $\mathbf{x}$. Equations (4.93) and (4.94) give the residuals

$$\mathbf{v}_1 = \mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{k}_1 \quad (4.98)$$

$$\mathbf{v}_2 = \mathbf{P}_2^{-1} \mathbf{B}_2^T \mathbf{k}_2 \quad (4.99)$$

Combining (4.98) and (4.96) yields

$$\mathbf{M}_1 \mathbf{k}_1 + \mathbf{A}_1 \hat{\mathbf{x}} + \mathbf{w}_1 = \mathbf{o} \quad (4.100)$$

where

$$\mathbf{M}_1 = \mathbf{B}_1 \mathbf{P}_1^{-1} \mathbf{B}_1^T \quad (4.101)$$

is an $r_1 \times r_1$ symmetric matrix. The Lagrange multiplier becomes

$$\mathbf{k}_1 = -\mathbf{M}_1^{-1} \mathbf{A}_1 \hat{\mathbf{x}} - \mathbf{M}_1^{-1} \mathbf{w}_1 \quad (4.102)$$

Equations (4.95) and (4.97) become, after combination with (4.102) and (4.99),

$$\mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \hat{\mathbf{x}} + \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 - \mathbf{A}_2^T \mathbf{k}_2 = \mathbf{o} \quad (4.103)$$

$$\mathbf{B}_2 \mathbf{P}_2^{-1} \mathbf{B}_2^T \mathbf{k}_2 + \mathbf{A}_2 \hat{\mathbf{x}} + \mathbf{w}_2 = \mathbf{o} \quad (4.104)$$

By using

$$\mathbf{M}_2 = \mathbf{B}_2 \mathbf{P}_2^{-1} \mathbf{B}_2^T \quad (4.105)$$

we can write both Equations (4.103) and (4.104) in matrix form:

$$\begin{bmatrix} \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 & \mathbf{A}_2^T \\ \mathbf{A}_2 & -\mathbf{M}_2 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}} \\ -\mathbf{k}_2 \end{bmatrix} = \begin{bmatrix} -\mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 \\ -\mathbf{w}_2 \end{bmatrix} \quad (4.106)$$

Equation (4.106) shows how the normal matrix of the first group must be augmented in order to find the solution of both groups. The whole matrix can be inverted in one step to give the solution for $\hat{\mathbf{x}}$ and $\mathbf{k}_2$. Alternatively, one can compute the inverse using the matrix partitioning techniques of Section A.3.5, giving

$$\hat{\mathbf{x}} = -\mathbf{Q}_{11} \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 - \mathbf{Q}_{12} \mathbf{w}_2 \quad (4.107)$$

$$\mathbf{k}_2 = \mathbf{Q}_{21} \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 + \mathbf{Q}_{22} \mathbf{w}_2 \quad (4.108)$$

Setting

$$\mathbf{N}_1 = \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \quad (4.109)$$

$$\mathbf{N}_2 = \mathbf{A}_2^T \mathbf{M}_2^{-1} \mathbf{A}_2 \quad (4.110)$$

then

$$\mathbf{Q}_x \equiv \mathbf{Q}_{11} = (\mathbf{N}_1 + \mathbf{N}_2)^{-1} = \mathbf{N}_1^{-1} - \mathbf{N}_1^{-1} \mathbf{A}_2^T [\mathbf{M}_2 + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T]^{-1} \mathbf{A}_2 \mathbf{N}_1^{-1} \quad (4.111)$$

$$\mathbf{Q}_{12} = \mathbf{Q}_{21}^T = \mathbf{N}_1^{-1} \mathbf{A}_2^T [\mathbf{M}_2 + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T]^{-1} \quad (4.112)$$

$$\mathbf{Q}_{22} = -[\mathbf{M}_2 + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T]^{-1} \quad (4.113)$$

Substituting $\mathbf{Q}_{11}$ and $\mathbf{Q}_{12}$ into (4.107) gives the sequential solution for the parameters. We denote the solution of the first group by an asterisk and the contribution of the second group by Δ . In that notation, the estimated parameters of the first group are denoted by $\hat{\mathbf{x}}^*$, which is simplified to $\mathbf{x}^*$. Thus,

$$\hat{\mathbf{x}} = \mathbf{x}^* + \Delta \mathbf{x} \quad (4.114)$$

Comparing (4.107) and (4.52) the sequential solution becomes

$$\mathbf{x}^* = -\mathbf{N}_1^{-1} \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 \quad (4.115)$$

and

$$\Delta \mathbf{x} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T [\mathbf{M}_2 + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T]^{-1} (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2) \quad (4.116)$$

Similarly, the expression for the Lagrange multiplier $\mathbf{k}_2$ is

$$\mathbf{k}_2 = -[\mathbf{M}_2 + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T]^{-1} (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2) \quad (4.117)$$

A different form for the solution of the augmented system (4.106) is obtained by using alternative relations of the matrix partitioning inverse Expressions (A.82) to (A.89). It follows readily that

$$\begin{aligned}\hat{\mathbf{x}} &= -(\mathbf{N}_1 + \mathbf{N}_2)^{-1} (\mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 + \mathbf{A}_2^T \mathbf{M}_2^{-1} \mathbf{w}_2) \\ &= -(\mathbf{N}_1 + \mathbf{N}_2)^{-1} (-\mathbf{N}_1 \mathbf{x}^* + \mathbf{A}_2^T \mathbf{M}_2^{-1} \mathbf{w}_2) \\ &= \mathbf{x}^* - (\mathbf{N}_1 + \mathbf{N}_2)^{-1} (\mathbf{N}_2 \mathbf{x}^* + \mathbf{A}_2^T \mathbf{M}_2^{-1} \mathbf{w}_2)\end{aligned}\quad (4.118)$$

The procedure implied by the first line in (4.118) is called the method of adding normal equations. The contributions of the new observations are simply added appropriately.

The cofactor matrix $\mathbf{Q}_x$ of the parameters can be written in sequential form as

$$\begin{aligned}\mathbf{Q}_x &= \mathbf{Q}_{x^*} - \mathbf{Q}_{x^*} \mathbf{A}_2^T [\mathbf{M}_2 + \mathbf{A}_2 \mathbf{Q}_{x^*} \mathbf{A}_2^T]^{-1} \mathbf{A}_2 \mathbf{Q}_{x^*} \\ &= \mathbf{Q}_{x^*} + \Delta \mathbf{Q}_x\end{aligned}\quad (4.119)$$

$\mathbf{Q}_{x^*}$ is the cofactor matrix of the first group of observations and equals $\mathbf{N}_1^{-1}$. The contribution of the second group of observations to the cofactor matrix is

$$\Delta \mathbf{Q}_x = -\mathbf{Q}_{x^*} \mathbf{A}_2^T [\mathbf{M}_2 + \mathbf{A}_2 \mathbf{Q}_{x^*} \mathbf{A}_2^T]^{-1} \mathbf{A}_2 \mathbf{Q}_{x^*}\quad (4.120)$$

The change $\Delta \mathbf{Q}_x$ can be computed without having the actual observations of the second group. This is relevant in simulation studies.

The computation of $\mathbf{v}^T \mathbf{P} \mathbf{v}$ proceeds as usual

$$\begin{aligned}\mathbf{v}^T \mathbf{P} \mathbf{v} &= \mathbf{v}_1^T \mathbf{P}_1 \mathbf{v}_1 + \mathbf{v}_2^T \mathbf{P}_2 \mathbf{v}_2 \\ &= -\mathbf{k}_1^T \mathbf{w}_1 - \mathbf{k}_2^T \mathbf{w}_2\end{aligned}\quad (4.121)$$

The second part of (4.121) follows from (4.95) to (4.99). Using (4.102) for $\mathbf{k}_1$, (4.114) for $\hat{\mathbf{x}}$, (4.116) for $\Delta \mathbf{x}$, and (4.117) for $\mathbf{k}_2$, then the sequential solution becomes

$$\begin{aligned}\mathbf{v}^T \mathbf{P} \mathbf{v} &= \mathbf{v}^T \mathbf{P} \mathbf{v}^* + \Delta \mathbf{v}^T \mathbf{P} \mathbf{v} \\ &= \mathbf{v}^T \mathbf{P} \mathbf{v}^* + (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2)^T [\mathbf{M}_2 + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T]^{-1} (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2)\end{aligned}\quad (4.122)$$

with $\mathbf{v}^T \mathbf{P} \mathbf{v}^*$ being obtained from (4.60) for the first group only.

The a posteriori variance of unit weight is computed in the usual way:

$$\hat{\sigma}_0^2 = \frac{\mathbf{v}^T \mathbf{P} \mathbf{v}}{r_1 + r_2 - u}\quad (4.123)$$

where r_1 and r_2 are the number of equations in (4.86) and (4.87), respectively. The letter u denotes, again, the number of parameters.

The second set of observations contributes to all residuals. From (4.98), (4.102), and (4.114) we obtain

$$\begin{aligned} \mathbf{v}_1 &= \mathbf{v}_1^* + \Delta \mathbf{v}_1 \\ &= -\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} (\mathbf{A}_1 \mathbf{x}^* + \mathbf{w}_1) - \mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \Delta \mathbf{x} \end{aligned} \quad (4.124)$$

The expression for $\mathbf{v}_2$ follows from Equations (4.99) and (4.117):

$$\mathbf{v}_2 = -\mathbf{P}_2^{-1} \mathbf{B}_2^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2) \quad (4.125)$$

where

$$\mathbf{T} = (\mathbf{M}_2 + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T)^{-1} \quad (4.126)$$

The cofactor matrices for the residuals follow, again, from the law of variance-covariance propagation. The residuals $\mathbf{v}_1$ are a function of $\mathbf{w}_1$ and $\mathbf{w}_2$, according to (4.124). Substituting the expressions for $\mathbf{x}^*$ and $\Delta \mathbf{x}$, we obtain, from (4.124)

$$\frac{\partial \mathbf{v}_1}{\partial \mathbf{w}_1} = -\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} (\mathbf{I} - \mathbf{A}_1 \mathbf{N}_1^{-1} \mathbf{A}_1^T \mathbf{M}_1^{-1} + \mathbf{A}_1 \mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_1^T \mathbf{M}_1^{-1}) \quad (4.127)$$

$$\frac{\partial \mathbf{v}_1}{\partial \mathbf{w}_2} = -\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} \quad (4.128)$$

Applying the law of covariance propagation to $\mathbf{w}_1$ and $\mathbf{w}_2$ of (4.91) and knowing that the observations are uncorrelated gives

$$\mathbf{Q}_{w_1, w_2} = \begin{bmatrix} \mathbf{M}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{M}_2 \end{bmatrix} \quad (4.129)$$

By using the partial derivatives (4.127) and (4.128), Expression (4.129), and the law of variance-covariance propagation, we obtain, after some algebraic computations, the cofactor matrices:

$$\mathbf{Q}_{v_1} = \mathbf{Q}_{v_1^*} + \Delta \mathbf{Q}_{v_1} \quad (4.130)$$

where

$$\mathbf{Q}_{v_1^*} = \mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} (\mathbf{P}_1^{-1} \mathbf{B}_1^T)^T - (\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1) \mathbf{N}_1^{-1} (\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1)^T \quad (4.131)$$

$$\Delta \mathbf{Q}_{v_1} = (\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \mathbf{N}_1^{-1} \mathbf{A}_2^T) \mathbf{T} (\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \mathbf{N}_1^{-1} \mathbf{A}_2^T)^T \quad (4.132)$$

The partial derivatives of $\mathbf{v}_2$ with respect to $\mathbf{w}_1$ and $\mathbf{w}_2$ follow from (4.125):

$$\frac{\partial \mathbf{v}_2}{\partial \mathbf{w}_1} = \mathbf{P}_2^{-1} \mathbf{B}_2^T \mathbf{T} \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_1^T \mathbf{M}_1^{-1} \quad (4.133)$$

$$\frac{\partial \mathbf{v}_2}{\partial \mathbf{w}_2} = -\mathbf{P}_2^{-1} \mathbf{B}_2^T \mathbf{T} \quad (4.134)$$

By using, again, the law of variance-covariance propagation and (4.129), we obtain the cofactor for $\mathbf{v}_2$:

$$\mathbf{Q}_{v_2} = \mathbf{P}_2^{-1} \mathbf{B}_2^T \mathbf{T} \mathbf{B}_2 \mathbf{P}_2^{-1} \quad (4.135)$$

The estimated variance-covariance matrix is

$$\hat{\Sigma}_{v_2} = \hat{\sigma}_0^2 \mathbf{Q}_{v_2} \quad (4.136)$$

The variance-covariance matrix of the adjusted observations is, as usual,

$$\Sigma_{\ell_a} = \Sigma_{\ell_b} - \Sigma_v \quad (4.137)$$

As for iterations, one has to make sure that all groups are evaluated for the same approximate parameters. If the first system is iterated, the approximate coordinates for the last iteration must be used as expansion points for the second group. Because there are no observations common to both groups, the iteration with respect to the observations can be done individually for each group.

Occasionally, it is desirable to remove a set of observations from an existing solution. Consider again the uncorrelated case in which the set of observations to be removed is not correlated with the other sets. The procedure is readily seen from (4.118), which shows how normal equations are added. When observations are removed, the respective parts of the normal matrix and the right-hand term must be subtracted. Equation (4.118) becomes

$$\begin{aligned} \hat{\mathbf{x}} &= -(\mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 - \mathbf{A}_2^T \mathbf{M}_2^{-1} \mathbf{A}_2)^{-1} (\mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 - \mathbf{A}_2^T \mathbf{M}_2^{-1} \mathbf{w}_2) \\ &= -[\mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 + \mathbf{A}_2^T (-\mathbf{M}_2^{-1}) \mathbf{A}_2]^{-1} [\mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 + \mathbf{A}_2^T (-\mathbf{M}_2^{-1}) \mathbf{w}_2] \end{aligned} \quad (4.138)$$

One only has to use a negative weight matrix of the group of observations that is being removed, because

$$-\mathbf{M}_2 = \mathbf{B}_2 (-\mathbf{P}_2^{-1}) \mathbf{B}_2^T \quad (4.139)$$

Observations can be removed sequentially following (4.116).

The sequential solution can be used in quite a general manner. One can add or remove any number of groups sequentially. A group may consist of a single observation. Given the solution for $i - 1$ groups, some of the relevant expressions that include all i groups of observations are,

$$\hat{\mathbf{x}}_i = \hat{\mathbf{x}}_{i-1} + \Delta \hat{\mathbf{x}}_i \quad (4.140)$$

$$\Delta \hat{\mathbf{x}}_i = -\mathbf{Q}_{i-1} \mathbf{A}_i^T (\mathbf{M}_i + \mathbf{A}_i \mathbf{Q}_{i-1} \mathbf{A}_i^T)^{-1} (\mathbf{A}_i \hat{\mathbf{x}}_{i-1} + \mathbf{w}_i) \quad (4.141)$$

$$\mathbf{v}^T \mathbf{P} \mathbf{v}_i = \mathbf{v}^T \mathbf{P} \mathbf{v}_{i-1} + \Delta \mathbf{v}^T \mathbf{P} \mathbf{v}_i \quad (4.142)$$

$$\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}_i = (\mathbf{A}_i \hat{\mathbf{x}}_{i-1} + \mathbf{w}_i)^T (\mathbf{M}_i + \mathbf{A}_i \mathbf{Q}_{i-1} \mathbf{A}_i^T)^{-1} (\mathbf{A}_i \hat{\mathbf{x}}_{i-1} + \mathbf{w}_i) \quad (4.143)$$

$$\mathbf{Q}_i = \mathbf{Q}_{i-1} - \mathbf{Q}_{i-1} \mathbf{A}_i^T (\mathbf{M}_i + \mathbf{A}_i \mathbf{Q}_{i-1} \mathbf{A}_i^T)^{-1} \mathbf{A}_i \mathbf{Q}_{i-1} \quad (4.144)$$

Every sequential solution is equivalent to a one-step adjustment that contains the same observations. The sequential solution requires the inverse of the normal matrix. Because computing the inverse of the normal matrix requires many more computations than merely solving the system of normal equations, one might sometimes prefer to use the one-step solution instead of the sequential approach.

4.7 WEIGHTED PARAMETERS AND CONDITIONS

The algorithms developed in the previous section can be used to incorporate exterior information about parameters. This includes weighted functions of parameters, weighted individual parameters, and conditions on parameters. The objective is to incorporate new types of observations that directly refer to the parameters, to specify parameters in order to avoid singularity of the normal equations, or to incorporate the results of prior adjustments. Evaluating conditions between the parameters is the basis for hypothesis testing. These cases are obtained by specifying the coefficient matrices $\mathbf{A}$ and $\mathbf{B}$ of the mixed model. For example, the mixed model (4.86) and (4.87) can be specified as

$$\mathbf{f}_1(\ell_{1a}, \mathbf{x}_a) = \mathbf{0} \quad (4.145)$$

$$\ell_{2a} = \mathbf{f}_2(\mathbf{x}_a) \quad (4.146)$$

The linearized form is

$$\mathbf{B}_1 \mathbf{v}_1 + \mathbf{A}_1 \mathbf{x} + \mathbf{w}_1 = \mathbf{0} \quad (4.147)$$

$$\mathbf{v}_2 = \mathbf{A}_2 \mathbf{x} + \ell_2 \quad (4.148)$$

The specifications are $\mathbf{B}_2 = -\mathbf{I}$ and $\ell_2 = \mathbf{w}_2$. For the observation equation model we obtain

$$\ell_{1a} = \mathbf{f}_1(\mathbf{x}_a) \quad (4.149)$$

$$\ell_{2a} = \mathbf{f}_2(\mathbf{x}_a) \quad (4.150)$$

with the linearized form being

$$\mathbf{v}_1 = \mathbf{A}_1 \mathbf{x} + \ell_1 \quad (4.151)$$

$$\mathbf{v}_2 = \mathbf{A}_2 \mathbf{x} + \ell_2 \quad (4.152)$$

The stochastic model is given by the matrices $\mathbf{P}_1$ and $\mathbf{P}_2$. With proper choice of the elements of $\mathbf{A}_2$ and $\mathbf{P}_2$, it is possible to introduce a variety of relations about the parameters.

As a first case, consider nonlinear relations between parameters. The design matrix $\mathbf{A}_2$ contains the partial derivatives, and ℓ_{2b} contains the observed value of the function. This is the case of weighted functions of parameters. Examples are the area or volume of geometric figures as computed from coordinates, angles in geodetic networks, and differences between parameters (coordinates). Each function contributes one equation to (4.148) or (4.152). The respective expressions are identical with those given in Table 4.2 and require no further discussion.

As a second case, consider information about individual parameters. This is a special case of the general method discussed above. Each row of $\mathbf{A}_2$ contains zeros with the exception of one position, which contains a 1. The number of rows in the $\mathbf{A}_2$ matrix corresponds to the number of weighted parameters. The expressions of Table 4.2 are still valid for this case. If information enters into the adjustment in this manner, one speaks of the method of weighted parameters. In the most general case, all parameters are observed and weighted, giving

$$\ell_{2a} = \mathbf{x}_a \quad (4.153)$$

$$\ell_{2b} = \mathbf{x}_b \quad (4.154)$$

$$\mathbf{A}_2 = \mathbf{I} \quad (4.155)$$

$$\ell_2 = \mathbf{f}_2(\mathbf{x}_0) - \ell_{2b} = \mathbf{x}_0 - \mathbf{x}_b \quad (4.156)$$

The symbols $\mathbf{x}_b$ and $\mathbf{x}_0$ denote the observed parameters and approximate parameters. During the iterations, $\mathbf{x}_0$ converges toward the solution, whereas $\mathbf{x}_b$ remains unchanged just as does the vector ℓ_{2b} . As a special case, the vector ℓ_2 can be zero, which implies that the current values for the approximate parameters also serve as observations of the parameters. This can generally be done if the intent is to define the coordinate system by assigning weights to the current approximate parameters. Table 4.3 summarizes the solution for weighted parameters for the observation equation model. The parameters are weighted simply by adding the respective weights to the diagonal elements of the normal matrix. The parameters not weighted have zeros in the respective diagonal elements of $\mathbf{P}_2$. This is a convenient way of weighting a subset of parameters. Parameters can be fixed by assigning a large weight.

It is not necessary that the second group of observations represent the observed parameters. Table 4.4 shows the case in which the first group consists of the observed parameters. This approach has the unique feature that all observations can be added to the adjustment in a sequential manner; the first solution is a nonredundant one based solely on the values of the observed parameters. It is important, once again, to distinguish the roles of the observed parameters $\mathbf{x}_b$ and the approximations $\mathbf{x}_0$.

TABLE 4.2 Sequential Adjustment Models

	Mixed Model	Observation Model
Nonlinear model	$\mathbf{f}_1(\ell_{1a}, \mathbf{x}_a) = \mathbf{o}$ $\mathbf{f}_2(\ell_{2a}, \mathbf{x}_a) = \mathbf{o}$ $\mathbf{P} = \begin{bmatrix} \mathbf{P}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{P}_2 \end{bmatrix}$	$\ell_{1a} = \mathbf{f}_1(\mathbf{x}_a)$ $\ell_{2a} = \mathbf{f}_2(\mathbf{x}_a)$ $\mathbf{P} = \begin{bmatrix} \mathbf{P}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{P}_2 \end{bmatrix}$
Linear model	$\mathbf{B}_1 \mathbf{v}_1 + \mathbf{A}_1 \mathbf{x} + \mathbf{w}_1 = \mathbf{o}$ $\mathbf{B}_2 \mathbf{v}_2 + \mathbf{A}_2 \mathbf{x} + \mathbf{w}_2 = \mathbf{o}$	$\mathbf{v}_1 = \mathbf{A}_1 \mathbf{x} + \ell_1$ $\mathbf{v}_2 = \mathbf{A}_2 \mathbf{x} + \ell_2$
Normal equation elements	$\mathbf{M}_1 = \mathbf{B}_1 \mathbf{P}_1^{-1} \mathbf{B}_1^T \quad \mathbf{M}_2 = \mathbf{B}_2 \mathbf{P}_2^{-1} \mathbf{B}_2^T$ $\mathbf{N}_1 = \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \quad \mathbf{N}_2 = \mathbf{A}_2^T \mathbf{M}_2^{-1} \mathbf{A}_2$ $\mathbf{u}_1 = \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1 \quad \mathbf{u}_2 = \mathbf{A}_2^T \mathbf{M}_2^{-1} \mathbf{w}_2$	$\mathbf{M}_1 = \mathbf{P}_1^{-1} \quad \mathbf{M}_2 = \mathbf{P}_2^{-1}$ $\mathbf{N}_1 = \mathbf{A}_1^T \mathbf{P}_1 \mathbf{A}_1 \quad \mathbf{N}_2 = \mathbf{A}_2^T \mathbf{P}_2 \mathbf{A}_2$ $\mathbf{u}_1 = \mathbf{A}_1^T \mathbf{P}_1 \ell_1 \quad \mathbf{u}_2 = \mathbf{A}_2^T \mathbf{P}_2 \ell_2$
Minimum $\mathbf{v}^T \mathbf{P} \mathbf{v}$	$\mathbf{v}^T \mathbf{P} \mathbf{v} = \mathbf{v}^T \mathbf{P} \mathbf{v}^* + \Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ $\mathbf{v}^T \mathbf{P} \mathbf{v}^* = -\mathbf{u}_1^T \mathbf{N}_1^{-1} \mathbf{u}_1 + \mathbf{w}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1$ $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v} = (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2)^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2)$	$\mathbf{v}^T \mathbf{P} \mathbf{v} = \mathbf{v}^T \mathbf{P} \mathbf{v}^* + \Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ $\mathbf{v}^T \mathbf{P} \mathbf{v}^* = -\mathbf{u}_1^T \mathbf{N}_1^{-1} \mathbf{u}_1 + \ell_1^T \mathbf{P}_1 \ell_1$ $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v} = (\mathbf{A}_2 \mathbf{x}^* + \ell_2)^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \ell_2)$
Estimated parameters	$\hat{\mathbf{x}} = \mathbf{x}^* + \Delta \mathbf{x}$ $\mathbf{x}^* = -\mathbf{N}_1^{-1} \mathbf{u}_1$ $\mathbf{T} = (\mathbf{M}_2 + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T)^{-1}$ $\Delta \mathbf{x} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2)$	$\hat{\mathbf{x}} = \mathbf{x}^* + \Delta \mathbf{x}$ $\mathbf{x}^* = -\mathbf{N}_1^{-1} \mathbf{u}_1$ $\mathbf{T} = (\mathbf{P}_2^{-1} + \mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T)^{-1}$ $\Delta \mathbf{x} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \ell_2)$
Estimated residuals	$\mathbf{v}_1 = \mathbf{v}_1^* + \Delta \mathbf{v}_1$ $\mathbf{v}_1^* = -\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} (\mathbf{A}_1 \mathbf{x}^* + \mathbf{w}_1)$ $\Delta \mathbf{v}_1 = -\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \Delta \mathbf{x}$	$\mathbf{v}_1 = \mathbf{v}_1^* + \Delta \mathbf{v}_1$ $\mathbf{v}_1^* = \mathbf{A}_1 \mathbf{x}^* + \ell_1$ $\Delta \mathbf{v}_1 = \mathbf{A}_1 \Delta \mathbf{x}$
Estimated variance of unit weight	$\hat{\sigma}_0^2 = \frac{\mathbf{v}^T \mathbf{P} \mathbf{v}}{r_1 + r_2 - u}$	$\hat{\sigma}_0^2 = \frac{\mathbf{v}^T \mathbf{P} \mathbf{v}}{n_1 + n_2 - u}$
Estimated parameter cofactor matrix	$\mathbf{Q}_x = \mathbf{Q}_{x^*} + \Delta \mathbf{Q}$ $\mathbf{Q}_{x^*} = \mathbf{N}_1^{-1}$ $\Delta \mathbf{Q} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} \mathbf{A}_2 \mathbf{N}_1^{-1}$	$\mathbf{Q}_x = \mathbf{Q}_{x^*} + \Delta \mathbf{Q}$ $\mathbf{Q}_{x^*} = \mathbf{N}_1^{-1}$ $\Delta \mathbf{Q} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} \mathbf{A}_2 \mathbf{N}_1^{-1}$

Because, in most cases the $\mathbf{P}_1$ matrix will be diagonal, no matrix inverse computation is required. The size of the matrix $\mathbf{T}$ equals the number of observations in the second group. Thus, if one observation is added at a time, only a 1×1 matrix must be inverted. The residuals can be computed directly from the mathematical model as desired.

A third case pertains to the role of the weight matrix of the parameters. The weight matrix expresses the quality of the information known about the observed parameters.

TABLE 4.3 Observed Parameters

$$\ell_{1a} = \mathbf{f}_1(\mathbf{x}_a) \quad \ell_{2a} = \mathbf{x}_a \quad \mathbf{P} = \begin{bmatrix} \mathbf{P}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{P}_2 \end{bmatrix}$$

$$\mathbf{v}_1 = \mathbf{A}_1 \mathbf{x} + \ell_1 \quad \mathbf{v}_2 = \mathbf{x} + \ell_2 \quad \ell_2 = \mathbf{x}_0 - \mathbf{x}_b$$

$$\mathbf{N}_1 = \mathbf{A}_1^T \mathbf{P}_1 \mathbf{A}_1 \quad \mathbf{N}_2 = \mathbf{P}_2$$

$$\mathbf{u}_1 = \mathbf{A}_1^T \mathbf{P}_1 \ell_1 \quad \mathbf{u}_2 = \mathbf{P}_2 \ell_2$$

$$\hat{\mathbf{x}} = -(\mathbf{N}_1 + \mathbf{P}_2)^{-1} (\mathbf{u}_1 + \mathbf{P}_2 \ell_2)$$

$$\mathbf{Q}_x = (\mathbf{N}_1 + \mathbf{P}_2)^{-1}$$

Note: Case of observation equation model.

For the adjustment to be meaningful, one must make every attempt to obtain a weight matrix that truly reflects the quality of the additional information. Low weights, or, equivalently, large variances, imply low precision. Even low-weighted parameters can have, occasionally, a positive effect on the quality of the least-squares solution. If the parameters or functions of the parameters are introduced with an infinitely large weight, one speaks of conditions between parameters. The only specifications for implementing conditions are:

$$\mathbf{P}_2^{-1} = \mathbf{O} \quad (4.157)$$

and

$$\mathbf{P}_2 = \infty \quad (4.158)$$

The respective mathematical models are

$$\mathbf{f}(\ell_{1a}, \mathbf{x}_a) = \mathbf{o} \quad (4.159)$$

$$\mathbf{g}(\mathbf{x}_a) = \mathbf{o} \quad (4.160)$$

with

$$\mathbf{B}_1 \mathbf{v}_1 + \mathbf{A}_1 \mathbf{x} + \mathbf{w}_1 = \mathbf{o} \quad (4.161)$$

$$\mathbf{A}_2 \mathbf{x} + \ell_2 = \mathbf{o} \quad (4.162)$$

and

$$\ell_{1a} = \mathbf{f}(\mathbf{x}_a) \quad (4.163)$$

$$\mathbf{g}(\mathbf{x}_a) = \mathbf{o} \quad (4.164)$$

TABLE 4.4 Sequential Solution without Inverting the Normal Matrix

$$\begin{aligned}\ell_{1a} &= \mathbf{x}_a \\ \ell_{2a} &= \mathbf{f}_{2a}(\mathbf{x}_a)\end{aligned}\quad \mathbf{P} = \begin{bmatrix} \mathbf{P}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{P}_2 \end{bmatrix}$$

$$\mathbf{v}_1 = \mathbf{x} + \ell_1$$

$$\ell_1 = \mathbf{x}_0 - \mathbf{x}_b$$

$$\mathbf{v}_2 = \mathbf{A}_2 \mathbf{x} + \ell_2$$

$$\mathbf{N}_1 = \mathbf{P}_1 \quad \mathbf{N}_2 = \mathbf{A}_2^T \mathbf{P}_2 \mathbf{A}_2$$

$$\mathbf{u}_1 = \mathbf{P}_1 \ell_1 \quad \mathbf{u}_2 = \mathbf{A}_2^T \mathbf{P}_2 \ell_2$$

$$\hat{\mathbf{x}}_1 = -(\mathbf{x}_0 - \mathbf{x}_b)$$

$$\mathbf{Q}_1 = \mathbf{P}_1^{-1}$$

$$\mathbf{v}^T \mathbf{P} \mathbf{v}_1 = 0$$

$$\hat{\mathbf{x}}_i = \hat{\mathbf{x}}_{i-1} + \Delta \hat{\mathbf{x}}_{i-1}$$

$$\mathbf{v}^T \mathbf{P} \mathbf{v}_i = \mathbf{v}^T \mathbf{P} \mathbf{v}_{i-1} + \Delta \mathbf{v}^T \mathbf{P} \mathbf{v}_{i-1}$$

$$\mathbf{Q}_i = \mathbf{Q}_{i-1} + \Delta \mathbf{Q}_{i-1}$$

$$\mathbf{T} = \left(\mathbf{P}_i^{-1} + \mathbf{A}_i \mathbf{Q}_{i-1} \mathbf{A}_i^T \right)^{-1}$$

$$\Delta \mathbf{x}_{i-1} = -\mathbf{Q}_{i-1} \mathbf{A}_i^T \mathbf{T} (\mathbf{A}_i \hat{\mathbf{x}}_{i-1} + \ell_i)$$

$$\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}_{i-1} = (\mathbf{A}_i \hat{\mathbf{x}}_{i-1} + \ell_i)^T \mathbf{T} (\mathbf{A}_i \hat{\mathbf{x}}_{i-1} + \ell_i)$$

$$\Delta \mathbf{Q}_{i-1} = -\mathbf{Q}_{i-1} \mathbf{A}_i^T \mathbf{T} \mathbf{A}_i \mathbf{Q}_{i-1}$$

Note: Case of observation equation model.

with

$$\mathbf{v}_1 = \mathbf{A}_1 \mathbf{x} + \ell_1 \quad (4.165)$$

$$\mathbf{A}_2 \mathbf{x} + \ell_2 = \mathbf{o} \quad (4.166)$$

Table 4.5 contains the expression of the sequential solution with conditions between parameters. If (4.158) is used to impose the conditions, the largest numbers that can still be represented in the computer should be used. In most situations, it will be readily clear what constitutes a large weight; the weight must simply be large enough so that the respective observations or parameters do not change during the adjustment. For sequential solution, the solution of the first group must exist. Conditions cannot

TABLE 4.5 Conditions on Parameters

	Mixed Model with Conditions	Observation Model with Conditions
Nonlinear model	$\mathbf{f}_1(\ell_{1a}, \mathbf{x}_a) = \mathbf{0}$ $\mathbf{g}(\mathbf{x}_a) = \mathbf{0} \quad \mathbf{P}_1$	$\ell_{1a} = \mathbf{f}_1(\mathbf{x}_a)$ $\mathbf{g}(\mathbf{x}_a) = \mathbf{0} \quad \mathbf{P}_1$
Linear model	$\mathbf{B}_1 \mathbf{v}_1 + \mathbf{A}_1 \mathbf{x} + \mathbf{w}_1 = \mathbf{0}$ $\mathbf{A}_2 \mathbf{x} + \ell_2 = \mathbf{0}$	$\mathbf{v}_1 = \mathbf{A}_1 \mathbf{x} + \ell_1$ $\mathbf{A}_2 \mathbf{x} + \ell_2 = \mathbf{0}$
Normal equation elements	$\mathbf{M}_1 = \mathbf{B}_1 \mathbf{P}_1^{-1} \mathbf{B}_1^T$ $\mathbf{N}_1 = \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1$ $\mathbf{u}_1 = \mathbf{A}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1$	$\mathbf{M}_1 = \mathbf{P}_1^{-1}$ $\mathbf{N}_1 = \mathbf{A}_1^T \mathbf{P}_1 \mathbf{A}_1$ $\mathbf{u}_1 = \mathbf{A}_1^T \mathbf{P}_1 \ell_1$
Minimum $\mathbf{v}^T \mathbf{P} \mathbf{v}$	$\mathbf{v}^T \mathbf{P} \mathbf{v} = \mathbf{v}^T \mathbf{P} \mathbf{v}^* + \Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ $\mathbf{v}^T \mathbf{P} \mathbf{v}^* = -\mathbf{u}_1^T \mathbf{N}_1^{-1} \mathbf{u}_1 + \mathbf{w}_1^T \mathbf{M}_1^{-1} \mathbf{w}_1$ $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v} = (\mathbf{A}_2 \mathbf{x}^* + \ell_2)^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \ell_2)$	$\mathbf{v}^T \mathbf{P} \mathbf{v} = \mathbf{v}^T \mathbf{P} \mathbf{v}^* + \Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ $\mathbf{v}^T \mathbf{P} \mathbf{v}^* = -\mathbf{u}_1^T \mathbf{N}_1^{-1} \mathbf{u}_1 + \ell_1^T \mathbf{P}_1 \ell_1$ $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v} = (\mathbf{A}_2 \mathbf{x}^* + \ell_2)^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \ell_2)$
Estimated parameters	$\hat{\mathbf{x}} = \mathbf{x}^* + \Delta \mathbf{x}$ $\mathbf{x}^* = -\mathbf{N}_1^{-1} \mathbf{u}_1$ $\mathbf{T} = (\mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T)^{-1}$ $\Delta \mathbf{x} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \mathbf{w}_2)$	$\hat{\mathbf{x}} = \mathbf{x}^* + \Delta \mathbf{x}$ $\mathbf{x}^* = -\mathbf{N}_1^{-1} \mathbf{u}_1$ $\mathbf{T} = (\mathbf{A}_2 \mathbf{N}_1^{-1} \mathbf{A}_2^T)^{-1}$ $\Delta \mathbf{x} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \ell_2)$
Estimated residuals	$\mathbf{v}_1 = \mathbf{v}_1^* + \Delta \mathbf{v}_1$ $\mathbf{v}_1^* = -\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} (\mathbf{A}_1 \mathbf{x}^* + \mathbf{w}_1)$ $\Delta \mathbf{v}_1 = -\mathbf{P}_1^{-1} \mathbf{B}_1^T \mathbf{M}_1^{-1} \mathbf{A}_1 \Delta \mathbf{x}$	$\mathbf{v}_1 = \mathbf{v}_1^* + \Delta \mathbf{v}_1$ $\mathbf{v}_1^* = \mathbf{A}_1 \mathbf{x}^* + \ell_1$ $\Delta \mathbf{v}_1 = \mathbf{A}_1 \Delta \mathbf{x}$
Estimated variance of unit weight	$\hat{\sigma}_0^2 = \frac{\mathbf{v}^T \mathbf{P} \mathbf{v}}{r_1 + r_2 - u}$	$\hat{\sigma}_0^2 = \frac{\mathbf{v}^T \mathbf{P} \mathbf{v}}{n_1 + n_2 - u}$
Estimated parameter cofactor matrix	$\mathbf{Q}_x = \mathbf{Q}_{x^*} + \Delta \mathbf{Q}$ $\mathbf{Q}_{x^*} = \mathbf{N}_1^{-1}$ $\Delta \mathbf{Q} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} \mathbf{A}_2 \mathbf{N}_1^{-1}$	$\mathbf{Q}_x = \mathbf{Q}_{x^*} + \Delta \mathbf{Q}$ $\mathbf{Q}_{x^*} = \mathbf{N}_1^{-1}$ $\Delta \mathbf{Q} = -\mathbf{N}_1^{-1} \mathbf{A}_2^T \mathbf{T} \mathbf{A}_2 \mathbf{N}_1^{-1}$

be imposed sequentially to eliminate a singularity in the first group; e.g., conditions should not be used sequentially to define the coordinate system. A one-step solution is given by (4.118).

The a posteriori variance of unit weight is always computed from the final set of residuals. The degree of freedom increases by 1 for every observed parameter function, weighted parameter, or condition. In nonlinear adjustments the linearized condition must always be evaluated for the current point of expansion, i.e., the point of expansion of the last iteration (current solution).

The expressions in Table 4.2 and Table 4.5 are almost identical. The only difference is that the matrix $\mathbf{T}$ contains the matrix $\mathbf{M}_2$ in Table 4.2.

4.8 MINIMAL AND INNER CONSTRAINTS

This section deals with the implementation of minimal and inner constraints to the observation equation model. The symbol r denotes the rank of the design matrix, $R({}_n\mathbf{A}_u) = R(\mathbf{A}^T \mathbf{P} \mathbf{A}) = r \leq u$. Note that the use of the symbol r in this context is entirely different from its use in the mixed model, where r denotes the number of equations. The rank deficiency of $u - r$ is generally caused by a lack of coordinate system definition. For example, a network of distances is invariant with respect to translation and rotation, a network of angles is invariant with respect to translation, rotation, and scaling, and a level network (consisting of measured height differences) is invariant with respect to a translation in the vertical. The rank deficiency is dealt with by specifying $u - r$ conditions of the parameters. Much of the theory of inner and minimal constraint solution is discussed by Pope (1971). The main reason for dealing with minimal and inner constraint solutions is that this type of adjustment is important for the quality control of observations. Inner constraint solutions have the additional advantage that the standard ellipses (ellipsoids) represent the geometry as implied by the $\mathbf{A}$ and $\mathbf{P}$ matrices.

The formulation of the least-squares adjustment for the observation equation model in the presence of a rank deficiency is

$${}_n\mathbf{v}_1 = {}_n\mathbf{A}_u \mathbf{x}_B + {}_n\ell_1 \quad (4.167)$$

$$\mathbf{P} = \sigma_0^2 \boldsymbol{\Sigma}_{\ell_b}^{-1} \quad (4.168)$$

$${}_{u-r}\mathbf{B}_u \mathbf{x}_B = \mathbf{o} \quad (4.169)$$

The subscript B indicates that the solution of the parameters $\mathbf{x}$ depends on the special condition implied by the $\mathbf{B}$ matrix in (4.169). This is the observation equation model with conditions between the parameters that was treated in Section 4.7. The one-step solution is given by (4.106):

$$\begin{bmatrix} \mathbf{A}^T \mathbf{P} \mathbf{A} & \mathbf{B}^T \\ \mathbf{B} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}}_B \\ -\hat{\mathbf{k}}_2 \end{bmatrix} = \begin{bmatrix} -\mathbf{A}^T \mathbf{P} \ell \\ \mathbf{o} \end{bmatrix} \quad (4.170)$$

The matrix on the left side of (4.170) is a nonsingular matrix if the conditions (4.169) are linearly independent; i.e., the $(u - r) \times u$ matrix $\mathbf{B}$ has full row rank, and the rows are linear-independent of the rows of the design matrix $\mathbf{A}$. A general expression for the inverse is obtained from

$$\begin{bmatrix} \mathbf{A}^T \mathbf{P} \mathbf{A} & \mathbf{B}^T \\ \mathbf{B} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{Q}_B & \mathbf{S}^T \\ \mathbf{S} & \mathbf{R} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{O} \\ \mathbf{O} & \mathbf{I} \end{bmatrix} \quad (4.171)$$

This matrix equation gives the following four equations of submatrices:

$$\mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_B + \mathbf{B}^T \mathbf{S} = \mathbf{I} \quad (4.172)$$

$$\mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{S}^T + \mathbf{B}^T \mathbf{R} = \mathbf{O} \quad (4.173)$$

$$\mathbf{B} \mathbf{Q}_B = \mathbf{O} \quad (4.174)$$

$$\mathbf{B} \mathbf{S}^T = \mathbf{I} \quad (4.175)$$

The solution of these equations requires the introduction of the $(u - r) \times u$ matrix $\mathbf{E}$, whose rows span the null space of the design matrix $\mathbf{A}$ or the null space of the normal matrix. According to (A.53), there is a matrix $\mathbf{E}$ such that

$$(\mathbf{A}^T \mathbf{P} \mathbf{A}) \mathbf{E}^T = \mathbf{O} \quad (4.176)$$

or

$$\mathbf{A} \mathbf{E}^T = \mathbf{O} \quad \text{or} \quad \mathbf{E} \mathbf{A}^T = \mathbf{O} \quad (4.177)$$

Because the rows of $\mathbf{B}$ are linearly independent of the rows of $\mathbf{A}$, the $(u - r) \times (u - r)$ matrix $\mathbf{B} \mathbf{E}^T$ has full rank and thus can be inverted. Multiplying (4.172) by $\mathbf{E}$ from the left and using (4.177), we get

$$\mathbf{S} = (\mathbf{E} \mathbf{B}^T)^{-1} \mathbf{E} \quad (4.178)$$

This expression also satisfies (4.175). Substituting $\mathbf{S}$ into (4.173) gives

$$\mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{E}^T (\mathbf{B} \mathbf{E}^T)^{-1} + \mathbf{B}^T \mathbf{R} = \mathbf{O} \quad (4.179)$$

Because of (4.176), this expression becomes

$$\mathbf{B}^T \mathbf{R} = \mathbf{O} \quad (4.180)$$

Because $\mathbf{B}$ has full rank, it follows that the matrix $\mathbf{R} = \mathbf{O}$. Thus,

$$\begin{bmatrix} \mathbf{A}^T \mathbf{P} \mathbf{A} & \mathbf{B}^T \\ \mathbf{B} & \mathbf{O} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{Q}_B & \mathbf{E}^T (\mathbf{B} \mathbf{E}^T)^{-1} \\ (\mathbf{E} \mathbf{B}^T)^{-1} \mathbf{E} & \mathbf{O} \end{bmatrix} \quad (4.181)$$

Substituting Expression (4.178) for $\mathbf{S}$ into (4.172) gives the nonsymmetric matrix

$$\mathbf{T}_B \equiv \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_B = \mathbf{I} - \mathbf{B}^T (\mathbf{E} \mathbf{B}^T)^{-1} \mathbf{E} \quad (4.182)$$

This expression is modified with the help of (4.174), (4.176), and (4.182):

$$(\mathbf{A}^T \mathbf{P} \mathbf{A} + \mathbf{B}^T \mathbf{B}) \left[\mathbf{Q}_B + \mathbf{E}^T (\mathbf{B} \mathbf{E}^T)^{-1} (\mathbf{E} \mathbf{B}^T)^{-1} \mathbf{E} \right] = \mathbf{I} \quad (4.183)$$

It can be solved for $\mathbf{Q}_B$:

$$\mathbf{Q}_B = (\mathbf{A}^T \mathbf{P} \mathbf{A} + \mathbf{B}^T \mathbf{B})^{-1} - \mathbf{E}^T (\mathbf{E} \mathbf{B}^T \mathbf{B} \mathbf{E}^T)^{-1} \mathbf{E} \quad (4.184)$$

The least-squares solution of $\hat{\mathbf{x}}_B$ subject to condition (4.169) is, according to (4.170), (4.171), and (4.181),

$$\hat{\mathbf{x}}_B = -\mathbf{Q}_B \mathbf{A}^T \mathbf{P} \ell \quad (4.185)$$

The cofactor matrix of the parameters follows from the law of variance-covariance propagation

$$\mathbf{Q}_{x_B} = \mathbf{Q}_B \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_B = \mathbf{Q}_B \quad (4.186)$$

The latter part of (4.186) follows from (4.182) upon multiplying from the left by $\mathbf{Q}_B$ and using (4.174). Multiplying (4.182) from the right by $\mathbf{A}^T \mathbf{P} \mathbf{A}$ and using (4.177) gives

$$\mathbf{A}^T \mathbf{P} \mathbf{A} = \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_B \mathbf{A}^T \mathbf{P} \mathbf{A} \quad (4.187)$$

The relation implied in (4.186) is

$$\mathbf{Q}_B \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_B = \mathbf{Q}_B \quad (4.188)$$

$u - r$ conditions are necessary to solve the least-squares problem; i.e., the minimal number of conditions is equal to the rank defect of the design (or normal) matrix. Any solution derived in this manner is called a minimal constraint solution. There are obviously many different sets of minimal constraints possible for the same adjustment. The only prerequisite on the $\mathbf{B}$ matrix is that it have full row rank and that its rows be linearly independent of $\mathbf{A}$. Assume that

$$\mathbf{C} \mathbf{x}_C = \mathbf{o} \quad (4.189)$$

is an alternative set of conditions. The solution $\hat{\mathbf{x}}_C$ follows from the expressions given by simply replacing the matrix $\mathbf{B}$ by $\mathbf{C}$. The pertinent expressions are

$$\hat{\mathbf{x}}_C = -\mathbf{Q}_C \mathbf{A}^T \mathbf{P} \ell \quad (4.190)$$

$$\mathbf{Q}_C = (\mathbf{A}^T \mathbf{P} \mathbf{A} + \mathbf{C}^T \mathbf{C})^{-1} - \mathbf{E}^T (\mathbf{E} \mathbf{C}^T \mathbf{C} \mathbf{E}^T)^{-1} \mathbf{E} \quad (4.191)$$

$$\mathbf{T}_C \equiv \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_C = \mathbf{I} - \mathbf{C}^T (\mathbf{E} \mathbf{C}^T)^{-1} \mathbf{E} \quad (4.192)$$

$$\mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_C \mathbf{A}^T \mathbf{P} \mathbf{A} = \mathbf{A}^T \mathbf{P} \mathbf{A} \quad (4.193)$$

$$\mathbf{Q}_C \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_C = \mathbf{Q}_C \quad (4.194)$$

The solutions pertaining to the various alternative sets of conditions are all related. In particular,

$$\hat{\mathbf{x}}_B = \mathbf{T}_B^T \hat{\mathbf{x}}_C \quad (4.195)$$

$$\mathbf{Q}_B = \mathbf{T}_B^T \mathbf{Q}_C \mathbf{T}_B \quad (4.196)$$

$$\hat{\mathbf{x}}_C = \mathbf{T}_C^T \hat{\mathbf{x}}_B \quad (4.197)$$

$$\mathbf{Q}_C = \mathbf{T}_C^T \mathbf{Q}_B \mathbf{T}_C \quad (4.198)$$

Equations (4.195) to (4.198) constitute the transformation of minimal control; i.e., they relate the adjusted parameters and the covariance matrix for different minimal constraints. These transformation expressions are readily proven. For example, by using (4.190), (4.182), (4.192), and (4.177), we obtain

$$\begin{aligned} \mathbf{T}_B^T \hat{\mathbf{x}}_C &= -\mathbf{T}_B^T \mathbf{Q}_C \mathbf{A}^T \mathbf{P} \boldsymbol{\ell} \\ &= -\mathbf{Q}_B \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_C \mathbf{A}^T \mathbf{P} \boldsymbol{\ell} \\ &= -\mathbf{Q}_B \left[\mathbf{I} - \mathbf{C}^T (\mathbf{E} \mathbf{C}^T)^{-1} \mathbf{E} \right] \mathbf{A}^T \mathbf{P} \boldsymbol{\ell} \\ &= -\mathbf{Q}_B \mathbf{A}^T \mathbf{P} \boldsymbol{\ell} \\ &= \hat{\mathbf{x}}_B \end{aligned} \quad (4.199)$$

With (4.192), (4.187), and (4.194), it follows that

$$\begin{aligned} \mathbf{T}_C^T \mathbf{Q}_B \mathbf{T}_C &= \mathbf{Q}_C \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_B \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_C \\ &= \mathbf{Q}_C \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_C \\ &= \mathbf{Q}_C \end{aligned} \quad (4.200)$$

Instead of using the general condition (4.189), we can use the condition

$$\mathbf{E} \mathbf{x}_P = \mathbf{o} \quad (4.201)$$

The rows of $\mathbf{E}$ are linearly independent of $\mathbf{A}$ because of (4.177). Thus, replacing the matrix $\mathbf{C}$ by $\mathbf{E}$ in Equations (4.190) through (4.198) gives this special solution:

$$\hat{\mathbf{x}}_P = -\mathbf{Q}_P \mathbf{A}^T \mathbf{P} \ell \quad (4.202)$$

$$\mathbf{Q}_P = (\mathbf{A}^T \mathbf{P} \mathbf{A} + \mathbf{E}^T \mathbf{E})^{-1} - \mathbf{E}^T (\mathbf{E} \mathbf{E}^T \mathbf{E} \mathbf{E}^T)^{-1} \mathbf{E} \quad (4.203)$$

$$\mathbf{T}_P \equiv \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_P = \mathbf{I} - \mathbf{E}^T (\mathbf{E} \mathbf{E}^T)^{-1} \mathbf{E} \quad (4.204)$$

$$\mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_P \mathbf{A}^T \mathbf{P} \mathbf{A} = \mathbf{A}^T \mathbf{P} \mathbf{A} \quad (4.205)$$

$$\mathbf{Q}_P \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{Q}_P = \mathbf{Q}_P \quad (4.206)$$

$$\hat{\mathbf{x}}_B = \mathbf{T}_B^T \hat{\mathbf{x}}_P \quad (4.207)$$

$$\mathbf{Q}_B = \mathbf{T}_B^T \mathbf{Q}_P \mathbf{T}_B \quad (4.208)$$

$$\hat{\mathbf{x}}_P = \mathbf{T}_P^T \hat{\mathbf{x}}_B \quad (4.209)$$

$$\mathbf{Q}_P = \mathbf{T}_P^T \mathbf{Q}_B \mathbf{T}_P \quad (4.210)$$

The solution (4.202) is called the inner constraint solution. The matrix $\mathbf{T}_P$ in (4.204) is symmetric. The matrix $\mathbf{Q}_P$ is a generalized inverse, called the pseudoinverse of the normal matrix; the following notation is used:

$$\mathbf{Q}_P = \mathbf{N}^+ = (\mathbf{A}^T \mathbf{P} \mathbf{A})^+ \quad (4.211)$$

The pseudoinverse of the normal matrix is computed from available algorithms of generalized matrix inverses or, equivalently, by finding the $\mathbf{E}$ matrix and using Equation (4.203). For typical applications in surveying, the matrix $\mathbf{E}$ can be readily identified. Because of (4.177), the solution (4.202) can also be written as

$$\hat{\mathbf{x}}_P = -(\mathbf{A}^T \mathbf{P} \mathbf{A} + \mathbf{E}^T \mathbf{E})^{-1} \mathbf{A}^T \mathbf{P} \ell \quad (4.212)$$

Note that the covariance matrix of the adjusted parameters is

$$\Sigma_x = \hat{\sigma}_0^2 \mathbf{Q}_{B,C,P} \quad (4.213)$$

depending on whether constraint (4.169), (4.189), or (4.201) is used.

The inner constraint solution is yet another minimal constraint solution, although it has some special features. It can be shown that among all possible minimal constraint solutions, the inner constraint solution also minimizes the sum of the squares of the parameters, i.e.,

$$\mathbf{x}^T \mathbf{x} = \text{minimum} \quad (4.214)$$

This property can be used to obtain a geometric interpretation of the inner constraints. For example, it can be shown that the approximate parameters $\mathbf{x}_0$ and the adjusted parameters $\hat{\mathbf{x}}_P$ can be related by a similarity transformation whose least-squares estimates of translation and rotation are zero. For inner constraint solutions, the standard ellipses show the geometry of the network and are not affected by the definition of the coordinate system. It can also be shown that the trace of $\mathbf{Q}_P$ is the smallest compared to the trace of the other cofactor matrices. All minimal constraint solutions yield the same adjusted observations, a posteriori variance of unit weight, covariance matrices for residuals, and the same values for estimable functions of the parameters and their variances. The next section presents a further explanation of quantities invariant with respect to changes in minimal constraints.

4.9 STATISTICS IN LEAST-SQUARES ADJUSTMENT

Statistics completes the theory of adjustments, because it allows one to make objective statements about the data. The basic requirements, however, are that the mathematical model and the stochastic model be correct and that the observations have a multivariate normal distribution. Statistics cannot guarantee the right decision, but it can be helpful in gaining deeper insight into often unconscious motives that lead to certain decisions.

4.9.1 Multivariate Normal Distribution

This section contains a brief introduction to multivariate normal distribution. A few theorems are given that will be helpful in subsequent derivations. The multivariate normal distribution is especially pleasing, because the marginal distributions derived from multivariate normal distributions are also normally distributed. An extensive treatment of this distribution is found in the standard statistical literature. To simplify notation, the tilde is not used to identify random variables. The random nature of variables can be readily deduced from the context.

Let $\mathbf{x}$ be a vector with n random components with a mean of

$$E(\mathbf{x}) = \boldsymbol{\mu} \quad (4.215)$$

and a covariance matrix of

$$E[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] = {}_n \boldsymbol{\Sigma}_n \quad (4.216)$$

If $\mathbf{x}$ has a multivariate normal distribution, then the multivariate density function is

$$f(x_1, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} e^{-(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})/2} \quad (4.217)$$

The mean and the covariance matrix completely describe the multivariate normal distribution. The notation

$${}_n\mathbf{x}_1 \sim N_n({}_n\boldsymbol{\mu}_1, {}_n\boldsymbol{\Sigma}_n) \quad (4.218)$$

is used. The dimension of the distribution is n .

In the following, some theorems on multivariate normal distributions are given without proofs. These theorems are useful in deriving the distribution of $\mathbf{v}^T \mathbf{P} \mathbf{v}$ and some of the basic statistical tests in least-squares adjustments.

Theorem 1 If $\mathbf{x}$ is multivariate normal

$$\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (4.219)$$

and

$$\mathbf{z} = {}_m\mathbf{D}_n \mathbf{x} \quad (4.220)$$

is a linear function of the random variable, where $\mathbf{D}$ is a $m \times n$ matrix of rank $m \leq n$, then

$$\mathbf{z} \sim N_m(\mathbf{D} \boldsymbol{\mu}, \mathbf{D} \boldsymbol{\Sigma} \mathbf{D}^T) \quad (4.221)$$

is a multivariate normal distribution of dimension m . The mean and variance of the random variable $\mathbf{z}$ follow from the laws for propagating the mean (4.33) and variance-covariances (4.34).

Theorem 2 If $\mathbf{x}$ is multivariate normal $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the marginal distribution of any set of components of $\mathbf{x}$ is multivariate normal with means, variances, and covariances obtained by taking the proper component of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. For example, if

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} \sim N \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \right) \quad (4.222)$$

then the marginal distribution of $\mathbf{x}_2$ is

$$\mathbf{x}_2 \sim N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22}) \quad (4.223)$$

The same law holds, of course, if the set contains only one component, say x_i . The marginal distribution of x_i is then

$$x_i \sim n(\mu_i, \sigma_i^2) \quad (4.224)$$

Theorem 3 If $\mathbf{x}$ is multivariate normal, a necessary and sufficient condition that two subsets of the random variables are stochastically independent is that the covariances be zero. For example, if

$$\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} \sim N \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \mathbf{O} \\ \mathbf{O} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \right) \quad (4.225)$$

then $\mathbf{x}_1$ and $\mathbf{x}_2$ are stochastically independent. If one set of normally distributed random variables is uncorrelated with the remaining variables, the two sets are independent. The proof of the above theorem follows from the fact that the density function can be written as a product of $f_1(\mathbf{x}_1)$ and $f_2(\mathbf{x}_2)$ because of the special form of the density function (4.217).

4.9.2 Distribution of $\mathbf{v}^T \mathbf{P} \mathbf{v}$

The derivation of the distribution is based on the assumption that the observations have a multivariate normal distribution. The dimension of the distribution equals the number of observations. In the subsequent derivations the observation equation model is used. However, these statistical derivations could just as well have been carried out with the mixed model.

The observation equations are

$$\begin{aligned} \mathbf{v} &= \mathbf{A} \mathbf{x} + \boldsymbol{\ell}_0 - \boldsymbol{\ell}_b \\ &= \mathbf{A} \mathbf{x} + \boldsymbol{\ell} \end{aligned} \quad (4.226)$$

A first assumption is that the residuals are randomly distributed, i.e., the probability for a positive or negative residual of the equal magnitude is the same. From this assumption it follows that

$$E(\mathbf{v}) = \mathbf{0} \quad (4.227)$$

Because $\mathbf{x}$ and $\boldsymbol{\ell}_0$ are constant vectors, it further follows that the mean and variance-covariance matrix, respectively, are

$$E(\boldsymbol{\ell}_b) = \boldsymbol{\ell}_0 + \mathbf{A} \mathbf{x} \quad (4.228)$$

$$E(\mathbf{v} \mathbf{v}^T) = E\{[\boldsymbol{\ell}_b - E(\boldsymbol{\ell}_b)][\boldsymbol{\ell}_b - E(\boldsymbol{\ell}_b)]^T\} = \boldsymbol{\Sigma}_{\boldsymbol{\ell}_b} = \sigma_0^2 \mathbf{P}^{-1} \quad (4.229)$$

The second basic assumption refers to the type of distribution of the observations. It is assumed that the distribution is multivariate normal. Using the mean (4.228) and the covariance matrix (4.229), the n -dimensional multivariate normal distribution of $\boldsymbol{\ell}_b$ is written as

$$\boldsymbol{\ell}_b \sim N_n(\boldsymbol{\ell}_0 + \mathbf{A} \mathbf{x}, \boldsymbol{\Sigma}_{\boldsymbol{\ell}_b}) \quad (4.230)$$

Alternative expressions are

$$\boldsymbol{\ell} \sim N_n(-\mathbf{A} \mathbf{x}, \boldsymbol{\Sigma}_{\boldsymbol{\ell}_b}) \quad (4.231)$$

$$\mathbf{v} \sim N_n(\mathbf{o}, \Sigma_{\ell_b}) = N_n(\mathbf{o}, \sigma_0^2 \mathbf{P}^{-1}) \quad (4.232)$$

Applying two orthogonal transformations we can conveniently derive $\mathbf{v}^T \mathbf{P} \mathbf{v}$. If Σ_{ℓ_b} is nondiagonal, one can always find observations that are stochastically independent and have a unit variate normal distribution. As discussed in Appendix A, for a positive definite matrix $\mathbf{P}$ there exists a nonsingular matrix $\mathbf{D}$ such that the following is valid,

$$\mathbf{D} = \mathbf{E} \Lambda^{-1/2} \quad (4.233)$$

$$\mathbf{D}^T \mathbf{P}^{-1} \mathbf{D} = \mathbf{I} \quad (4.234)$$

$$\mathbf{D}^T \mathbf{v} = \mathbf{D}^T \mathbf{A} \mathbf{x} + \mathbf{D}^T \ell \quad (4.235)$$

$$\bar{\mathbf{v}} = \bar{\mathbf{A}} \mathbf{x} + \bar{\ell} \quad (4.236)$$

$$\bar{\ell} = \mathbf{D}^T \ell_0 - \mathbf{D}^T \ell_b = \bar{\ell}_0 - \bar{\ell}_b \quad (4.237)$$

$$E(\bar{\mathbf{v}}) = \mathbf{D}^T E(\mathbf{v}) = \mathbf{o} \quad (4.238)$$

$$\Sigma_{\bar{\mathbf{v}}} = \sigma_0^2 \mathbf{D}^T \mathbf{P}^{-1} \mathbf{D} = \sigma_0^2 \mathbf{I} \quad (4.239)$$

$$\bar{\mathbf{v}} \sim N_n(\mathbf{o}, \sigma_0^2 \mathbf{I}) \quad (4.240)$$

The columns of the orthogonal matrix $\mathbf{E}$ consist of the normalized eigenvectors of $\mathbf{P}^{-1}$; Λ is a diagonal matrix having the eigenvalues of $\mathbf{P}^{-1}$ at the diagonal. The quadratic form $\mathbf{v}^T \mathbf{P} \mathbf{v}$ remains invariant under this transformation because

$$R \equiv \mathbf{v}^T \mathbf{P} \mathbf{v} = \bar{\mathbf{v}}^T \Lambda^{1/2} \mathbf{E}^T \mathbf{P} \mathbf{E} \Lambda^{1/2} \bar{\mathbf{v}} = \bar{\mathbf{v}}^T \Lambda^{1/2} \Lambda^{-1} \Lambda^{1/2} \bar{\mathbf{v}} = \bar{\mathbf{v}}^T \bar{\mathbf{v}} \quad (4.241)$$

If the covariance matrix Σ_{ℓ_b} has a rank defect, then one could use matrix $\mathbf{F}$ of (A.52) for the transformation. The dimension of the transformed observations $\bar{\ell}_b$ equals the rank of the covariance matrix.

In the next step, the parameters are transformed to a new set that is stochastically independent. To keep the generality, let the matrix $\bar{\mathbf{A}}$ in (4.236) have less than full column rank, i.e., $R(\bar{\mathbf{A}}) = r < u$. Let the matrix $\mathbf{F}$ be an $n \times r$ matrix whose columns constitute an orthonormal basis for the column space of $\bar{\mathbf{A}}$. One such choice for the columns of $\mathbf{F}$ may be to take the normalized eigenvectors of $\bar{\mathbf{A}} \bar{\mathbf{A}}^T$. Let $\mathbf{G}$ be an $n \times (n - r)$ matrix, such that $[\mathbf{F} \ \mathbf{G}]$ is orthogonal and such that the columns of $\mathbf{G}$ constitute an orthonormal basis to the $n - r$ -dimensional null space of $\bar{\mathbf{A}} \bar{\mathbf{A}}^T$. Such a matrix always exists. There is no need to compute this matrix explicitly. With these specifications we obtain

$$\begin{bmatrix} \mathbf{F}^T \\ \mathbf{G}^T \end{bmatrix} [\mathbf{F} \ \mathbf{G}] = \begin{bmatrix} \mathbf{F}^T \mathbf{F} & \mathbf{F}^T \mathbf{G} \\ \mathbf{G}^T \mathbf{F} & \mathbf{G}^T \mathbf{G} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_{n-r} \end{bmatrix} \quad (4.242)$$

$$[\mathbf{F} \quad \mathbf{G}][\mathbf{F} \quad \mathbf{G}]^T = \mathbf{F}\mathbf{F}^T + \mathbf{G}\mathbf{G}^T = \mathbf{I} \quad (4.243)$$

$$\bar{\mathbf{A}}^T \mathbf{G} = \mathbf{O} \quad (4.244)$$

$$\mathbf{G}^T \bar{\mathbf{A}} = \mathbf{O} \quad (4.245)$$

The required transformation is

$$\begin{bmatrix} \mathbf{F}^T \\ \mathbf{G}^T \end{bmatrix} \bar{\mathbf{v}} = \begin{bmatrix} \mathbf{F}^T \\ \mathbf{G}^T \end{bmatrix} \bar{\mathbf{A}}\mathbf{x} + \begin{bmatrix} \mathbf{F}^T \\ \mathbf{G}^T \end{bmatrix} \bar{\ell} \quad (4.246)$$

or, equivalently,

$$\begin{bmatrix} \mathbf{F}^T \bar{\mathbf{v}} \\ \mathbf{G}^T \bar{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} \mathbf{F}^T \bar{\mathbf{A}}\mathbf{x} \\ \mathbf{O} \end{bmatrix} + \begin{bmatrix} \mathbf{F}^T \bar{\ell} \\ \mathbf{G}^T \bar{\ell} \end{bmatrix} \quad (4.247)$$

Labeling the newly transformed observations by $\mathbf{z}$, i.e.,

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{F}^T \bar{\ell} \\ \mathbf{G}^T \bar{\ell} \end{bmatrix} \quad (4.248)$$

we can write (4.247) as

$$\bar{\mathbf{v}}_z = \begin{bmatrix} \bar{\mathbf{v}}_{z_1} \\ \bar{\mathbf{v}}_{z_2} \end{bmatrix} = \begin{bmatrix} \mathbf{F}^T \bar{\mathbf{A}}\mathbf{x} \\ \mathbf{O} \end{bmatrix} + \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{bmatrix} \quad (4.249)$$

There are r random variables in $\mathbf{z}_1$ and $n - r$ random variables in $\mathbf{z}_2$. The quadratic form again remains invariant under the orthogonal transformation, since

$$\begin{aligned} \bar{\mathbf{v}}_z^T \bar{\mathbf{v}}_z &= \bar{\mathbf{v}}^T (\mathbf{F}\mathbf{F}^T + \mathbf{G}\mathbf{G}^T) \bar{\mathbf{v}} \\ &= \bar{\mathbf{v}}^T \bar{\mathbf{v}} = R \end{aligned} \quad (4.250)$$

according to (4.243). The actual quadratic form is obtained from (4.249):

$$R = \bar{\mathbf{v}}_z^T \bar{\mathbf{v}}_z = (\mathbf{F}^T \bar{\mathbf{A}}\mathbf{x} + \mathbf{z}_1)^T (\mathbf{F}^T \bar{\mathbf{A}}\mathbf{x} + \mathbf{z}_1) + \mathbf{z}_2^T \mathbf{z}_2 \quad (4.251)$$

The least-squares solution requires that R be minimized by variation of the parameters. Generally, equating partial derivatives with respect to $\mathbf{x}$ to zero and solving the resulting equations gives the minimum. The special form of (4.251) permits a much simpler approach. The expressions on the right side of Equation (4.251) consist of the sum of two positive terms (sum of squares). Because only the first term is a function of the parameters $\mathbf{x}$, the minimum is achieved if the first term is zero, i.e.,

$$-{}_r \mathbf{F}_n^T \bar{\mathbf{A}}_u \hat{\mathbf{x}}_1 = \mathbf{z}_1 \quad (4.252)$$

Note that the caret identifies the estimated parameters. Consequently, the estimate of the quadratic form is

$$\hat{R} = \mathbf{z}_2^T \mathbf{z}_2 \quad (4.253)$$

Because there are $r < u$ equations for the u parameters in (4.252), there always exists a solution for $\hat{\mathbf{x}}$. The simplest approach is to equate $u - r$ parameters to zero. This would be identical to having these $u - r$ parameters treated as constants in the adjustment. They could be left out when setting up the design matrix and thus, the singularity problem would be avoided altogether. Equation (4.252) can be solved subject to $u - r$ general conditions between the parameters. The resulting solution is a minimal constraint solution. If the particular condition (4.201) is applied, one obtains the inner constraint solution. If $\bar{\mathbf{A}}$ has no rank defect, then the system (4.252) consists of u equations for u unknowns.

The estimate for the quadratic form (4.253) does not depend on the parameters $\mathbf{x}$ and, thus, is invariant with respect to the selection of the minimal constraints for finding the least-square estimate of $\mathbf{x}$. Moreover, the residuals themselves are independent of the minimal constraints. Substituting the solution (4.252) into (4.247) gives

$$\begin{bmatrix} \mathbf{F}^T \\ \mathbf{G}^T \end{bmatrix} \hat{\mathbf{v}} = \begin{bmatrix} \mathbf{0} \\ \mathbf{G}^T \bar{\ell} \end{bmatrix} \quad (4.254)$$

Since the matrix $[\mathbf{F} \ \mathbf{G}]$ is orthonormal, the expression for the residuals becomes

$$\hat{\mathbf{v}} = [\mathbf{F} \ \mathbf{G}] \begin{bmatrix} \mathbf{0} \\ \mathbf{G}^T \bar{\ell} \end{bmatrix} = \mathbf{G} \mathbf{G}^T \bar{\ell} \quad (4.255)$$

Thus, the residuals are independent of the specific solution for $\hat{\mathbf{x}}$. The matrix $\mathbf{G}$ depends only on the structure of the design matrix $\bar{\mathbf{A}}$. By applying the law of variance-covariance propagation to (4.255), we clearly see that the covariance matrix of the adjusted residuals, and thus the covariance matrix of the adjusted observations, does not depend on the specific set of minimal constraints. Note that the transformation (4.235) does not invalidate these statements, since the $\mathbf{D}$ matrix is not related to the parameters.

Returning to the derivation of the distribution of $\mathbf{v}^T \mathbf{P} \mathbf{v}$, we find from (4.248) that

$$\mathbf{E}(\mathbf{z}) = \begin{bmatrix} -\mathbf{F}^T \bar{\mathbf{A}} \mathbf{x} \\ \mathbf{0} \end{bmatrix} \quad (4.256)$$

using (4.245) and the fact that $\mathbf{E}(\bar{\ell}) = -\bar{\mathbf{A}} \mathbf{x}$ according to (4.236). Making use of (4.240) the covariance matrix is

$$\Sigma_z = \sigma_0^2 \begin{bmatrix} \mathbf{F}^T \\ \mathbf{G}^T \end{bmatrix} \mathbf{I} [\mathbf{F} \ \mathbf{G}] = \sigma_0^2 \begin{bmatrix} \mathbf{F}^T \mathbf{F} & \mathbf{F}^T \mathbf{G} \\ \mathbf{G}^T \mathbf{F} & \mathbf{G}^T \mathbf{G} \end{bmatrix} = \sigma_0^2 \begin{bmatrix} \mathbf{I} & \mathbf{O} \\ \mathbf{O} & \mathbf{I} \end{bmatrix} \quad (4.257)$$

Since a linear transformation of a random variable with multivariate normal distribution results in another multivariate normal distribution according to Theorem 1, it follows that $\mathbf{z}$ is distributed as

$$\mathbf{z} \sim N_n \left(\begin{bmatrix} -\mathbf{F}^T \bar{\mathbf{A}} \mathbf{x} \\ \mathbf{o} \end{bmatrix}, \sigma_0^2 \begin{bmatrix} {}_r \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & {}_{n-r} \mathbf{I}_{n-r} \end{bmatrix} \right) \quad (4.258)$$

The random variables $\mathbf{z}_1$ and $\mathbf{z}_2$ are stochastically independent, as are the individual components according to Theorem 3. From Theorem 2 it follows that

$$\mathbf{z}_2 \sim N_{n-r}(\mathbf{o}, \sigma_0^2 \mathbf{I}) \quad (4.259)$$

Thus

$$z_{2i} \sim n(0, \sigma_0^2) \quad (4.260)$$

$$\frac{z_{2i}}{\sigma_0} \sim n(0, 1) \quad (4.261)$$

are unit variate normal distributed. As listed in Appendix A5, the square of a standardized normal distributed variable has a chi-square distribution with one degree of freedom. In addition, the sum of chi-square distributed variables is also a chi-square distribution with a degree of freedom equal to the sum of the individual degrees of freedom. Using these functions of random variables, it follows that $\mathbf{v}^T \mathbf{P} \mathbf{v}$

$$\frac{\hat{R}}{\sigma_0^2} = \frac{\mathbf{z}_2^T \mathbf{z}_2}{\sigma_0^2} = \sum_{i=1}^{n-r} \frac{z_{2i}^2}{\sigma_0^2} \sim \chi_{n-r}^2 \quad (4.262)$$

has a chi-square distribution with $n - r$ degrees of freedom.

4.9.3 Testing $\mathbf{v}^T \mathbf{P} \mathbf{v}$ and $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$

Combining the result of (4.262) with the expression for the a posteriori variance of unit weight of Table 4.1, we obtain the formulation for a fundamental statistical test in least-squares estimation:

$$\frac{\mathbf{v}^T \mathbf{P} \mathbf{v}}{\sigma_0^2} = \frac{\hat{\sigma}_0^2}{\sigma_0^2} (n - r) \sim \chi_{n-r}^2 \quad (4.263)$$

Note that $n - r$ is the degree of freedom of the adjustment. If there is no rank deficiency in the design matrix, the degree of freedom is $n - u$. Based on the statistics (4.263), the test can be performed to find out whether the adjustment is distorted. The formulation of the hypothesis is as follows:

$$H_0: \sigma_0^2 = \hat{\sigma}_0^2 \quad (4.264)$$

$$H_1: \sigma_0^2 \neq \hat{\sigma}_0^2 \tag{4.265}$$

The zero hypothesis states that the a priori variance of unit weight statistically equals the a posteriori variance of unit weight. Recall that the a posteriori variance of unit weight is a random variable; the adjustment makes a sample value available for this quantity on the basis of the observations (the samples). Both variances of unit weight do not have to be numerically equal; but they should be statistically equal in the sense of (4.67). If the zero hypothesis is accepted, the adjustment is judged to be correct. If the numerical value

$$\chi^2 = \frac{\hat{\sigma}_0^2}{\sigma_0^2}(n - r) = \frac{\mathbf{v}^T \mathbf{P} \mathbf{v}}{\sigma_0^2} \tag{4.266}$$

is such that

$$\chi^2 < \chi_{n-r, 1-\alpha/2}^2 \tag{4.267}$$

$$\chi^2 > \chi_{n-r, \alpha/2}^2 \tag{4.268}$$

then the zero hypothesis is rejected. The significance level α , i.e., the probability of a type-I error, or the probability of rejecting the zero hypothesis even though it is true, is generally fixed to 0.05. Here the significance level is the sum of the probabilities in both tails. Table 4.6 lists selected values from the chi-square distribution $\chi_{n-r, \alpha}^2$. Rejection of the zero hypothesis is taken to indicate that something is wrong with the adjustment. The cause for rejection remains to be clarified. Figure 4.3 shows the limits for the posteriori variance of unit weight as a function of the degree of freedom given the significance level $\alpha = 0.05$.

The probability β of the type-II error, i.e., the probability of rejecting the alternative hypothesis and accepting the zero hypothesis even though the alternative hypothesis is true, is generally not computed. Type-II errors are considered in Section 4.10.2 in regards to reliability and in Section 7.8.3 in regards to discernibility of estimated ambiguity sets.

TABLE 4.6 Selected Values for Chi-Square

Degree of Freedom (DF)	Probability α			
	0.975	0.950	0.050	0.025
1	0.00	0.00	3.84	5.02
5	0.83	1.15	11.07	12.83
10	3.25	3.94	18.31	20.48
20	9.59	10.85	31.41	34.17
50	32.36	34.76	67.50	71.42
100	74.22	77.93	124.34	129.56

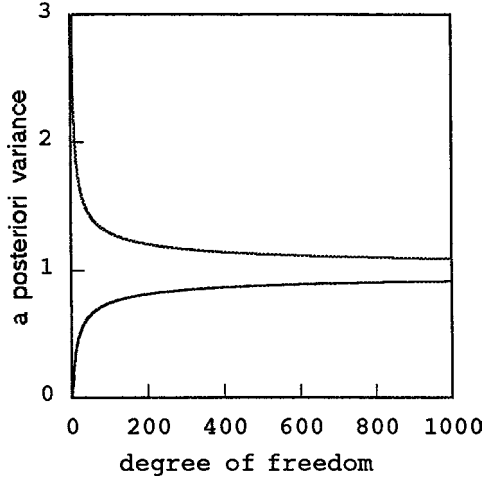


Figure 4.3 Limits on the a posteriori variance of unit weight. The figure refers to $\alpha = 0.05$.

The test statistics for testing groups of observations is based on $\mathbf{v}^T \mathbf{P} \mathbf{v}^*$ and the change $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$. According to Table 4.2 we have

$$\begin{aligned} \Delta \mathbf{v}^T \mathbf{P} \mathbf{v} &= (\mathbf{A}_2 \mathbf{x}^* + \ell_2)^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \ell_2) \\ &= \mathbf{z}_3^T \mathbf{T} \mathbf{z}_3 \end{aligned} \quad (4.269)$$

The new random variable $\mathbf{z}_3$ is a function of observations ℓ_1 and ℓ_2 . Applying the laws of propagation of mean and variance, one finds

$$E(\mathbf{z}_3) = \mathbf{A}_2 E(\mathbf{x}^*) + E(\ell_2) = \mathbf{A}_2 \mathbf{x} - \mathbf{A}_2 \mathbf{x} = \mathbf{0} \quad (4.270)$$

$$\Sigma_{\mathbf{z}_3} = \mathbf{T}^{-1} \quad (4.271)$$

$$\mathbf{z}_3 \sim N(\mathbf{0}, \sigma_0^2 \mathbf{T}^{-1}) \quad (4.272)$$

Carrying out the orthonormal transformation yields a random vector whose components are stochastically independent and normally distributed. By standardizing these distributions and summing the squares of these random variables, it follows that

$$\frac{\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}}{\sigma_0^2} = \frac{\mathbf{z}_3^T \mathbf{T} \mathbf{z}_3}{\sigma_0^2} \sim \chi_{n_2}^2 \quad (4.273)$$

has a chi-square distribution with n_2 degrees of freedom, where n_2 equals the number of observations in the second group. The random variables (4.273) and (4.263) are stochastically independent. To prove this, consider the new random variable $\mathbf{z} =$

$[\mathbf{z}_1 \ \mathbf{z}_2 \ \mathbf{z}_3]^T$, which is a linear function of the random variables ℓ (first group) and ℓ_2 , according to Equations (4.235), (2.248), and (4.269). By using the covariance matrix (4.88) and applying variance-covariance propagation, we find that the covariances between the $\mathbf{z}_i$ are zero. Because the distribution of the $\mathbf{z}$ is multivariate normal, it follows that the random variables $\mathbf{z}_i$ are stochastically independent. Since $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ is a function of $\mathbf{z}_3$ only, it follows that $\mathbf{v}^T \mathbf{P} \mathbf{v}$ in (4.263), which is only a function of $\mathbf{z}_2$, and $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ in (4.273) are stochastically independent. Thus, it is permissible to form the following ratio of random variables:

$$\frac{\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}(n_1 - r)}{\mathbf{v}^T \mathbf{P} \mathbf{v}^*(n_2)} \sim F_{n_2, n_1 - r} \tag{4.274}$$

which has an F distribution.

Thus the fundamental test in sequential adjustment is based on the F distribution. The zero hypothesis states that the second group of observations does not distort the adjustment, or that there is no indication that something is wrong with the second group of observations. The alternative hypothesis states that there is an indication that the second group of observations contains errors. The zero hypothesis is rejected, and the alternative hypothesis is accepted if

$$F < F_{n_2, n_1 - r, 1 - \alpha/2} \tag{4.275}$$

$$F > F_{n_2, n_1 - r, \alpha/2} \tag{4.276}$$

Table 4.7 lists selected values from the F distribution as a function of the degrees of freedom and probability. The tabulation refers to the parameters as specified in $F_{n_1, n_2, 0.05}$.

4.9.4 General Linear Hypothesis

The general linear hypothesis deals with linear conditions between parameters. Non-linear conditions are first linearized. The basic idea is to test the change $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ for its statistical significance. Any of the three adjustment models can be used to carry out

TABLE 4.7 Selected Values for F

n_2	n_1			
	1	2	3	4
5	6.61	5.79	5.41	5.19
10	4.96	4.10	3.71	3.48
20	4.35	3.49	3.10	2.87
60	4.00	3.15	2.76	2.53
120	3.92	3.07	2.68	2.45
∞	3.84	3.00	2.60	2.37

the general linear hypothesis test. For the observation equation model with additional conditions between the parameters, one has

$$\mathbf{v}_1 = \mathbf{A}_1 \mathbf{x} + \ell_1 \quad (4.277)$$

$$H_0: \mathbf{A}_2 \mathbf{x} + \ell_2 = \mathbf{0} \quad (4.278)$$

Equation (4.278) expresses the zero hypothesis H_0 . The solution of the combined adjustment is found in Table 4.5. Adjusting (4.277) alone results in $\mathbf{v}^T \mathbf{P} \mathbf{v}^*$, which has a chi-square distribution with $n - r$ degrees of freedom according to (4.273). The change $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ resulting from the condition (4.278) is

$$\Delta \mathbf{v}^T \mathbf{P} \mathbf{v} = (\mathbf{A}_2 \mathbf{x}^* + \ell_2)^T \mathbf{T} (\mathbf{A}_2 \mathbf{x}^* + \ell_2) \quad (4.279)$$

The expression in (4.279) differs from (4.269) in two respects. First, the matrix $\mathbf{T}$ differs; i.e., the matrix $\mathbf{T}$ in (4.279) does not contain the $\mathbf{P}_2$ matrix. Second, the quantity ℓ_2 is not a random variable. These differences, however, do not matter in the proof of stochastic independence of $\mathbf{v}^T \mathbf{P} \mathbf{v}^*$ and $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$. Analogously to (4.269), we can express the change $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ in (4.279) as a function of a new random variable $\mathbf{z}_3$. The proof for stochastic independence follows the same lines of thought as given before (for the case of additional observations). Thus, just as (4.274) is the basis for testing two groups of observations, the basic test for the general linear hypothesis (4.278) is

$$\frac{\Delta \mathbf{v}^T \mathbf{P} \mathbf{v} (n_1 - r)}{\mathbf{v}^T \mathbf{P} \mathbf{v}^* / n_2} \sim F_{n_2, n_1 - r} \quad (4.280)$$

A small $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ implies that the null hypothesis (4.278) is acceptable; i.e., the conditions are in agreement with the observations. The conditions do not impose any distortions on the adjustment. The rejection criterion is based on the one-tail test at the upper end of the distribution. Thus, reject H_0 at a $100\alpha\%$ significance level if

$$F > F_{n_2, n_1 - r, \alpha} \quad (4.281)$$

The general formulation of the null hypothesis in (4.278) makes it possible to test any hypothesis on the parameters, so long as the hypothesis can be expressed in a mathematical equation. Nonlinear hypotheses must first be linearized. Simple hypotheses could be used to test whether an individual parameter has a certain numerical value, whether two parameters are equal, whether the distance between two stations has a certain length, whether an angle has a certain size, etc. For example, consider the hypothesis

$$H_0: \mathbf{x} - \mathbf{x}_T = \mathbf{0} \quad (4.282)$$

$$H_1: \mathbf{x} - \mathbf{x}_T \neq \mathbf{0} \quad (4.283)$$

The zero hypothesis states that the parameters equal a certain (true) value $\mathbf{x}_T$. From (4.278) it follows that $\mathbf{A}_2 = \mathbf{I}$ and $\ell_2 = -\mathbf{x}_T$. Using these specifications we can use $\mathbf{T} = \mathbf{N}$ in (4.279), and the statistic (4.280) becomes

$$\frac{(\hat{\mathbf{x}}^* - \mathbf{x}_T)^T \mathbf{N} (\hat{\mathbf{x}}^* - \mathbf{x}_T)}{\hat{\sigma}_0^2 r} \sim F_{r, n_1 - r, \alpha} \quad (4.284)$$

where the a posteriori variance of unit weight (first group only) has been substituted for $\mathbf{v}^T \mathbf{P} \mathbf{v}^*$. Once the adjustment of the first group (4.277) is completed, the values for the adjusted parameters and the a posteriori variance of unit weight are entered in (4.284), and the fraction is computed and compared with the F value (taking the proper degrees of freedom and the desired significance level into account). Rejection or acceptance of the zero hypothesis follows rule (4.281).

Note that one of the degrees of freedom in (4.284) is $r = R(\mathbf{N}) < u$, instead of u , which equals the number of parameters, even though Equation (4.282) expresses u conditions. Because of the possible rank defect of the normal matrix $\mathbf{N}$, the distribution of $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$ in (4.279) is a chi-square distribution with r degrees of freedom. Consider the derivation leading to (4.273). The u components of $\mathbf{z}_3$ are transformed to r stochastically independent unit variate normal distributions that are then squared and summed to yield the distribution of $\Delta \mathbf{v}^T \mathbf{P} \mathbf{v}$. The interpretation is that (4.282) represents one hypothesis on all parameters $\mathbf{x}$, and not u hypotheses on the u components on $\mathbf{x}$.

Expression (4.284) can be used to define the r -dimensional confidence region. Replace the particular $\mathbf{x}_T$ by the unknown parameter $\mathbf{x}$, and drop the asterisk; then

$$P \left[\frac{(\hat{\mathbf{x}} - \mathbf{x})^T \mathbf{N} (\hat{\mathbf{x}} - \mathbf{x})}{\hat{\sigma}_0^2 r} \leq F_{r, n_1 - r, \alpha} \right] = \int_0^{F_{r, n_1 - r, \alpha}} F_{r, n_1 - r} dF = 1 - \alpha \quad (4.285)$$

The probability region described by the expression on the left side of Equation (4.285) is an $R(\mathbf{N})$ -dimensional ellipsoid. The probability region is an ellipsoid, because the normal matrix $\mathbf{N}$ is positive definite or, at least, semipositive definite. If one identifies the center of the ellipsoid with $\hat{\mathbf{x}}$, then there is $(1 - \alpha)$ probability that the unknown point $\mathbf{x}$ lies within the ellipsoid. The orientation and the size of this ellipsoid are a function of the eigenvectors and eigenvalues of the normal matrix, the rank of the normal matrix, and the degree of freedom. Consider the orthonormal transformation

$$\mathbf{z} = \mathbf{F}^T (\mathbf{x} - \hat{\mathbf{x}}) \quad (4.286)$$

with $\mathbf{F}$ as specified in (A.52) and containing the normalized eigenvectors of $\mathbf{N}$, then

$$\mathbf{F}^T \mathbf{N} \mathbf{F} = \mathbf{\Lambda} \quad (4.287)$$

with $\mathbf{\Lambda}$ containing the r eigenvalues of $\mathbf{N}$, and

$$(\hat{\mathbf{x}} - \mathbf{x})^T \mathbf{N} (\hat{\mathbf{x}} - \mathbf{x}) = \mathbf{z}^T \mathbf{\Lambda} \mathbf{z} = \sum_{i=1}^r z_i^2 \lambda_i = \sum_{i=1}^r \frac{z_i^2}{(1/\sqrt{\lambda_i})^2} \quad (4.288)$$

Combining Equations (4.285) and (4.288), we can write the r -dimensional ellipsoid, or the r -dimensional confidence region, in the principal axes form:

$$P \left[\frac{z_1^2}{(\hat{\sigma}_0 \sqrt{r F_{r,n-r,\alpha}/\lambda_1})^2} + \cdots + \frac{z_r^2}{(\hat{\sigma}_0 \sqrt{r F_{r,n-r,\alpha}/\lambda_r})^2} \leq 1 \right] = 1 - \alpha \quad (4.289)$$

The confidence region is centered at $\hat{\mathbf{x}}$. Whenever the zero hypothesis H_0 of (4.282) is accepted, the point $\mathbf{x}_T$ falls within the confidence region. The probability that the ellipsoid contains the true parameters $\mathbf{x}_T$, is $1 - \alpha$. For these reasons, one naturally would like the ellipsoid to be small. Equation (4.289) shows that the semimajor axes are proportional to the inverse of the eigenvalues of the normal matrix. It is exactly this relationship that makes us choose the eigenvalues of $\mathbf{N}$ as large as possible, provided that we have a choice through appropriate network design variation. As an eigenvalue approaches zero, the respective axis of the confidence ellipsoid approaches infinity; this is an undesirable situation, both from a statistical point of view and because of the numerical difficulties encountered during the inversion of the normal matrix.

4.9.5 Ellipses as Confidence Regions

Confidence ellipses are statements of precision. They are frequently used in connection with two-dimensional networks in order to make the directional precision of station location visible. Ellipses of confidence follow from Section 4.9.4 simply by limiting the hypothesis (4.282) to two parameters, i.e., the Cartesian coordinates of a station. Of course, in a three-dimensional network one can compute three-dimensional ellipsoids or several ellipses, e.g., one for the horizontal and others for the vertical. Confidence ellipses or ellipsoids are not limited to the specific application of networks. However, in networks the confidence regions can be referenced with respect to the coordinate system of the network and thus can provide an integrated view of the geometry of the confidence regions and the network.

Consider the following hypothesis:

$$H_0: \mathbf{x}_i - \mathbf{x}_{i,T} = \mathbf{0} \quad (4.290)$$

where the notation

$$\mathbf{x}_i = [x_1 \quad x_2]^T \quad (4.291)$$

is used. The symbols x_1 and x_2 denote the Cartesian coordinates of a two-dimensional network station P_i . The test of this hypothesis follows the outline given in the previous section. The $\mathbf{A}_2$ matrix is of size $2 \times u$ because there are two separate equations in the hypothesis and u components in $\mathbf{x}$. The elements of $\mathbf{A}_2$ are zero except those elements

of rows 1 and 2, which correspond to the respective positions of x_1 and x_2 in $\mathbf{x}$. With these specifications it follows that

$$\mathbf{Q}_i = \mathbf{A}_2 \mathbf{N}^{-1} \mathbf{A}_2^T = \begin{bmatrix} q_{x_1} & q_{x_1, x_2} \\ q_{x_2, x_1} & q_{x_2} \end{bmatrix} \quad (4.292)$$

$\mathbf{Q}_i$ contains the respective elements of the inverse of the normal matrix. With these specifications $\mathbf{T} = \mathbf{Q}_i^{-1}$ and Expression (4.280) becomes

$$\frac{1}{2\hat{\sigma}_0^2} (\hat{\mathbf{x}}_i - \mathbf{x}_{i,T})^T \mathbf{Q}_i^{-1} (\hat{\mathbf{x}}_i - \mathbf{x}_{i,T}) \sim F_{2,n-r} \quad (4.293)$$

Given the significance level α , the hypothesis test can be carried out. The two-dimensional confidence region is

$$P \left[\frac{(\hat{\mathbf{x}}_i - \mathbf{x}_i)^T \mathbf{Q}_i^{-1} (\hat{\mathbf{x}}_i - \mathbf{x}_i)}{2\hat{\sigma}_0^2} \leq F_{2,n-r,\alpha} \right] = \int_0^{F_{2,n-r,\alpha}} F_{2,n-r} dF = 1 - \alpha \quad (4.294)$$

The size of the confidence ellipses defined by (4.294) depends on the degree of freedom of the adjustment and the significance level. The ellipses are centered at the adjusted position and delimit the $(1 - \alpha)$ probability area for the true position. The principal axis form of (4.294) is obtained through orthogonal transformation. Let $\mathbf{R}_i$ denote the matrix whose rows are the orthonormal eigenvectors of $\mathbf{Q}_i$, then

$$\mathbf{R}_i^T \mathbf{Q}_i^{-1} \mathbf{R}_i = \mathbf{\Lambda}_i^{-1} \quad (4.295)$$

according to (A.48). The matrix $\mathbf{\Lambda}_i$ is diagonal and contains the eigenvalues λ_1^Q and λ_2^Q of $\mathbf{Q}_i$. With

$$\mathbf{z}_i = \mathbf{R}_i^T (\hat{\mathbf{x}}_i - \mathbf{x}_i) \quad (4.296)$$

Expression (4.294) becomes

$$P \left\{ \left[\frac{z_1^2}{\left(\hat{\sigma}_0 \sqrt{\lambda_1^Q} 2F_{2,n-r,\alpha} \right)^2} + \frac{z_2^2}{\left(\hat{\sigma}_0 \sqrt{\lambda_2^Q} 2F_{2,n-r,\alpha} \right)^2} \right] \leq 1 \right\} = \int_0^{F_{2,n-r,\alpha}} F_{2,n-r} dF = 1 - \alpha \quad (4.297)$$

For $F_{2,n-r,\alpha} = 1/2$, the ellipse is called the *standard ellipse* or the *error ellipse*. Thus, the probability enclosed by the standard ellipse is a function of the degree of freedom $n - r$ and is computed as follows:

$$P(\text{standard ellipse}) = \int_0^{1/2} F_{2,n-r} dF \quad (4.298)$$

TABLE 4.8 Magnification Factor for Standard Ellipses

$n - r$	Probability $1 - \alpha$		
	95%	98%	99%
1	20.00	50.00	100.00
2	6.16	9.90	14.10
3	4.37	6.14	7.85
4	3.73	4.93	6.00
5	3.40	4.35	5.15
6	3.21	4.01	4.67
8	2.99	3.64	4.16
10	2.86	3.44	3.89
12	2.79	3.32	3.72
15	2.71	3.20	3.57
20	2.64	3.09	3.42
30	2.58	2.99	3.28
50	2.52	2.91	3.18
100	2.49	2.85	3.11
∞	2.45	2.80	3.03

The magnification factor, $\sqrt{2F_{2,n-r,\alpha}}$, as a function of the probability and the degree of freedom, is shown in Table 4.8. The table shows immediately that a small degree of freedom requires a large magnification factor to obtain, e.g., 95% probability. It is seen that in the range of small degrees of freedom, an increase in the degree of freedom rapidly decreases the magnification factor, whereas with a large degree of freedom, any additional observations cause only a minor reduction of the magnification factor. After a degree of freedom of about 8 or 10, the decrease in the magnification factor slows down noticeably. Thus, based on the speed of decreasing magnification factor, a degree of 10 appears optimal, considering the expense of additional observations and the little gain derived from them in the statistical sense. For a degree of freedom of 10, the magnification factor is about 3 to cover 95% probability.

The hypothesis (4.290) can readily be generalized to three dimensions encompassing the Cartesian coordinates of a three-dimensional network station. The magnification factor of the respective *standard ellipsoid* is $\sqrt{3F_{3,n-r,\alpha}}$ for it to contain $(1 - \alpha)$ probability. Similarly, the standard deviation of an individual coordinate is converted to a $(1 - \alpha)$ probability confidence interval by multiplication with $\sqrt{F_{1,n-r,\alpha}}$. These magnification factors are shown in Figure 4.4 for $\alpha = 0.05$. For higher degrees of freedom, the magnification factors converge toward the respective chi-square values because of the relationship $rF_{r,\infty} = \chi_r^2$.

For drawing the confidence ellipse at station P_i , we need the rotation angle φ between the (x_i) and (z_i) coordinate systems as well as the semimajor and semiminor axis of the ellipse. Let (y_i) denotes the translated (x_i) coordinate system through the adjusted point $\hat{\mathbf{x}}_i$, then Equation (4.296) becomes

$$\mathbf{z}_i = \mathbf{R}_i^T \mathbf{y}_i \quad (4.299)$$

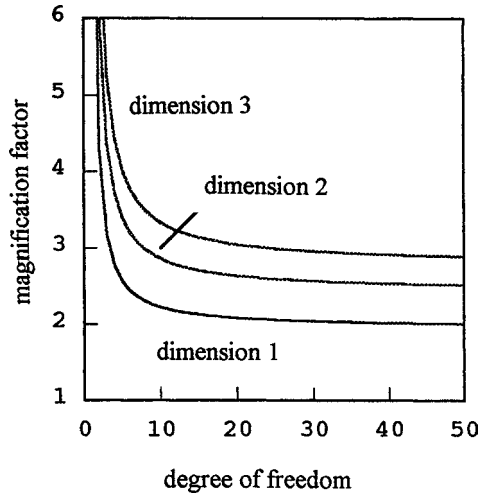


Figure 4.4 Magnification factors for confidence regions. The values refer to $\alpha = 0.05$.

The eigenvectors of $\mathbf{Q}_i$ determine the directions of the semiaxes, and the eigenvalues determine their lengths. Rather than computing the vectors explicitly, we choose to compute the rotation angle φ by comparing coefficients from quadratic forms. Figure 4.5 shows the rotational relation

$$\mathbf{z}_i = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix} \mathbf{y}_i \tag{4.300}$$

and Equations (4.295) and (4.299) give the two quadratic forms

$$\mathbf{y}_i^T \mathbf{Q}_i \mathbf{y}_i = \mathbf{z}_i^T \mathbf{\Lambda}_i \mathbf{z}_i \tag{4.301}$$

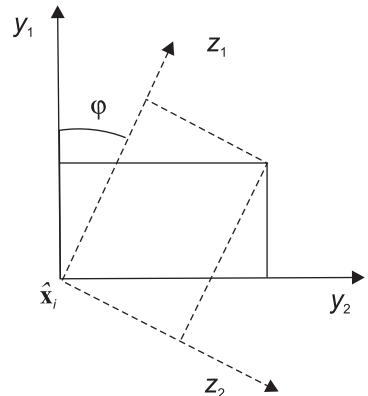


Figure 4.5 Rotation of the principal axis coordinate system.

We substitute (4.300) into the right-hand side of (4.301) and the matrix elements of $\mathbf{Q}_i$ of (4.292) into the left-hand side and compare the coefficient of $y_1 y_2$ on both sides, giving

$$\sin 2\varphi = \frac{2q_{x_1, x_2}}{\lambda_1^Q - \lambda_2^Q} \quad (4.302)$$

The eigenvalues follow directly from the characteristic equation

$$|\mathbf{Q}_i - \lambda^Q \mathbf{I}| = \begin{vmatrix} q_{x_1} - \lambda^Q & q_{x_1, x_2} \\ q_{x_1, x_2} & q_{x_2} - \lambda^Q \end{vmatrix} = (q_{x_1} - \lambda^Q)(q_{x_2} - \lambda^Q) - q_{x_1, x_2}^2 = 0 \quad (4.303)$$

The solution of the quadratic equation is given in (4.304) to (4.308). The terms $\sin 2\varphi$ and $\cos 2\varphi$ determine the quadrant of φ .

$$\lambda_1^Q = \frac{q_{x_1} + q_{x_2}}{2} + \frac{1}{2}W \quad (4.304)$$

$$\lambda_2^Q = \frac{q_{x_1} + q_{x_2}}{2} - \frac{1}{2}W \quad (4.305)$$

$$W = \sqrt{(q_{x_1} - q_{x_2})^2 + 4q_{x_1, x_2}^2} \quad (4.306)$$

$$\sin 2\varphi = \frac{2q_{x_1, x_2}}{W} \quad (4.307)$$

$$\cos 2\varphi = \frac{q_{x_1} - q_{x_2}}{W} \quad (4.308)$$

Figure 4.6 shows the defining elements of the standard ellipse. Recall Equation (4.297) regarding the interpretation of the standard ellipses as a confidence region. In any adjustment, any two parameters can comprise $\mathbf{x}_i$, regardless of the geometric meaning of the parameters. Examples are the intercept and slope in the fitting of a straight line or ambiguity parameters in the case of GPS carrier phase solutions. The components $\mathbf{x}_i$ can always be interpreted as Cartesian coordinates for drawing the standard ellipse and thus can give a graphical display of the covariance. In surveying networks, the vectors $\mathbf{x}_i$ contain coordinates of stations in a well-defined coordinate system. If $\mathbf{x}_i$ represents latitude and longitude or northing and easting, the horizontal standard ellipse is computed. If $\mathbf{x}_i$ contains the vertical coordinate and easting, then the standard ellipse in the prime vertical is obtained.

Because the shape of the standard ellipses and ellipsoids depends on the geometry of the network through the design matrix and the weight matrix, the geometric interpretation is enhanced if the network and the standard ellipses are displayed together. Occasionally, users prefer to compute coordinate differences and their covariance matrix and plot relative standard ellipses.

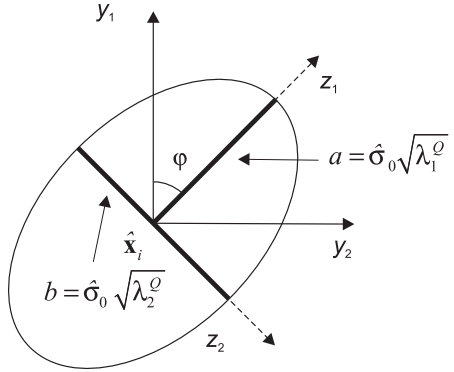


Figure 4.6 Defining elements of standard ellipse.

4.9.6 Properties of Standard Ellipses

The positional error p of a station is directly related to the standard ellipse, as seen in Figure 4.7. The positional error is the standard deviation of a station in a certain direction, say ψ . It is identical with the standard deviation of the distance to a known (fixed) station along the same direction ψ , as computed from the linearized distance equation and variance-covariance propagation. The linear function is

$$r = z_1 \cos \psi + z_2 \sin \psi \tag{4.309}$$

Because of Equations (4.295) and (4.296), the distribution of the random variable $\mathbf{z}_i$ is multivariate normal with

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \hat{\sigma}_0^2 \begin{bmatrix} \lambda_1^Q & 0 \\ 0 & \lambda_2^Q \end{bmatrix} \right) = N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} a^2 & 0 \\ 0 & b^2 \end{bmatrix} \right) \tag{4.310}$$

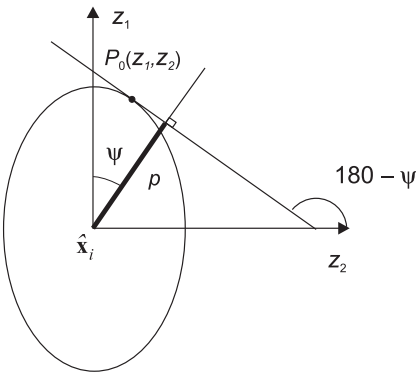


Figure 4.7 Position error.

The variance of the random variable r follows from the law of variance-covariance propagation:

$$\sigma_r^2 = a^2 \cos^2 \psi + b^2 \sin^2 \psi \quad (4.311)$$

The variance (4.311) is geometrically related to the standard ellipse. Let the ellipse be projected onto the direction ψ . The point of tangency is denoted by P_0 . Because the equation of the ellipse is

$$\frac{z_1^2}{a^2} + \frac{z_2^2}{b^2} = 1 \quad (4.312)$$

the slope of the tangent is

$$\frac{dz_1}{dz_2} = -\frac{z_2 a^2}{z_1 b^2} = -\tan \psi \quad (4.313)$$

See Figure 4.7 regarding the relation of the slope of the tangent and the angle ψ . The second part of (4.313) yields

$$\frac{z_{01}}{a^2} \sin \psi - \frac{z_{02}}{b^2} \cos \psi = 0 \quad (4.314)$$

This equation relates the coordinates of the point of tangency P_0 to the slope of the tangent. The length p of the projection of the ellipse is, according to Figure 4.7,

$$p = z_{01} \cos \psi + z_{02} \sin \psi \quad (4.315)$$

Next, (4.314) is squared and then multiplied with $a^2 b^2$, and the result is added to the square of (4.315), giving

$$p^2 = a^2 \cos^2 \psi + b^2 \sin^2 \psi \quad (4.316)$$

By comparing this expression with (4.311), it follows that $\hat{\sigma}_r = p$; i.e., the standard deviation in a certain direction is equal to the projection of the standard ellipse onto that direction. Therefore, the standard ellipse is not a standard deviation curve. Figure 4.8 shows the continuous standard deviation curve. We see that for narrow ellipses there are only small segments of the standard deviations that are close to the length of the semiminor axis. The standard deviation increases rapidly as the direction ψ moves away from the minor axis. Therefore, an extremely narrow ellipse is not desirable if the overall accuracy for the station position is important.

As a by-product of the property discussed, we see that the standard deviations of the parameter x_1 and x_2

$$\hat{\sigma}_{x_1} = \hat{\sigma}_0 \sqrt{q_{x_1}} \quad (4.317)$$

$$\hat{\sigma}_{x_2} = \hat{\sigma}_0 \sqrt{q_{x_2}} \quad (4.318)$$

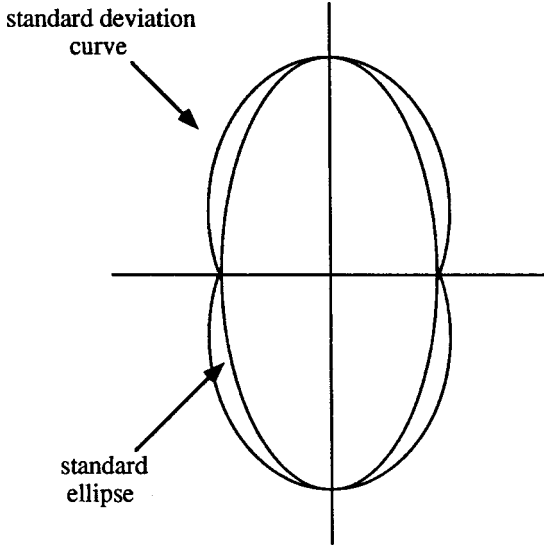


Figure 4.8 Standard deviation curve.

are the projections of the ellipse in the directions of the x_1 and x_2 axes. This is shown in Figure 4.9. Equations (4.317) and (4.318) follow from the fact that the diagonal elements of the covariance matrix are the variances of the respective parameters. Equation (4.316) confirms for $\psi = 0$ and $\psi = 90^\circ$ that the axes a and b equal the maximum and minimum standard deviations, respectively. The rectangle formed by the semisides $\hat{\sigma}_{x_1}$ and $\hat{\sigma}_{x_2}$ encloses the ellipse. This rectangle can be used as an approximation for the ellipses. The diagonal itself is sometimes referred to as the mean position error $\hat{\sigma}$,

$$\hat{\sigma} = \sqrt{\hat{\sigma}_{x_1}^2 + \hat{\sigma}_{x_2}^2} = \hat{\sigma}_0 \sqrt{q_{x_1} + q_{x_2}} \quad (4.319)$$

The points of contact between the ellipse and the rectangle in Figure 4.9 are functions of the correlation coefficients. For these points, the tangent on the ellipse is either horizontal or vertical in the (y_i) coordinate system. The equation of the ellipse in the (y) system is, according to (4.294),

$$\begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} q_{x_1} & q_{x_1, x_2} \\ q_{x_1, x_2} & q_{x_2} \end{bmatrix}^{-1} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \hat{\sigma}_0^2 \quad (4.320)$$

By replacing the matrix by its inverse, the expression becomes

$$\begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} q_{x_2} & -q_{x_1, x_2} \\ -q_{x_1, x_2} & q_{x_1} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = (q_{x_1} q_{x_2} - q_{x_1, x_2}^2) \hat{\sigma}_0^2 \quad (4.321)$$

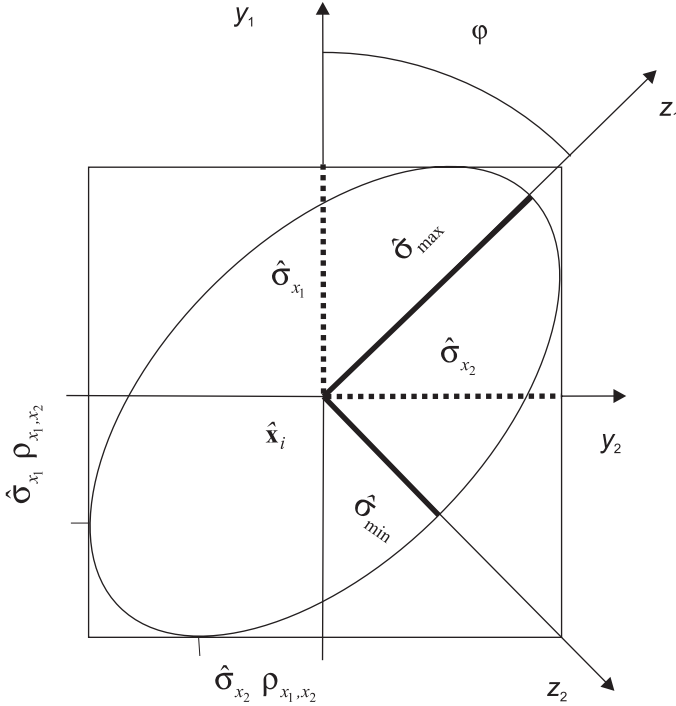


Figure 4.9 Characteristics of the standard deviation ellipse.

Evaluating the left-hand side and dividing both sides by $q_{x_1}q_{x_2}$ gives

$$\frac{y_1^2}{q_{x_1}} + \frac{y_2^2}{q_{x_2}} - \frac{2y_1y_2q_{x_1,x_2}}{q_{x_1}q_{x_2}} = \text{constant} \quad (4.322)$$

from which it follows that

$$\frac{dy_1}{dy_2} = \frac{(2y_2/q_{x_2}) - (2y_1\rho_{x_1,x_2}/\sqrt{q_{x_1}q_{x_2}})}{(2y_2\rho_{x_1,x_2}/\sqrt{q_{x_1}q_{x_2}}) - (2y_1/q_{x_1})} \quad (4.323)$$

Consider the tangent for which the slope is infinity. The equation of this tangent line is

$$y_2 = \hat{\sigma}_0 \sqrt{q_{x_2}} \quad (4.324)$$

Substituting this expression into the denominator of (4.323) and equating it to zero gives

$$\frac{\hat{\sigma}_0 \sqrt{q_{x_2}} \rho_{x_1,x_2}}{\sqrt{q_{x_1}q_{x_2}}} = \frac{y_1}{q_{x_1}} \quad (4.325)$$

which yields the y_1 coordinate for the point of tangency:

$$y_1 = \hat{\sigma}_0 \sqrt{q_{x_1}} \rho_{x_1, x_2} = \hat{\sigma}_{x_1} \rho_{x_1, x_2} \quad (4.326)$$

The equation for the horizontal tangent is

$$y_1 = \hat{\sigma}_0 \sqrt{q_{x_1}} \quad (4.327)$$

It follows from the numerator of (4.323) that

$$y_2 = \hat{\sigma}_0 \sqrt{q_{x_2}} \rho_{x_1, x_2} = \hat{\sigma}_{x_2} \rho_{x_1, x_2} \quad (4.328)$$

Figure 4.9 shows that the standard ellipse becomes narrower the higher the correlation. For correlation plus or minus 1 (linear dependence), the ellipse degenerates into the diagonal of the rectangle. The ellipse becomes a circle if $a = b$, or $\sigma_{x_1} = \sigma_{x_2}$ and $\rho_{x_1, x_2} = 0$.

4.9.7 Other Measures of Precision

In surveying and geodesy, the most popular measure of precision is the standard deviation. The confidence regions are usually expressed in terms of ellipses and ellipsoids of standard deviation. These figures are often scaled to contain 95% probability or higher. Because GPS is a popular tool for both surveying and navigation, several of the measures of precision used in navigation are becoming increasingly popular in surveying. Examples include the dilution of precision (DOP) numbers. The DOPs are discussed in detail in Section 7.4.1. Other single-number measures refer to circular or spherical confidence regions for which the eigenvalues of the cofactor matrix have the same magnitude. In these cases, the standard deviations of the coordinates and the semiaxes are of the same size. See Equation (4.297). When the standard deviations are not equal, these measures become a function of the ratio of the semiaxes. The derivation of the following measures and additional interpretation are given in Greenwalt and Shultz (1962).

The radius of a circle that contains 50% probability is called the circular error probable (CEP). This function is usually approximated by segments of straight lines. The expression

$$\text{CEP} = 0.5887 (\hat{\sigma}_{x_1} + \hat{\sigma}_{x_2}) \quad (4.329)$$

is, strictly speaking, valid in the region $\sigma_{\min}/\sigma_{\max} \geq 0.2$, but it is the function used most often. The 90% probability region

$$\text{CMAS} = 1.8227 \times \text{CEP} \quad (4.330)$$

is called the circular map accuracy standard. The mean position error (4.319) is also called the mean square positional error (MSPE), or the distance root mean square (DRMS), i.e.,

$$\text{DRMS} = \sqrt{\hat{\sigma}_{x_1}^2 + \hat{\sigma}_{x_2}^2} \quad (4.331)$$

This measure contains 64% to 77% probability. The related measure

$$2 \text{ DRMS} = 2 \times \text{DRMS} \quad (4.332)$$

contains about 95% to 98% probability.

The three-dimensional equivalent of CEP is the spherical error probable (SEP), defined as

$$\text{SEP} = 0.5127 (\hat{\sigma}_{x_1} + \hat{\sigma}_{x_2} + \hat{\sigma}_{x_3}) \quad (4.333)$$

Expression (4.333) is, strictly speaking, valid in the region $\sigma_{\min}/\sigma_{\max} \geq 0.35$. The corresponding 90% probability region,

$$\text{SAS} = 1.626 \times \text{SEP} \quad (4.334)$$

is called the spherical accuracy standard (SAS). The mean radial spherical error (MRSE) is defined as

$$\text{MRSE} = \sqrt{\hat{\sigma}_{x_1}^2 + \hat{\sigma}_{x_2}^2 + \hat{\sigma}_{x_3}^2} \quad (4.335)$$

and contains about 61% probability.

These measures of precision are sometimes used to capture the achieved or anticipated precision conveniently using single numbers. However, the geometry of the adjustment seldom produces covariance matrices that yield circular distribution. Consequently, the probability levels contained in these measures of precision inevitably are a function of the correlations between the parameters.

4.10 RELIABILITY

Small residuals are not necessarily an indication of a quality adjustment. Equally important is the knowledge that all blunders in the data have been identified and removed and that remaining small blunders in the observations do not adversely impact the adjusted parameters. Reliability refers to the controllability of observations, i.e., the ability to detect blunders and to estimate the effects that undetected blunders may have on a solution. The theory outlined here follows that of Baarda (1967, 1968), and Kok (1984).

4.10.1 Redundancy Numbers

Following the expressions in Table 4.1 the residuals for the observation equation model are

$$\hat{\mathbf{v}} = \mathbf{Q}_v \mathbf{P} \ell \quad (4.336)$$

with a cofactor matrix for the residuals

$$\mathbf{Q}_v = \mathbf{P}^{-1} - \mathbf{A}\mathbf{N}^{-1}\mathbf{A}^T \quad (4.337)$$

Compute the trace

$$\begin{aligned} \text{Tr}(\mathbf{Q}_v \mathbf{P}) &= \text{Tr}(\mathbf{I} - \mathbf{A}\mathbf{N}^{-1}\mathbf{A}^T \mathbf{P}) \\ &= n - \text{Tr}(\mathbf{N}^{-1}\mathbf{A}^T \mathbf{P} \mathbf{A}) \\ &= n - u \end{aligned} \quad (4.338)$$

A more general expression is obtained by noting that the matrix $\mathbf{A}\mathbf{N}^{-1}\mathbf{A}^T \mathbf{P}$ is idempotent. The trace of an idempotent matrix equals the rank of that matrix. Thus,

$$\text{Tr}(\mathbf{A}\mathbf{N}^{-1}\mathbf{A}^T \mathbf{P}) = R(\mathbf{A}^T \mathbf{P} \mathbf{A}) = R(\mathbf{A}) = r \leq u \quad (4.339)$$

Thus, from Equations (4.338) and (4.339)

$$\text{Tr}(\mathbf{Q}_v \mathbf{P}) = \text{Tr}(\mathbf{P} \mathbf{Q}_v) = n - R(\mathbf{A}) \quad (4.340)$$

By denoting the diagonal element of the matrix $\mathbf{Q}_v \mathbf{P}$ by r_i , we can write

$$\sum_{i=1}^n r_i = n - R(\mathbf{A}) \quad (4.341)$$

The sum of the diagonal elements of $\mathbf{Q}_v \mathbf{P}$ equals the degree of freedom. The element r_i is called the redundancy number for the observation i . It is the contribution of the i th observation to the degree of freedom. If the weight matrix $\mathbf{P}$ is diagonal, which is usually the case when original observations are adjusted, then

$$r_i = q_i p_i \quad (4.342)$$

where q_i is the diagonal element of the cofactor matrix $\mathbf{Q}_v$, and p_i denotes the weight of the i th observation. Equation (4.337) implies the inequality

$$0 \leq q_i \leq \frac{1}{p_i} \quad (4.343)$$

Multiplying by p_i gives the bounds for the redundancy numbers,

$$0 \leq r_i \leq 1 \quad (4.344)$$

Considering the general relation

$$\mathbf{Q}_{\ell_a} = \mathbf{Q}_{\ell_b} - \mathbf{Q}_v \quad (4.345)$$

given in Table 4.1 and the specification (4.342) for the redundancy number r_i as the diagonal element of $\mathbf{Q}_v\mathbf{P}$, it follows that if the redundancy number is close to 1, then the variance of the residuals is close to the variance of the observations, and the variance of the adjusted observations is close to zero. If the redundancy number is close to zero, then the variance of the residuals is close to zero, and the variance of the adjusted observations is close to the variance of the observations.

Intuitively, it is expected that the variance of the residuals and the variance of the observations are close; for this case, the noise in the residuals equals that of the observations, and the adjusted observations are determined with high precision. Thus the case of r_i close to 1 is preferred, and it is said that the gain of the adjustment is high. If r_i is close to zero, one expects the noise in the residuals to be small. Thus, small residuals as compared to the expected noise of the observations are not necessarily desirable. Because the inequality (4.344) is a result of the geometry as represented by the design matrix $\mathbf{A}$, small residuals can be an indication of a weak part of the network.

Because the weight matrix P is considered diagonal, i.e.,

$$p_i = \frac{\sigma_0^2}{\sigma_i^2} \quad (4.346)$$

it follows that

$$\hat{\sigma}_{v_i} = \hat{\sigma}_0 \sqrt{q_i} = \hat{\sigma}_0 \sqrt{\frac{r_i}{p_i}} = \hat{\sigma}_0 \sqrt{\frac{r_i \sigma_i^2}{\sigma_0^2}} = \frac{\hat{\sigma}_0}{\sigma_0} \sigma_i \sqrt{r_i} \quad (4.347)$$

From (4.341) it follows that the average redundancy number is

$$r_{av} = \frac{n - R(\mathbf{A})}{n} \quad (4.348)$$

The higher the degree of freedom, the closer the average redundancy number is to 1. However, as seen from Table 4.8, the gain, in terms of probability enclosed by the standard ellipses, reduces noticeably after a certain degree of freedom.

4.10.2 Controlling Type-II Error for a Single Blunder

Baarda's (1967) development of the concept of reliability of networks is based on un-Studentized hypothesis tests, which means that the a priori variance of unit weight is assumed to be known. Consequently, the a priori variance of unit weight (not the a posteriori variance of unit weight) is used in this section. The alternative hypothesis H_a specifies that the observations contain one blunder, that the blunder be located at observation i , and that its magnitude be ∇_i . Thus the adjusted residuals for the case of the alternative hypothesis are

$$\hat{\mathbf{v}}|H_a = \hat{\mathbf{v}} - \mathbf{Q}_v\mathbf{P}\mathbf{e}_i\nabla_i \quad (4.349)$$

where

$$\mathbf{e}_i = [0 \quad \cdots \quad 0 \quad 1 \quad 0 \quad \cdots \quad 0]^T \quad (4.350)$$

denotes an $n \times 1$ vector containing 1 in position i and zeros elsewhere. The expected value and the covariance matrix are

$$E(\hat{\mathbf{v}}|H_a) = -\mathbf{Q}_v \mathbf{P} \mathbf{e}_i \nabla_i \quad (4.351)$$

$$\Sigma_{v|H_a} = \hat{\Sigma}_v = \sigma_0^2 \mathbf{Q}_v \quad (4.352)$$

It follows from Theorem 1 of Section 4.9.1 that

$$\hat{\mathbf{v}}|H_a \sim N(-\mathbf{Q}_v \mathbf{P} \mathbf{e}_i \nabla_i, \sigma_0^2 \mathbf{Q}_v) \quad (4.353)$$

Since $\mathbf{P}$ is a diagonal matrix, the individual residuals are distributed as

$$\hat{v}_i|H_a \sim n(-q_i p_i \nabla_i, \sigma_0^2 q_i) \quad (4.354)$$

according to Theorem 2. Standardizing gives

$$\begin{aligned} w_a|H_a &= \frac{\hat{v}_i|H_a}{\sigma_0 \sqrt{q_i}} \sim n\left(\frac{-q_i p_i \nabla_i}{\sigma_0 \sqrt{q_i}}, 1\right) \\ &= n\left(\frac{-\sqrt{q_i} p_i \nabla_i}{\sigma_0}, 1\right) \end{aligned} \quad (4.355)$$

or

$$H_a: w_a = \frac{\hat{v}_i|H_a}{\sigma_{v_i}} \sim n\left(\frac{-\nabla_i p_i \sqrt{q_i}}{\sigma_0}, 1\right) \quad (4.356)$$

The zero hypothesis, which states that there is no blunder, is

$$H_0: w_0 = \frac{\hat{v}_i|H_0}{\sigma_{v_i}} \sim n(0, 1) \quad (4.357)$$

The noncentrality parameter in (4.356), i.e., the mean of the noncentral normal distribution, is denoted by δ_i and is

$$\delta_i = \frac{-\nabla_i p_i \sqrt{q_i}}{\sigma_0} = \frac{-\nabla_i \sqrt{r_i}}{\sigma_i} \quad (4.358)$$

The parameter δ_i is a translation parameter of the normal distribution. The situation is shown in Figure 4.10. The probability of committing an error of the first kind, i.e., of accepting the alternative hypothesis, equals the significance level α of the test

$$P(|w_0| \leq t_{\alpha/2}) = \int_{-t_{\alpha/2}}^{t_{\alpha/2}} n(0, 1) dx = 1 - \alpha \quad (4.359)$$

or

$$P(|w_0| \geq t_{\alpha/2}) = \int_{-\infty}^{t_{1-\alpha/2}} n(0, 1) dx + \int_{t_{\alpha/2}}^{\infty} n(0, 1) dx = \alpha \quad (4.360)$$

In $100\alpha\%$ of the cases, the observations are rejected and remeasurement or investigations for error sources are performed, even though the observations are correct (they do not contain a blunder). From Figure 4.10 it is seen that the probability β_i of a type-II error, i.e., the probability of rejecting the alternative hypothesis (and accepting the zero hypothesis) even though the alternative hypothesis is correct, depends on the noncentrality factor δ_i . *Because the blunder ∇_i is not known, the noncentrality factor is not known either.* As a practical matter one can proceed in the reverse: one can assume an acceptable probability β_0 for the error of the second kind and compute the respective noncentrality parameter δ_0 . This parameter in turn is used to compute the lower limit for the blunder, which can still be detected. Figure 4.10 shows that

$$P(|w_a| \leq t_{\alpha/2}) = \int_{-t_{\alpha/2}}^{t_{\alpha/2}} n(\delta_i, 1) dx \geq \beta_0 \quad (4.361)$$

if

$$\delta_i \leq \delta_0 \quad (4.362)$$

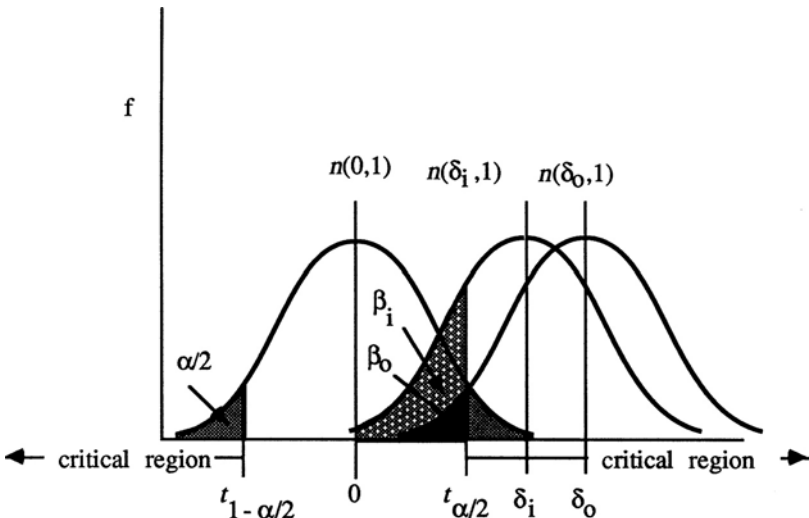


Figure 4.10 Defining the noncentrality.

Substituting Equation (4.358) into (4.362) gives the limit for the marginally detectable blunder, given the probability levels α and β_0 :

$$|\nabla_{0i}| \geq \frac{\delta_0}{\sqrt{r_i}} \sigma_i \tag{4.363}$$

Equations (4.361) and (4.363) state that in $100(1 - \beta_0)\%$ of the cases, blunders greater than those given in (4.363) are detected. In $100\beta_0\%$ of the cases, blunders greater than those given in (4.363) remain undetected. The larger the redundancy number, the smaller is the marginally detectable blunder (for the same δ_0 and σ_i). It is important to recognize that the marginally detectable blunders (4.363) are based on adopted probabilities of type-I and type-II errors for the normal distribution. The probability levels α and β_0 refer to the one-dimensional test (4.357) of the individual residual v_i , with the noncentrality being δ_0 . The assumption is that only one blunder at a time is present. The geometry is shown in Figure 4.10. It is readily clear that there is a simple functional relationship $\delta_0 = \delta_n(\alpha, \beta_0)$ between two normal distributions. Table 4.9 contains selected probability levels and the respective δ_0 values.

The chi-square test (4.263) of the a posteriori variance of unit weight $\hat{\sigma}_0^2$ is also sensitive to the blunder ∇_i . In fact, the blunder will cause a noncentrality of δ_i for the chi-square distribution of the alternative hypothesis. One can choose the probabilities α_{chi} and β_{chi} for this multidimensional chi-square test such that $\delta_0 = \delta_{\text{chi}}(\alpha_{\text{chi}}, \beta_{\text{chi}}, n - u)$. The factor δ_0 depends on the degree of freedom because the chi-square distribution depends on it. Baarda's B method suggests equal traceability of errors through one-dimensional tests of individual residuals, v_i , and the multidimensional test of the a posteriori variance of unit weight $\hat{\sigma}_0^2$. This is achieved by requiring that the one-dimensional test and the multidimensional test have the same type-II error, i.e., $\beta_0 = \beta_{\text{chi}}$. Under this condition there exists a relationship between the probability of type-II error, the significance levels, and the degree of freedom expressed symbolically by $\delta_0 = \delta_n(\alpha, \beta_0) = \delta_{\text{chi}}(\alpha_{\text{chi}}, \beta_0, n - r)$. The B method assures equal traceability but implies different significance levels for the

**TABLE 4.9 Selected Probability Levels
in Reliability**

α	β_0	δ_0
0.05	0.20	2.80
0.025	0.20	3.1
0.001	0.20	4.12
0.05	0.10	3.24
0.025	0.10	3.52
0.001	0.10	4.57

one-dimensional and multidimensional tests. For details see Baarda (1968, p. 25). In practical applications one chooses the factor δ_0 on the basis of a reasonable value for α and β_0 from Table 4.9.

4.10.3 Internal Reliability

Even though the one-dimensional test is based on the assumption that only one blunder exists in a set of observations, the limit (4.363) is usually computed for all observations. The marginally detectable errors, computed for all observations, are viewed as a measure of the capability of the network to detect blunders with probability $(1 - \beta_0)$. They constitute the internal reliability of the network. Because the marginally detectable errors (4.363) do not depend on the observations or on the residuals, they can be computed as soon as the configuration of the network and the stochastic model are known. If the limits (4.363) are of about the same size, the observations are equally well checked, and the internal reliability is said to be consistent. The emphasis is then on the variability of the marginally detectable blunders rather on their magnitude. A typical value is $\delta_0 = 4$.

4.10.4 Absorption

According to (4.336) the residuals in the presence of one blunder are:

$$\mathbf{v} = \mathbf{Q}_v \mathbf{P} (\ell - \mathbf{e}_i \nabla_i) \quad (4.364)$$

The impact on the residual of observation i is

$$\nabla v_i = -r_i \nabla_i \quad (4.365)$$

Equation (4.365) is used to estimate the blunders that might cause large residuals. Solving for ∇_i gives

$$\nabla_i = -\frac{\nabla v_i}{r_i} \approx -\frac{v_i^* + \nabla v_i}{r_i} \approx -\frac{v_i}{r_i} \quad (4.366)$$

because $v_i^* \ll \nabla v_i$, where v_i^* denotes the residual without the effect of the blunder. The computation (4.366) provides only estimates of possible blunders. Because the matrix $\mathbf{Q}_v \mathbf{P}$ is not a diagonal matrix, a specific blunder has an impact on all residuals. If several blunders are present, their effects overlap and one blunder can mask others; a blunder may cause rejection of a good observation.

Equation (4.365) demonstrates that the residuals in least-squares adjustments are not robust with respect to blunders in the sense that the effect of a blunder on the residuals is smaller than the blunder itself, because r varies between 0 and 1. The absorption, i.e., the portion of the blunder that propagates into the estimated parameters and falsifies the solution, is

$$A_i = (1 - r_i) \nabla_i \quad (4.367)$$

The factor $(1 - r_i)$ is called the absorption number. The larger the redundancy number, the less is a blunder absorbed, i.e., the less falsification. If $r_i = 1$, the observation is called fully controlled, because the residual completely reflects the blunder. A zero redundancy implies uncontrolled observations in that a blunder enters into the solution with its full size. Observations with small redundancy numbers might have small residuals and instill false security in the analyst. Substituting ∇_i from (4.366) expresses the absorption as a function of the residuals:

$$A_i = -\frac{1 - r_i}{r_i} v_i \quad (4.368)$$

The residuals can be looked on as the visible parts of errors. The factor in (4.368) is required to compute the invisible part from the residuals.

4.10.5 External Reliability

A good and homogeneous internal reliability does not automatically guarantee reliable coordinates. What are the effects of undetectable blunders on the parameters? In deformation analysis, where changes in parameters between adjustments of different epochs indicate existing deformations, it is particularly important that the impact of blunders on the parameters be minimal. The influence of each of the marginally detectable errors on the parameters of the adjustment or on functions of the parameters is called external reliability. The estimated parameters in the presence of a blunder are, for the observation equation model,

$$\hat{\mathbf{x}} = -\mathbf{N}^{-1} \mathbf{A}^T \mathbf{P} (\ell - \mathbf{e}_i \nabla_i) \quad (4.369)$$

The effect of the blunder in observation i is

$$\nabla \mathbf{x} = \mathbf{N}^{-1} \mathbf{A}^T \mathbf{P} \mathbf{e}_i \nabla_i \quad (4.370)$$

The shifts $\nabla \mathbf{x}$ are sometimes called local external reliability. The blunder affects all parameters. The impact of the marginally detectable blunder ∇_{0i} is

$$\nabla \mathbf{x}_{0i} = \mathbf{N}^{-1} \mathbf{A}^T \mathbf{P} \mathbf{e}_i \nabla_{0i} \quad (4.371)$$

Because there are n observations, one can compute n vectors (4.371), showing the impact of each marginal detectable blunder on the parameters. Graphical representations of these effects can be very helpful in the analysis. The problem with (4.371) is that the effect on the coordinates depends on the definition (minimal constraints) of the coordinate system. Baarda (1968) suggested the following alternative expression:

$$\lambda_{0i}^2 = \frac{\nabla \mathbf{x}_{0i}^T \mathbf{N} \nabla \mathbf{x}_{0i}}{\sigma_0^2} \quad (4.372)$$

By substituting (4.371) and (4.363), we can write this equation as

$$\lambda_{0i}^2 = \frac{\nabla_{0i} \mathbf{e}_i^T \mathbf{P} \mathbf{A} \mathbf{N}^{-1} \mathbf{A}^T \mathbf{P} \mathbf{e}_i \nabla_{0i}}{\sigma_0^2} = \frac{\nabla_{0i}^2 \mathbf{e}_i^T \mathbf{P} (\mathbf{I} - \mathbf{Q}_v \mathbf{P}) \mathbf{e}_i}{\sigma_0^2} = \frac{\nabla_{0i}^2 p_i (1 - r_i)}{\sigma_0^2} \quad (4.373)$$

or

$$\lambda_{0i}^2 = \frac{1 - r_i}{r_i} \delta_0^2 \quad (4.374)$$

The values λ_{0i} are a measure of global external reliability. There is one such value for each observation. If the λ_{0i} are the same order of magnitude, the network is homogeneous with respect to external reliability. If r_i is small, the external reliability factor becomes large and the global falsification caused by a blunder can be significant. It follows that very small redundancy numbers are not desirable. The global external reliability number (4.374) and the absorption number (4.368) have the same dependency on the redundancy numbers.

4.10.6 Correlated Cases

The derivations for detectable blunders, internal reliability, absorption, and external reliability assumes uncorrelated observations for which the covariance matrix Σ_{ℓ_b} is diagonal. Correlated observations are decorrelated by the transformation (4.235). It can be readily verified that the redundancy numbers for the decorrelated observations $\bar{\ell}$ are

$$\bar{r}_i = (\bar{\mathbf{Q}}_v \bar{\mathbf{P}})_{ii} = (\mathbf{I} - \mathbf{D}^T \mathbf{A} \mathbf{N}^{-1} \mathbf{A}^T \mathbf{D})_{ii} \quad (4.375)$$

In many applications, the covariance matrix Σ_{ℓ_b} is of block-diagonal form. For example, for GPS vector observations, this matrix consists of 3×3 full block-diagonal matrices if the correlations between the vectors are neglected. In this case, the matrix $\mathbf{D}$ is also block-diagonal and the redundancy numbers can be computed vector by vector from (4.375). The sum of the redundancy numbers for the three vector components varies between 0 and 3. Since, in general, the matrix $\mathbf{D}$ has a full rank, the degree of freedom ($n - r$) of the adjustment does not change. Once the redundancy numbers $\bar{r}_i$ are available, the marginal detectable blunders $\bar{\nabla}_{0i}$, the absorption numbers $\bar{A}_i$ and other reliability values can be computed for the decorrelated observations. These quantities, in turn, can be transformed back into the physical observation space by premultiplication with the matrix $(\mathbf{D}^T)^{-1}$.

4.11 BLUNDER DETECTION

Errors (blunders) made during the recording of field observations, data transfer, the computation, etc., can be costly and time-consuming to find and eliminate. Blunder

detection can be carried out before the adjustment or as part of the adjustment. Before the adjustment, the discrepancies (angle and/or distance of simple figures such as triangles and traverses) are analyzed. A priori blunder detection is helpful in detecting extralarge blunders caused by, e.g., erroneous station numbering. Blunder detection in conjunction with the adjustment is based on the analysis of the residuals. The problem with using least-squares adjustments when blunders are present is that the adjustments tend to hide (reduce) their impact and distribute their effects more or less throughout the entire network (see (4.364) and (4.365), noting that the redundancy number varies between zero and 1). The prerequisite for any blunder-detection procedure is the availability of a set of redundant observations. Only observations with redundancy numbers greater than zero can be controlled.

It is important to understand that if a residual does not pass a statistical test, this does not mean that there is a blunder in that observation. The observation is merely flagged so that it can be examined and a decision about its retention or rejection can be made. Blind rejection is never recommended. A blunder in one observation usually affects the residuals in other observations. Therefore, the tests will often flag other observations in addition to the ones containing blunders. If one or more observations are flagged, the search begins to determine if there is a blunder.

The first step is to check the field notes to confirm that no error occurred during the transfer of the observations to the computer file, and that all observations are reasonable “at face value.” If a blunder is not located, the network should be broken down into smaller networks, and each one should be adjusted separately. At the extreme, the entire network may be broken down into triangles or other simple geometric entities, such as traverses, and adjusted separately. Alternatively, the observations can be added sequentially, one at a time, until the blunder is found. This procedure starts with weights assigned to all parameters. The observations are then added sequentially. The sum of the normalized residuals squared is then inspected for unusually large variations. When searching for blunders, the coordinate system should be defined by minimal constraints.

Blunder detection in conjunction with the adjustment takes advantage of the total redundancy and the strength provided by the overall geometry of the network, and thus is more sensitive to smaller blunders. Only if the existence of a blunder is indicated does action need to be taken to locate the blunder. The flagged observations are the best hint where to look for errors and thus avoid unnecessary and disorganized searching of the whole observation data set.

4.11.1 The τ Test

The τ test was introduced by Pope (1976). The test belongs to the group of Studentized tests, which make use of the a posteriori variance of unit weight as estimated from the observations. The test statistic is

$$\tau_i = \frac{v_i}{\hat{\sigma}_{v_i}} = \frac{\sigma_0 v_i}{\hat{\sigma}_0 \sigma_i \sqrt{r_i}} \sim \tau_{n-r} \quad (4.376)$$

The symbol τ_{n-r} denotes the τ distribution with $n-r$ degrees of freedom. It is related to Student's t by

$$\tau_{n-r} = \frac{\sqrt{n-r} t_{n-r-1}}{\sqrt{n-r-1 + t_{n-r-1}^2}} \quad (4.377)$$

For an infinite degree of freedom the τ distribution converges toward the Student distribution or the standardized normal distribution, i.e., $\tau_{\infty} = t_{\infty} = n(0, 1)$.

Pope's blunder rejection procedure tests the hypothesis $v_i \sim n(0, \hat{\sigma}_{v_i}/\hat{\sigma}_0)$. The hypothesis is rejected, i.e., the observation is flagged for further investigation and possibly rejection, if

$$|\tau_i| \geq c \quad (4.378)$$

The critical value c is based on a preselected significance level. For large systems, the redundancy numbers are often replaced by the average value according to Equation (4.348), in order to reduce computation time; thus

$$\tau_i = \frac{\sigma_0}{\hat{\sigma}_0} \frac{v_i}{\sigma_i \sqrt{(n-r)/n}} \quad (4.379)$$

could be used instead of (4.376).

4.11.2 Data Snooping

Baarda's data snooping applies to the testing of individual residuals as well. The theory assumes that only one blunder be present in the set of observations. Applying a series of one-dimensional tests, i.e., testing consecutively all residuals, is called a data snooping strategy. Baarda's test belongs to the group of un-Studentized tests which assume that the a priori variance of unit weight is known. The zero hypothesis (4.357) is written as

$$n_i = \frac{v_i}{\sigma_0 \sqrt{q_i}} \sim n(0, 1) \quad (4.380)$$

At a significant level of 5%, the critical value is 1.96. The critical value for this test is not a function of the number of observations in the adjustment. The statistic (4.380) uses the a priori value σ_0 and not the a posteriori estimate $\hat{\sigma}_0$.

Both the τ and the data snooping procedures work best for iterative solutions. At each iteration step, the observation with the largest blunder should be removed. Since least-squares attempts to distribute blunders, several correct observations might receive large residuals and might be flagged mistakenly.

4.11.3 Changing Weights of Observations

This method, although not based on rigorous statistical theory, is an automated method whereby blunders are detected and their effects on the adjustment minimized (or even eliminated). The advantage that this method has, compared to previous methods, is that it locates and potentially eliminates the blunders automatically. The method examines the residuals per iteration. If the magnitude of a residual is outside a defined range, the weight of the corresponding observation is reduced. The process of reweighting and readjusting continues until the solution converges, i.e., no weights are being changed. The criteria for judging the residuals and choice for the reweighting function are somewhat arbitrary. For example, a simple strategy for selection of the new weights at iteration $k + 1$ could be

$$p_{k+1,i} = p_{k,i} \begin{cases} e^{-|v_{k,i}|/3\sigma_i} & \text{if } |v_{k,i}| > 3\sigma_i \\ 1 & \text{if } |v_{k,i}| \leq 3\sigma_i \end{cases} \quad (4.381)$$

where σ_i denotes the standard deviation of observation i .

The method works efficiently for networks with high redundancy. If the initial approximate parameters are inaccurate, it is possible that correct observations are deweighted after the first iteration because the nonlinearity of the adjustment can cause large residuals. To avoid unnecessary rejection and reweighting, one might not change the weights during the first iteration. Proper use of this method requires some experience. All observations whose weights are changed must be investigated, and the cause for the deweighting must be investigated.

4.12 EXAMPLES

In the following, we use plane two-dimensional networks to demonstrate the geometry of adjustments. As mentioned above, the geometry of a least-squares adjustment is the result of the combined effects of the stochastic model (weight matrix $\mathbf{P}$ —representing the quality of the observations) and the mathematical model (design matrix $\mathbf{A}$ —representing the geometry of the network and the spatial distribution of the observations). For the purpose of these examples, it is not necessary to be concerned about the physical realization of two-dimensional networks. The experienced reader might think of such networks as being located on the conformal mapping plane and all that it takes to compute the respective model observations. However, it is entirely sufficient here to stay simply within the area of plane geometry.

We will use the observation equation model summarized in Table 4.1. Assume there is a set of n observations, such as distances and angles that determine the points of a network. For a two-dimensional network of s stations, there could be as many as $u = 2s$ unknown coordinates. Let the parameter vector $\mathbf{x}_a$ consist of coordinates only, i.e., we do not parameterize refraction, centering errors, etc. To be specific, $\mathbf{x}_a$ contains only coordinates that are to be estimated. Coordinates of known stations are

constants and not included in $\mathbf{x}_a$. The mathematical model $\ell_a = \mathbf{f}(\mathbf{x}_a)$ is very simple in this case. The n components $\mathbf{f}$ will contain the functions:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (4.382)$$

$$a_{jik} = \tan^{-1} \frac{x_k - x_i}{y_k - y_i} - \tan^{-1} \frac{x_j - x_i}{y_j - y_i} \quad (4.383)$$

In these expressions the subscripts i , j , and k identify the network points. The notation a_{jik} implies that the angle is measured at station i , from j to k in a clockwise sense. The ordering of the components in $\mathbf{f}$ does not matter, as long as the same order is maintained with respect to the rows of $\mathbf{A}$ and diagonal elements of $\mathbf{P}$.

Although the $\mathbf{f}(\mathbf{x}_a)$ have been expressed in terms of $\mathbf{x}_a$, the components typically depend only on a subset of the coordinates. The relevant partial derivatives in a row of $\mathbf{A}$ are for distances and angles:

$$\left\{ \frac{-(y_k - y_i)}{d_{ik}}, \frac{-(x_k - x_i)}{d_{ik}}, \frac{y_k - y_i}{d_{ik}}, \frac{x_k - x_i}{d_{ik}} \right\} \quad (4.384)$$

$$\left\{ \frac{x_i - x_j}{d_{ij}^2}, -\frac{y_i - y_j}{d_{ij}^2}, \frac{x_k - x_j}{d_{kj}^2} - \frac{x_i - x_j}{d_{ij}^2}, -\frac{y_k - y_j}{d_{kj}^2} + \frac{y_i - y_j}{d_{ij}^2}, -\frac{x_k - x_j}{d_{kj}^2}, \frac{y_k - y_j}{d_{kj}^2} \right\} \quad (4.385)$$

Other elements are zero. The column location for these partials depends on the sequence in $\mathbf{x}_a$. In general, if α is the α -th component of ℓ_b and β the β -th component of $\mathbf{x}_a$, then the element $a_{\alpha,\beta}$ of $\mathbf{A}$ is

$$a_{\alpha,\beta} = \frac{\partial \ell_\alpha}{\partial x_\beta} \quad (4.386)$$

The partial derivatives and the discrepancy ℓ_0 must be evaluated for the approximate coordinates $\mathbf{x}_0$.

Example 1: This example demonstrates the impact of changes in the stochastic model. Figure 4.11 shows a traverse connecting two known stations. Three solutions

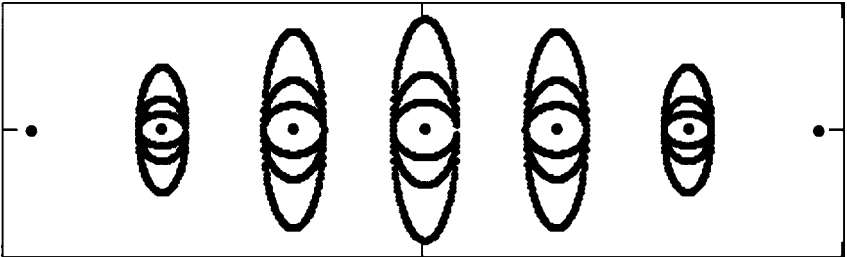


Figure 4.11 Impact of changing the stochastic model.

are given. In all cases, the distances are of the same length and observed with the same accuracy. The angle observations are 180° and measured with the same accuracy but are changed by a common factor for each solution. If we declare the solutions with the smallest ellipses in Figure 4.11 as the base solutions with observational standard deviation of σ_a then the other solutions use $2\sigma_a$ and $4\sigma_a$, respectively. The shape of the ellipses elongates as the standard deviation of the angles increases.

Example 2: This example demonstrates the impact of changing network geometry using a resection. Four known stations lie exactly on an imaginary circle with radius r . The coordinates of the new station are determined by angle measurements, i.e., no distances are involved. For the first solution, the unknown station is located at the center of the circle. In subsequent solutions its location moves to $0.5r$, $0.9r$, $1.1r$, and $1.5r$ from the center while retaining the same standard deviation for the angle observations in each case. Figure 4.12 shows that the ellipses become more elongated the closer the unknown station moves to the circle. The solution is singular if the new station is located exactly on the circle.

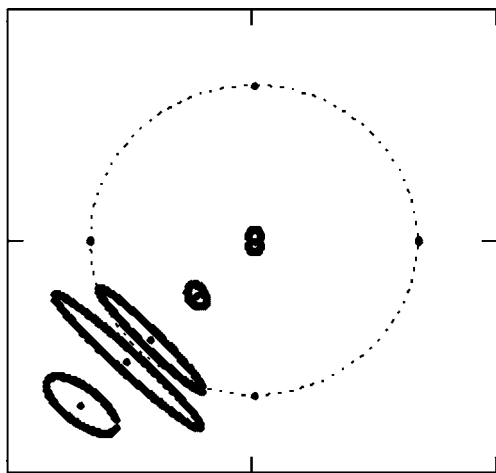


Figure 4.12 Impact of changing network geometry.

Example 3: Three cases are given that demonstrate how different definitions of the coordinate system affect the ellipses of standard deviation. All cases refer to the same plane network using the same observed angles and distances and the same respective standard deviations of the observations. A plane network that contains angle and distance observations requires three minimal constraints. Simply holding three coordinates fixed imposes such minimal constraints. The particular coordinates are constants and are not included in the parameter vector $\mathbf{x}_a$, and, consequently, there are no columns in the $\mathbf{A}$ matrix that pertain to these three coordinates. Inner constraints offer another possibility of defining the coordinate system.

Figure 4.13 shows the results of two different minimal constraints. The coordinates of station 2 are fixed in both cases. In the first case, we hold one of the coordinates of

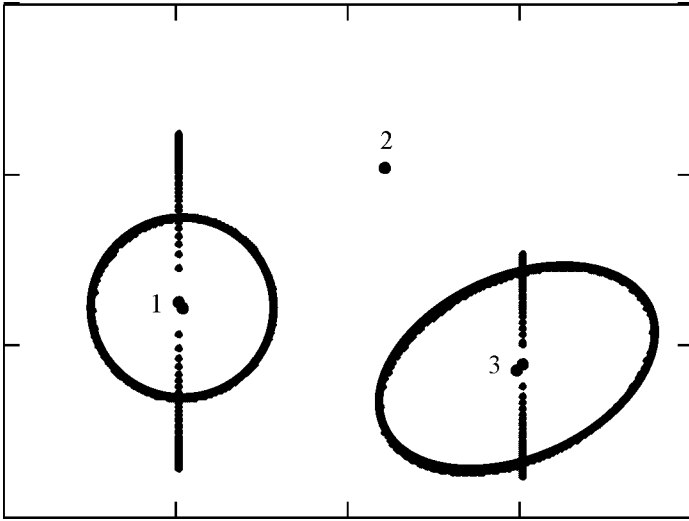


Figure 4.13 Changing minimal constraints.

station 1 fixed. This results in a degenerated ellipse (straight line) at station 1 and a regular ellipse at station 3. In the second case, we hold one of the coordinates of station 3 fixed. The result is a degenerated ellipse at station 3 and a regular ellipse at station 1. The ellipses of standard deviation change significantly due to the change in minimal constraints. Clearly, if one were to specify the quality of a survey in terms of ellipses of standard deviation, one must also consider the underlying minimal constraints. Figure 4.13 also shows that the adjusted coordinates for stations 1 and 3 differ in both cases, although the internal shape of the adjusted network 1-2-3 is the same.

The inner constraint solution, which is a special case of the minimal constraint solutions, has the property that no individual coordinates need to be held fixed. All coordinates become adjustable; for s stations of a plane network, the vector $\mathbf{x}_a$ contains $2s$ coordinate parameters. The ellipses reflect the geometry of the network, the distribution of the observations, and their standard deviations. Section 4.8 contains the theory of inner constraints. The elements for drawing the ellipses are taken from the cofactor matrix (4.203) and Equation (4.212) gives the adjusted parameters. A first step is to find a matrix $\mathbf{E}$ that fulfills $\mathbf{A}\mathbf{E}^T = \mathbf{O}$ according to (4.177). The number of rows of $\mathbf{E}$ equals the rank defect of $\mathbf{A}$. For trilateration networks with distances and angles we have

$$\mathbf{E} = \begin{bmatrix} \cdots & 1 & 0 & \cdots & 1 & 0 & \cdots \\ \cdots & 0 & 1 & \cdots & 0 & 1 & \cdots \\ \cdots & -y_i & x_i & \cdots & -y_k & x_k & \cdots \end{bmatrix} \quad (4.387)$$

Four constraints are required for triangulation networks that contain only angle observations. In addition to fixing translation and rotation, triangulation networks also require scaling information. The $\mathbf{E}$ matrix for such networks is

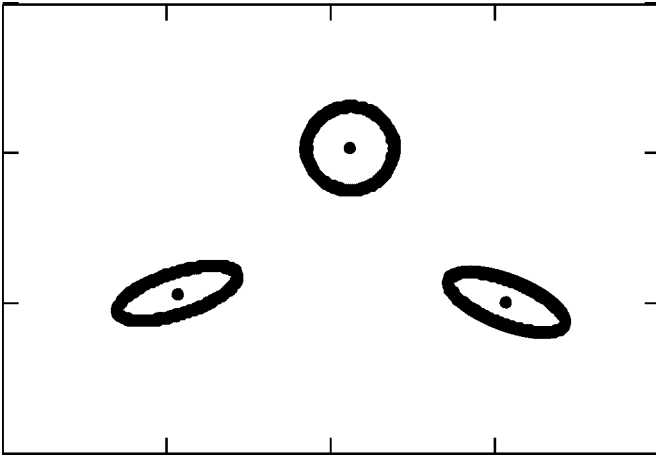


Figure 4.14 Inner constraint solution.

$$\mathbf{E} = \begin{bmatrix} \cdots & 1 & 0 & \cdots & 1 & 0 & \cdots & 1 & 0 & \cdots \\ \cdots & 0 & 1 & \cdots & 0 & 1 & \cdots & 0 & 1 & \cdots \\ \cdots & -y_i & x_i & \cdots & -y_j & x_j & \cdots & -y_k & x_j & \cdots \\ \cdots & x_i & y_i & \cdots & x_j & y_j & \cdots & x_k & y_k & \cdots \end{bmatrix} \quad (4.388)$$

The inner constraint solution is shown in Figure 4.14. Every station has an ellipse. The minimal constraint solutions and the inner constraint solution give the same estimates for residuals, a posteriori variance of unit weight, and redundancy numbers. While the estimated parameters (station coordinates) and their covariance matrix differ for these solutions, the same result is obtained when using these quantities in covariance propagation to compute other observables and their standard deviations.

Example 4: Weighting all approximate coordinates can also provide the coordinate system definition. Table 4.3 contains expressions that include a priori weights on the parameters. If the purpose of the adjustment is to control the quality of the observations, it is important that the weights of the approximate coordinate are small enough to allow observations to adjust freely. For example, if the approximate coordinates are accurate to 1 m, one can use a standard deviation of, say, 1–2 m, or even larger. Ideally, of course, the weight should reflect our knowledge of the approximate coordinates by using meaningful standard deviation. One may prefer to use large standard deviations just to make sure that the internal geometry of the network solution is not affected.

Figure 4.15 shows all ellipses for the case when each approximate station coordinate is assigned a standard deviation of 10 m. The ellipse at each network point is approximately circular. The size of the ellipses is in the range of the a priori coordinate standard deviations. The ellipses in Figure 4.15 imply a scale factor of 10^6 when compared to those in Figures 4.13 and 4.14, which roughly corresponds to the

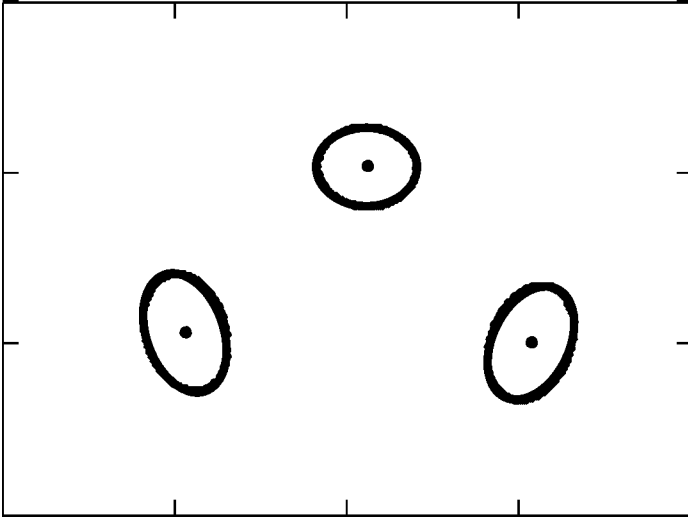


Figure 4.15 Weighting approximate coordinates to define the coordinate system.

ratio of the variances of the approximate coordinates over the average variance of the observations.

The weighted parameter approach is also a convenient way of imposing minimal constraints. Only a subset of three approximate coordinates needs to be weighted in case of a plane angle and distance network.

4.13 KALMAN FILTERING

Least-squares solutions are often applied to surveying networks whose network points refer to monuments that are fixed to the ground. When using the sequential least-squares approach (4.140) to (4.144), the parameters $\mathbf{x}$ are typically treated as a time invariant. The subscript i in these expressions identifies the set of additional observations added to the previous solution that contains the sets $1 \leq i \leq i - 1$. Each set of observations merely updates $\mathbf{x}$, resulting in a more accurate determination of the fixed monuments.

We generalize the sequential least-squares formulation by allowing the parameter vector $\mathbf{x}$ to change with time. For example, the vector $\mathbf{x}$ might now contain the three-dimensional coordinates of a moving receiver, the coordinates of satellites, tropospheric delay of signals, or other time-varying parameters. We assume that the dynamic model between parameters of adjacent epochs follows the system of linear equations

$$\mathbf{x}_k(-) = \Phi_{k-1} \mathbf{x}_{k-1} + \mathbf{w}_k \quad (4.389)$$

We have used the subscript k , instead of i , to emphasize that it now indicates the epoch. The matrix Φ_{k-1} is called the parameter transition matrix. The random vector

$\mathbf{w}_k$ is the system process noise and is distributed as $\mathbf{w}_k \sim N(\mathbf{0}, \mathbf{Q}_{w_k})$. The notation $(-)$ indicates the predicted value. Thus,

$$\hat{\mathbf{x}}_k(-) = \Phi_{k-1} \hat{\mathbf{x}}_{k-1} + \mathbf{w}_k \quad (4.390)$$

$\hat{\mathbf{x}}_k(-)$ is the predicted parameter vector at epoch k , based on the estimated parameter $\hat{\mathbf{x}}_{k-1}(-)$ from the previous epoch and the dynamic model. The solution that generated $\hat{\mathbf{x}}_{k-1}$ also generated the respective cofactor matrix $\mathbf{Q}_{k-1}$. The observation equations for epoch k are given in the familiar form

$$\mathbf{v}_k = \mathbf{A}_k \mathbf{x}_k + \ell_k \quad (4.391)$$

with $\mathbf{v}_k \sim N(\mathbf{0}, \mathbf{Q}_{\ell_k})$.

The first step in arriving at the Kalman filter formulation is to apply variance-covariance propagation to (4.389) to predict the parameter cofactor matrix at the next epoch,

$$\mathbf{Q}_k(-) = \Phi_{k-1} \mathbf{Q}_{k-1} \Phi_{k-1}^T + \mathbf{Q}_{w_k} \quad (4.392)$$

Expression (4.392) assumes that the random variables ℓ_k and $\mathbf{w}_k$ are uncorrelated. The various observation sets ℓ_k are also uncorrelated, as implied by (4.88). The second step involves updating the predicted parameters $\hat{\mathbf{x}}_k(-)$, based on the observations ℓ_k . Following the sequential least-squares formulation (4.140) to (4.144), we obtain

$$\mathbf{T}_k = [\mathbf{Q}_{\ell_k} + \mathbf{A}_k \mathbf{Q}_k(-) \mathbf{A}_k^T]^{-1} \quad (4.393)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k(-) - \mathbf{K}_k [\mathbf{A}_k \hat{\mathbf{x}}_k(-) + \ell_k] \quad (4.394)$$

$$\mathbf{Q}_k = [\mathbf{I} - \mathbf{K}_k \mathbf{A}_k] \mathbf{Q}_k(-) \quad (4.395)$$

$$\mathbf{v}^T \mathbf{P} \mathbf{v}_k = \mathbf{v}^T \mathbf{P} \mathbf{v}_{k-1} + [\mathbf{A}_k \hat{\mathbf{x}}_k(-) + \ell_k]^T \mathbf{T}_k [\mathbf{A}_k \hat{\mathbf{x}}_k(-) + \ell_k] \quad (4.396)$$

where the matrix

$$\mathbf{K}_k = \mathbf{Q}_k(-) \mathbf{A}_k^T \mathbf{T}_k \quad (4.397)$$

is called the Kalman gain matrix.

If the parameter x_{k+1} depends only on the past (previous) solution x_k , we speak of a first-order Markov process. If noise w_k has a normal distribution, we talk about a first-order Gauss-Markov process,

$$x_{k+1} = \varphi x_k + w_k \quad (4.398)$$

with $w_k \sim n(0, q_{w_k})$. In many applications a useful choice for φ is

$$\varphi = e^{-T/\tau} \quad (4.399)$$

which implies that the variable x is exponentially correlated, i.e., the autocorrelation function is decreasing exponentially (Gelb, 1974, p. 81). The symbol τ denotes the correlation time, and T denotes the time difference between epochs $k + 1$ and k . The variance of the process noise for correlation time τ is

$$q_{w_k} = E(w_k w_k) = \frac{\tau}{2} [1 - e^{-2T/\tau}] q_k \quad (4.400)$$

with q_k being the variance of the process noise (Gelb, 1974, p. 82). The quantities (τ, q_k) could be initially determined from data by fitting a sample mean and sample autocorrelation function.

As τ approaches zero, then $\varphi = 0$. This describes the pure white noise model with no correlation from epoch to epoch. x can be thought of as a random constant that is a nondynamic quantity.

As τ approaches infinity, we obtain the pure random walk. Applying l'Hospital rule for computing the limit or using series expansion, we obtain $\varphi = 1$ and $q_{w_k} = T q_k$. The random noises w_k are uncorrelated.

In general, both the dynamic model (4.389) and the observation model (4.391) are nonlinear. The extended Kalman filter formulation (Gelb, 1974, p. 187) applies to this general case. The reader is urged to consult that reference or other specialized literature for additional details on Kalman filtering.